TERM 1

Primary 6 Science

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Explore, experiment and discover the wonders of Science!

Chapter 1:
Conditions for Burning

Burning needs oxygen (O_2) and produces carbon dioxide (CO_2).

O_2

Oxygen from the air

CO_2

Carbon dioxide released into the air

Things that burn

Candle

Wood

Gas

Paper

Charcoal

Remember:
Burning happens when we have
FUEL + OXYGEN + HEAT!

Lesson 1-1: How DO Things Burn and Air

What's meant by combustion?

- Combustion means **burning.**

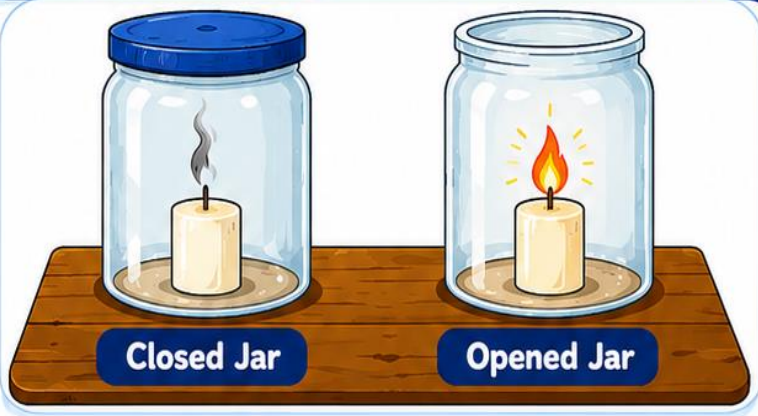
Conditions needed for combustion to occur and continue.

- There must be **Fresh Air (Oxygen)** for things to burn and keep burning.
- Fresh air must **constantly flow in and out** to allow combustion to occur and continue.
- If the air supply stops or its amount is not enough, **combustion stops.**

Experiment of burning things inside a jar

Steps

1. Get 2 jars.
2. Put a candle in each jar.
3. Light the candles, then cover one jar with a lid and leave the other one opened.



Observation:

- The flame of the candle placed in the closed jar goes out after a while.
- The flame of the candle in the opened jar continues.

Conclusion:

- Fresh air helps the candle in the open jar continue burning, whereas the flame in the closed jar goes out after a while because the oxygen in the air is gradually used up **(CONSUMED)** during combustion.

Activity: The Importance of Air During Combustion

Tools:

Steps:

1. Spread the clay on the wooden board.

2. Fix the candle on the clay and light it.

3. Cover it with the bottomless jar.

4. Put the lid on the jar.

Observation:
The candle flame goes out after short time.

5. Cut the clay to create a gap at the bottom and light the candle.

Observation:
The candle burns with a weaker flame.

6. Remove the metal lid completely.

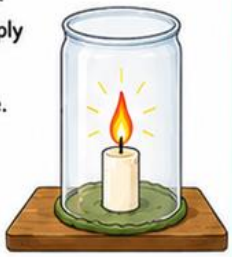
Observation:
The flame becomes brighter and stronger.

Conclusion:

- When the jar is covered with the lid, the air inside the jar is gradually consumed by the candle. As the oxygen level decreases, the flame eventually goes out and combustion stops.

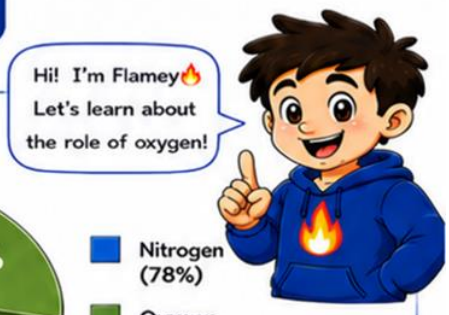
- When the bottom of the jar is slightly raised, a small amount of fresh air enters the jar, allowing limited combustion to continue.

- When the lid is removed completely, the air supply increases, allowing combustion to continue. As a result, the flame becomes stronger.

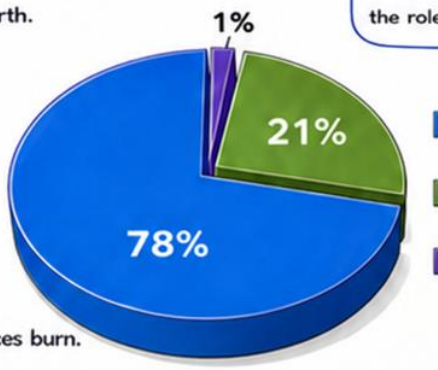


Lesson 1.2: The Role of Oxygen

What is air?



- Air is a mixture of different gases that surrounds the Earth.
- It is invisible, colorless, and odorless.
- Air is made of:
  - Nitrogen gas which represents 78%.
  - Oxygen gas which represents 21%.
  - Other gases which represents 1% including Carbon dioxide (0.04%), water vapor, ...etc



- Oxygen gas supports combustion because it helps substances burn.
- Nitrogen and carbon dioxide do not support combustion, but they have other important uses.

Note:

- ⭐ Oxygen represents 21% or 1/5 of the volume of air.
- ⭐ Oxygen is necessary for combustion (burning).
- ⭐ It is necessary for breathing of all living organisms.

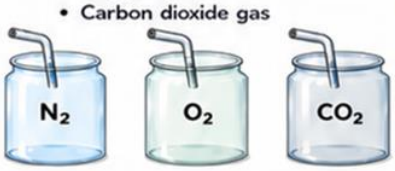
Activity: Gases That Help in Combustion

Tools:

Nitrogen gas cylinder | Oxygen gas cylinder | Carbon dioxide gas cylinder | 3 gas jars | 3 lids | Water basin | 3 candles | Candle holders | Bent tubes

Steps:

1. Prepare equal amounts of the following gases in 3 jars:
   - Nitrogen gas
   - Oxygen gas
   - Carbon dioxide gas



2. Place a lighted candle in each jar.

3. Observe the candle in each jar.

Observation:

The flame goes out in both the nitrogen and carbon dioxide jars, but it continues burning in the oxygen jar.

Conclusion:

- ✔ Oxygen gas helps in burning.
- ✔ Nitrogen and carbon dioxide gases don't help in burning.
- ✔ If you put a lit incense stick (glowing splint) into a jar of oxygen it lights up again while if put into a jar containing carbon dioxide it goes off without changing any thing.

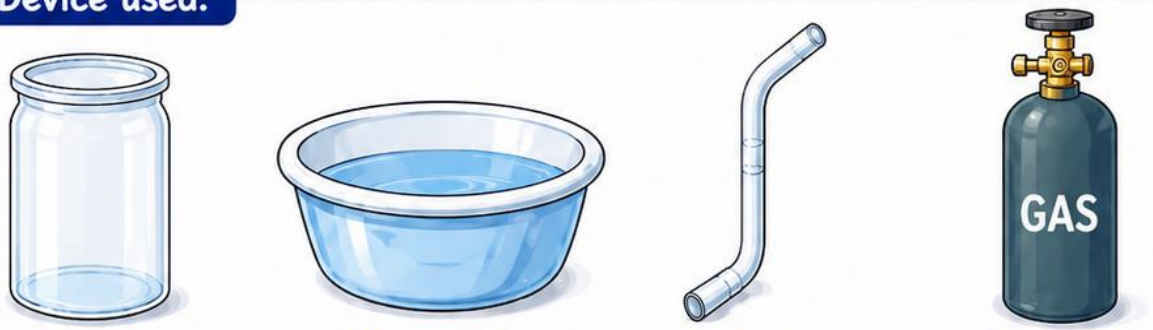
How can we collect a gas?

Any gas (oxygen, nitrogen or carbon dioxide) can be collected by the following method:

1. Fill a jar completely with water.
2. Invert the jar and place it in a water basin.
3. Open the gas cylinder to let the gas enter the jar and displaces the water pushing the water out.
4. About 70% of the jar should be filled with the gas before removing it from the water basin.

Gas is collected by Water Displacement.

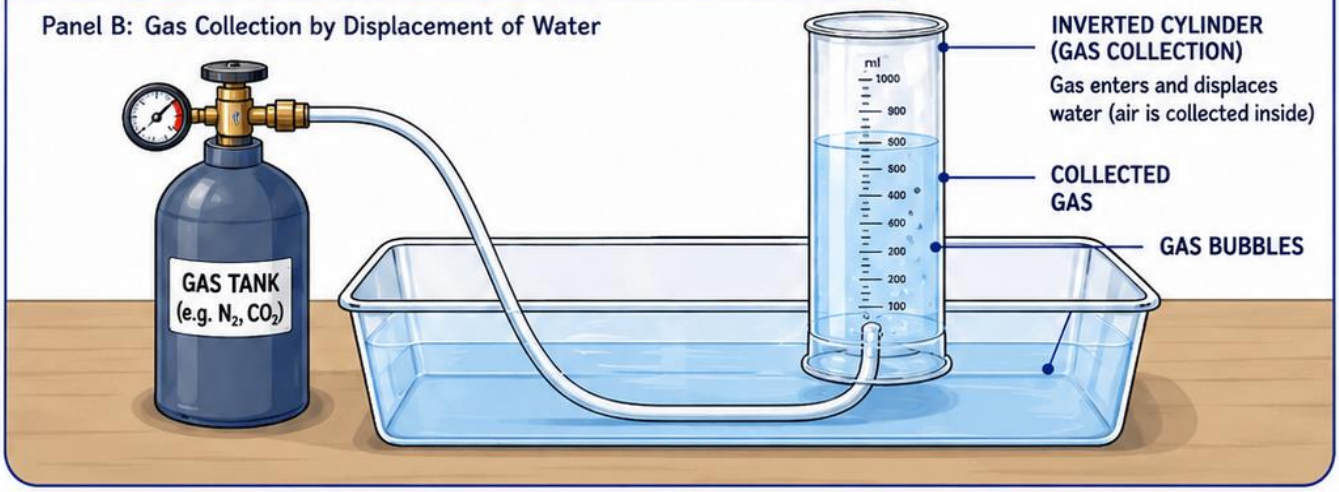
Device used:



Gas jar | Water basin | Bent tube | Gas cylinder

EXPERIMENTS IN GAS ANALYSIS AND COLLECTION

Panel B: Gas Collection by Displacement of Water

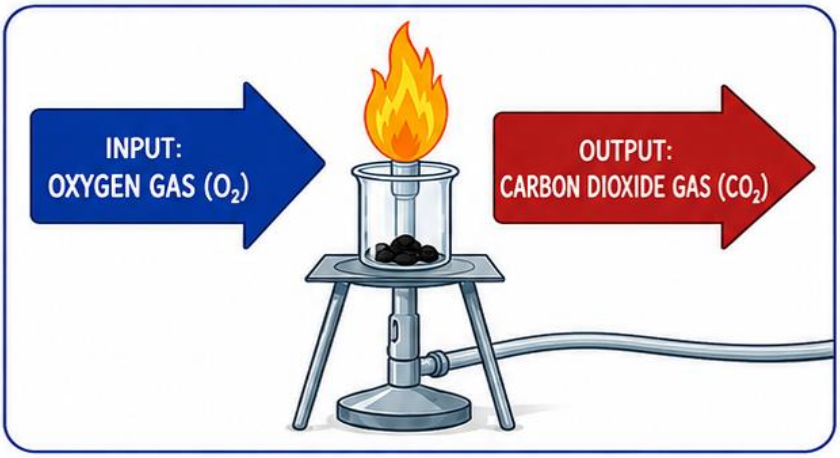


Lesson 1.3: Air After Burning

What kind of changes happen to air after things burn?

When things burn, a part of the oxygen gas in air is used (oxygen is necessary for burning), and carbon dioxide is produced, so:

- The amount of oxygen gas decreases.
- The amount of carbon dioxide gas increases.



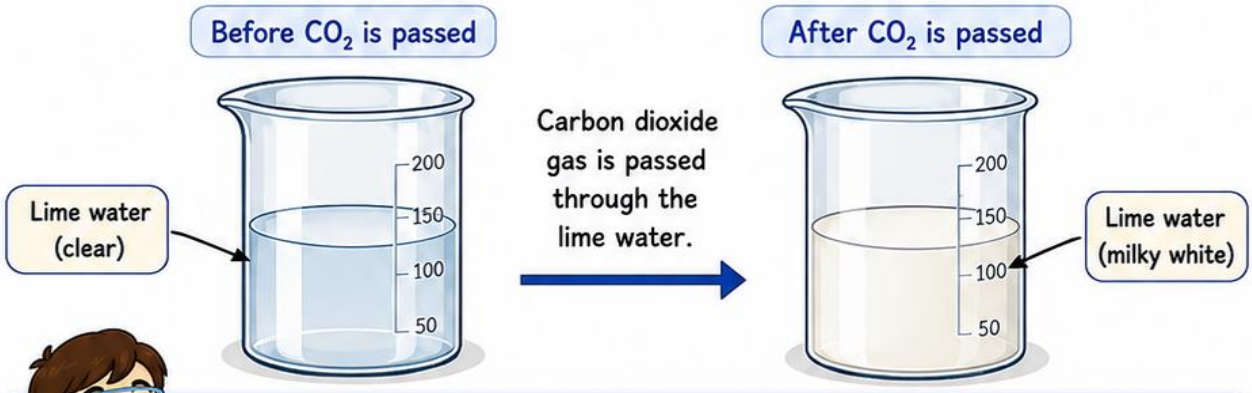
How to prove that carbon dioxide gas is produced during burning?

- Lime water is used to test for the presence of carbon dioxide.
- Lime water is colorless, but when carbon dioxide is passed through it, it turns milky white due to a reaction with the carbon dioxide gas.

When carbon dioxide reacts with lime water, it forms calcium carbonate (white solid), so the color becomes milky.

Testing Carbon Dioxide with Lime Water

Lime water turns milky white when carbon dioxide gas is passed through it.



The lime water changes from colorless to milky white because it reacts with carbon dioxide gas.

Let's find out what has increased in the air after burning.

1 Preparation:

- 2 gas jars

- Lids

- Candle

- Combustion spoon

- Limewater

- Safety goggles

- Gas lighter

2 Experiment:

1 Put limewater into each of the 2 jars.

2 Put a lit candle into one jar, put the lid on, and wait until the flame goes out.

3 After taking out the candle, shake the 2 jars.

Caution!

- Wear safety goggles so the limewater doesn't get in your eyes.

3 Compare:

Let's check the color change of the limewater in the 2 jars.

Jar with candle	Jar without candle
Limewater turns milky white.	Limewater remains colorless.

Conclusion:

The limewater in the jar with the candle turned milky white, while the limewater in the other jar remained colorless.

So, Carbon dioxide gas has increased in the air after burning.

Limewater turns milky white when carbon dioxide gas is present.

Tools Used to Detect Gases

Did You Know?

The levels and the kinds of gases in the air around us may be very different — so we use special tools to check them accurately.

Glass detection (indicator) tubes

Sampling pump (aspirator)

First: Glass tubes for detecting gases

They are small glass tubes containing chemical substances that react with certain gases only.

Importance of gas detection tubes

- They help us (industries, labs, workers, and others) give and demand-supply their concentration.

How gas detection tubes work

When the sample (tested) gas is drawn into the tube through the pump, the gas in the air or in the sample reacts inside the tube and changes to a specific color.

How to use gas detection tubes

1 Insert the tip of the tube carefully into one end of the holder and break it.

2 Pull the tube completely into the holder.

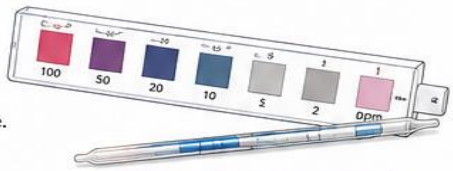
3 Break off one end of the tube at the opposite point.

4 Connect the pump (aspirator) has connected tightly.

5 Observe the color change inside the tube.

6 Remove the tube from the pump after use.

1. Insert one tip of the tube into the inlet of the hand pump.
2. Pump completely and repeat several times.
   This draws the sampled amount of test gas to be in contact with the gas inside the tube.
3. When the chemical reaction (inside the tube) reacts with the gas in the sample, the color of the measured scale (or a color change, for comparing it with the color chart on the tube packaging) gives the concentration (%) of the gas inside the sample.



The number on the chart indicates the percentage (%) of the gas inside the sample.

Using the right tools helps us know what gases are present in the air and in what amounts.

2 Second: Gas Monitoring Equipment: It is an electronic device with a sensor (detector) that shows the percentage (%) of the gas in the surrounding air on the screen as numbers.

Gas monitoring equipment

Note

- There are different types of gas monitoring devices that measure the percentage of oxygen gas and the percentage of carbon dioxide gas at the same time.
- This type of device is available in specialized hospitals in the intensive care units for patients who suffer from breathing problems.
- It is portable and easy to use.
- It does not expose the sensor to breakage.

Important Notes

- There are different types of gas monitoring devices that measure the percentage of oxygen gas and the percentage of carbon dioxide gas at the same time.
- This type of device is available in specialized hospitals in intensive care units for patients who suffer from breathing problems.
- It is portable and easy to use.
- It does not expose the sensor to breakage even when the device tube are bent.

Combustion and the Future of Energy

1 Discovering Fire and Ancient Egyptian Civilization:

- People discovered fire thousands of years ago in nature and gradually used it.
- They needed all things that required oxygen gas to burn.

Discovering fire

2 How Did the Ancient Egyptians Obtain Light?

- The ancient Egyptians used vegetable oils and animal fats to make lamps and torches.
- They used clever designs to transport the light of torches to perform the process of combustion.

Fire and light played a big role in the life of ancient people and helped them build the first steps of civilization.

1

How did the ancient Egyptians obtain oxygen for combustion?

- When the percentage of **oxygen** increases, combustion happens faster and the temperature rises.
- The ancient Egyptians used **bellows** (blowing tools) to push more air into the furnace.
- More **oxygen** in the air raised the **temperature** of the fire used in ancient Egyptian furnaces and tools for metal smelting.

Second: Reducing Earth's Temperature by Decreasing Carbon Dioxide Gas

The greenhouse effect (global warming)

- Carbon dioxide gas is produced from different combustion processes.
- Increasing the percentage of carbon dioxide in the air raises the temperature of the Earth, a phenomenon known as the **greenhouse effect (global warming).**

Concept

The **greenhouse effect (global warming)** is the rise in Earth's temperature caused by an increase in the percentage of carbon dioxide gas in the air.

Renewable Energy

Solar power plant

1. Renewable Energy:

- It is energy obtained from natural sources that do not run out, such as the sun, wind, water, and biomass.

Example The sun and wind, and their energy can be used to generate electricity.

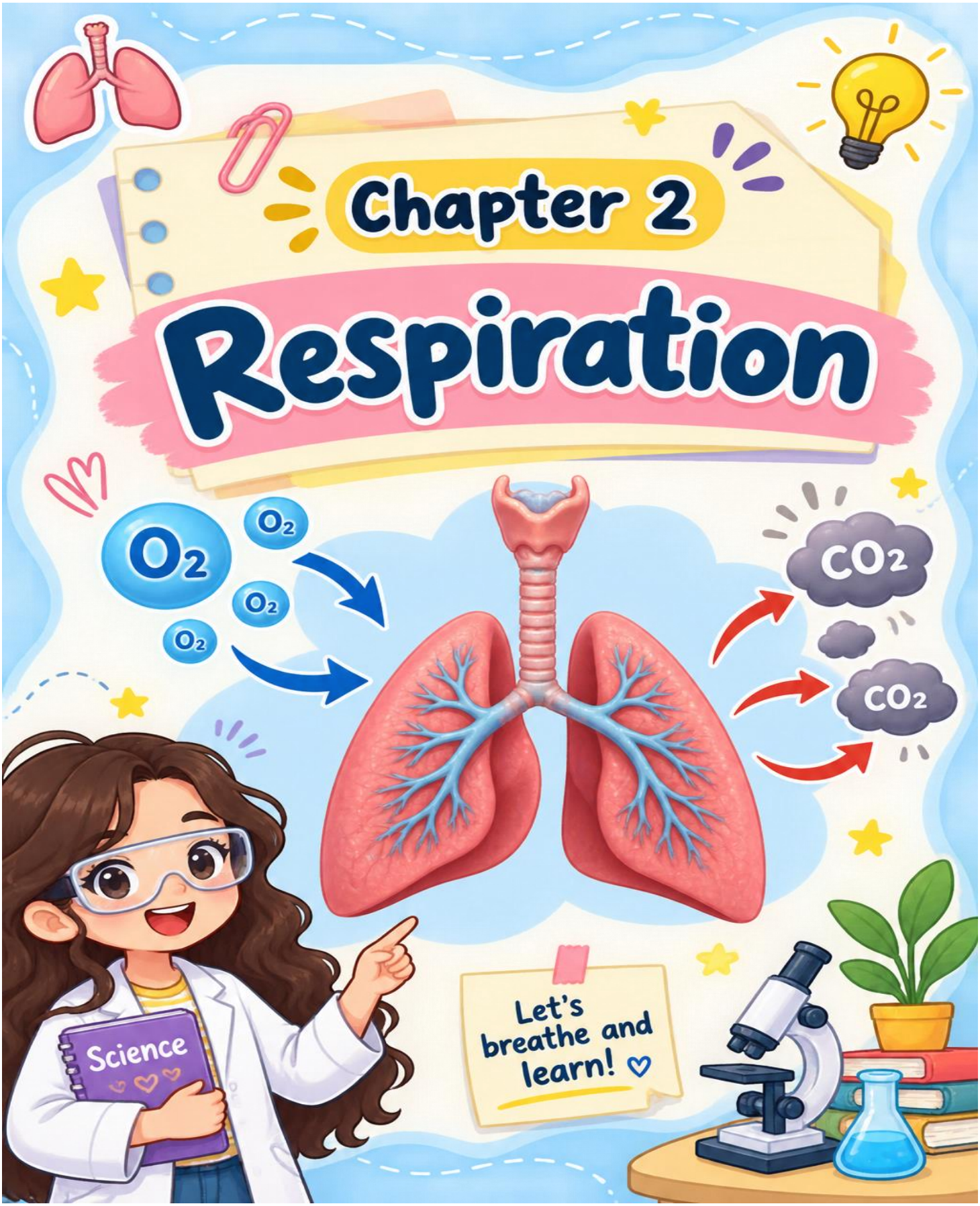
2. Solar Power Plant

- It is a power plant that uses sunlight to generate electricity.
- It is located near **Benban village**, about 20 km from Aswan city.

3. Benefits of Solar Power Plant:

- Produces clean electricity without burning fossil fuels, so it does not produce carbon dioxide emissions.
- This reduces the negative effects of carbon dioxide gas.

Using clean energy and reducing carbon dioxide helps protect our planet and build a better future.

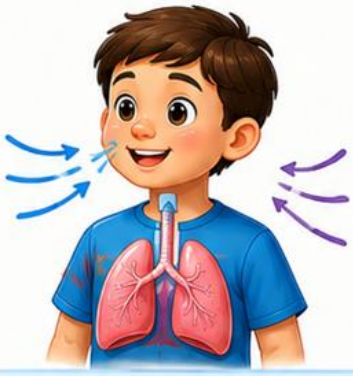

Chapter 2
Respiration
O2
O2
O2
O2
CO2
CO2
Science
Let's breathe and learn!

Lesson 2-1: How Breathing Works

Living organisms perform many important life processes that help them stay alive and healthy. These include digestion, respiration, excretion, growth, and more.

What's meant by breathing?

Breathing is how we take in oxygen from the air and let out carbon dioxide. This helps our body get the oxygen it needs to release energy from food.

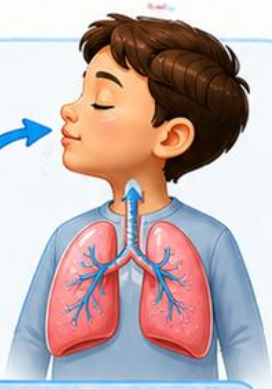


Our body needs energy to move, think, play and do everything!

Breathing has two main parts: inhalation and exhalation.

Inhalation

We breathe in air that contains oxygen (O_2) from the air.



O_2

Oxygen (O_2) goes into our lungs.

Exhalation

We breathe out air that contains carbon dioxide (CO_2) into the air.

CO_2

Carbon dioxide (CO_2) comes out of our lungs.

Remember!
Our body uses oxygen to release energy from the food we eat.

The breathing cycle:

Breathe in → Oxygen (O_2) → Breathe out → Carbon dioxide (CO_2)

⋆ Remember: Oxygen helps our body release energy from the food we eat.

Lesson 2-1: How Breathing Works

Activity: What is the air we breathe out?

We will do two simple experiments to compare the air we breathe out with the air around us.

Tools:

Plastic bags

Lime water

Gas detector tubes (for oxygen and carbon dioxide)

Safety goggles

Scissors

A. Test with Lime Water

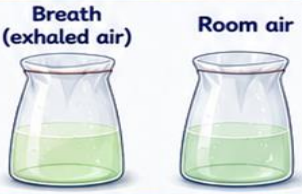
1 Take two plastic bags.

2 Breathe into one bag to fill it with your breath. Fill the other bag with normal room air.

3 Pour a small amount of lime water into each bag. Close the bags and shake them well.

Observation:

Lime water turns milky in the first bag (more quickly) than in the second bag.



Conclusion:

The air we breathe out contains more carbon dioxide. This gas reacts with lime water and makes it turn milky.

B. Test with a Gas Detector Tube

1 Do steps 1 and 2 again.

2 Use the gas detector tubes to check the amount of oxygen (O_2) and carbon dioxide (CO_2) in each bag.

Observation:

The gas detector tubes show more carbon dioxide and less oxygen in the first bag (exhaled air).

Conclusion:

The air we breathe out has a larger amount of carbon dioxide gas.

Remember!
Always wear safety goggles and be careful with tools.

We can use gas testing tubes!

Oxygen relights a glowing splint.

Carbon dioxide turns lime water milky.

Breathing helps our body get the oxygen it needs and get rid of carbon dioxide!

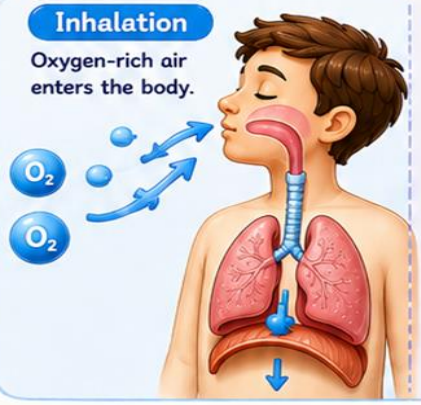
Lesson 2-2: How Do Lungs Work?

1 What is the function of the respiratory system?

The respiratory system brings oxygen (O_2) into the body during inhalation and removes carbon dioxide (CO_2) from the body during exhalation.

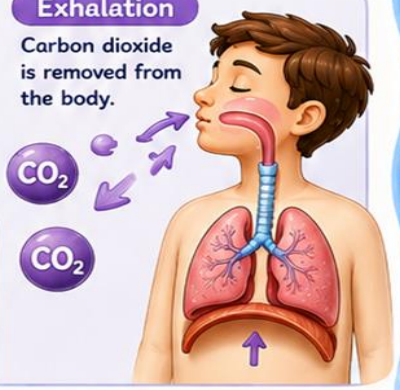
Inhalation

Oxygen-rich air enters the body.

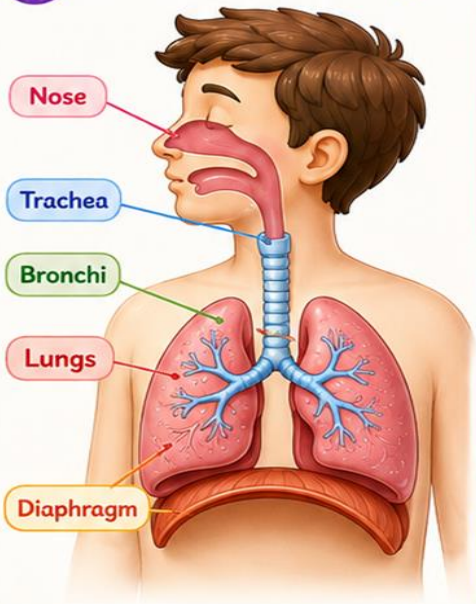


Exhalation

Carbon dioxide is removed from the body.



2 What are the main parts of the respiratory system?



1 Nose

The nose is the entrance of air. It filters, warms, and moistens the air we breathe in.

2 Trachea (Windpipe)

A strong tube that carries air from the nose or mouth down to the lungs.

3 Bronchi

Two main tubes that branch off from the trachea and carry air to each lung.

4 Lungs

The lungs are soft, spongy organs where gas exchange takes place.

5 Bronchioles

Tiny branches of the bronchi that carry air deeper into the lungs.

6 Alveoli

Very tiny air sacs where oxygen enters the blood and carbon dioxide leaves the blood.

7 Diaphragm

A dome-shaped muscle under the lungs. It moves up and down to help the lungs expand (inhalation) and relax (exhalation).

The path of air

Nose → Trachea → Bronchi → Bronchioles → Alveoli

Air follows this path until it reaches the tiny air sacs (alveoli) where gas exchange happens.

How it works

O_2 Oxygen from the air passes into the blood.

CO_2 Carbon dioxide from the blood passes into the air and is exhaled.

1 How do gases exchange inside the lungs?

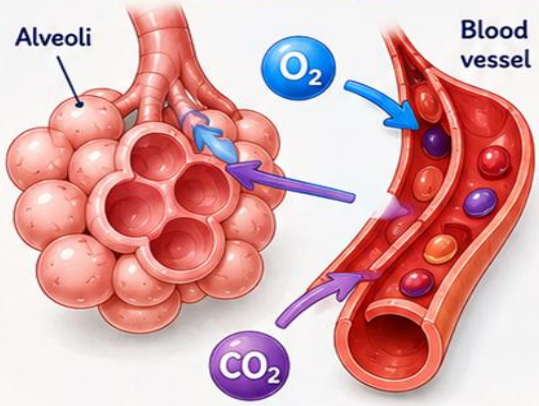
Gas exchange happens in the **alveoli**. Alveoli are tiny air sacs in the lungs.

O_2 **Oxygen (O_2)** moves from the air in the alveoli **into the blood.**

CO_2 Carbon dioxide (CO_2) moves from the blood **into the alveoli.**

Carbon dioxide is then removed from the body when we **breathe out.**

Inside the lungs



In simple words:

O_2 → alveoli → blood

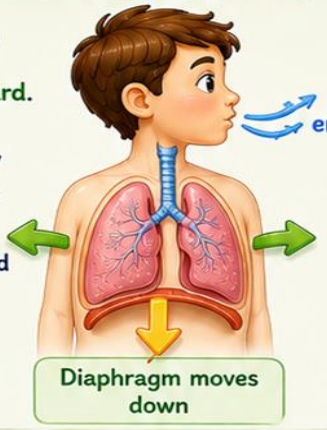
CO_2 → blood → alveoli → breathed out

2 How does your chest move during breathing in and out?

The movement of the **diaphragm** helps the lungs fill with air and empty it.

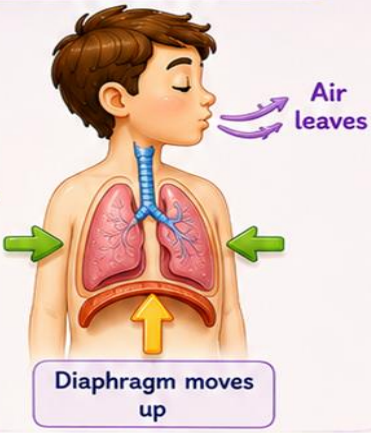
During inhalation (breathing in)

1. The diaphragm contracts and moves **downward.**
2. The chest cavity becomes **larger.**
3. The lungs expand and air **enters.**



During exhalation (breathing out)

1. The diaphragm relaxes and moves **upward.**
2. The chest cavity becomes **smaller.**
3. The lungs get smaller and air **leaves.**



Remember!

- Inhale → chest expands → air enters.
- Exhale → chest gets smaller → air leaves.

Breathe in (Inhalation)

Breathe out (Exhalation)

Our lungs are amazing! They help bring oxygen to our body and remove carbon dioxide to keep us healthy.

Lesson 2-2

How Do Lungs Work?

2-2

O_2 **Oxygen** is carried in the blood from the **lungs** to all body cells.

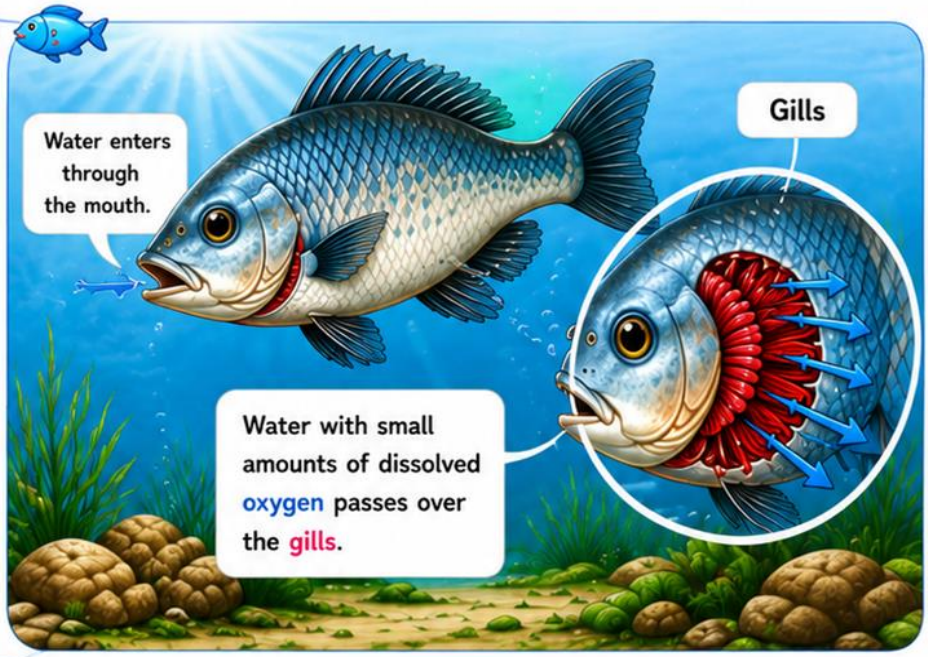
CO_2 **Carbon dioxide** is carried from the body cells back to the **lungs** and is breathed **out**.

Living things breathe in different ways depending on **where they live**.

1 Breathing in Fish

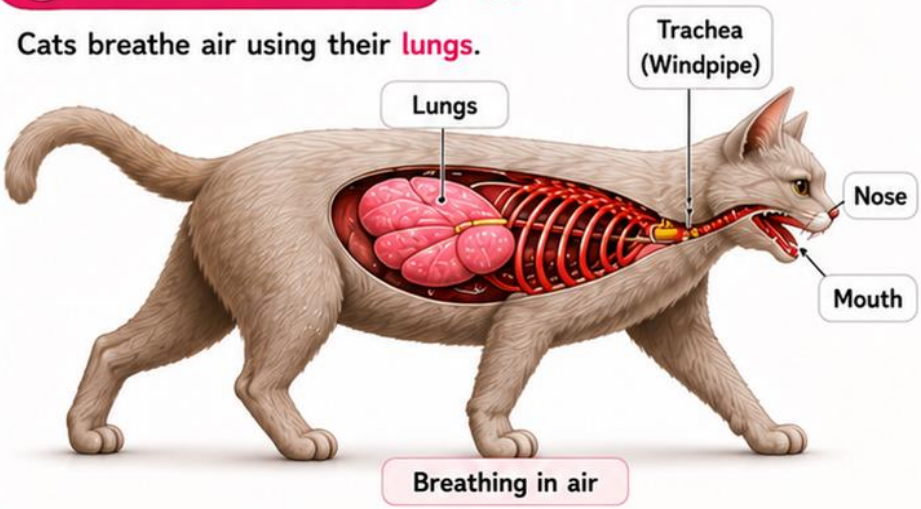
Fish take oxygen from water using their **gills**.

1. Water and **oxygen** dissolve in water and pass through the **gills** into the blood.
2. The blood delivers **oxygen** to the body cells.
3. The **gills** take **carbon dioxide** out of the blood, and the water carries it away when it flows out.



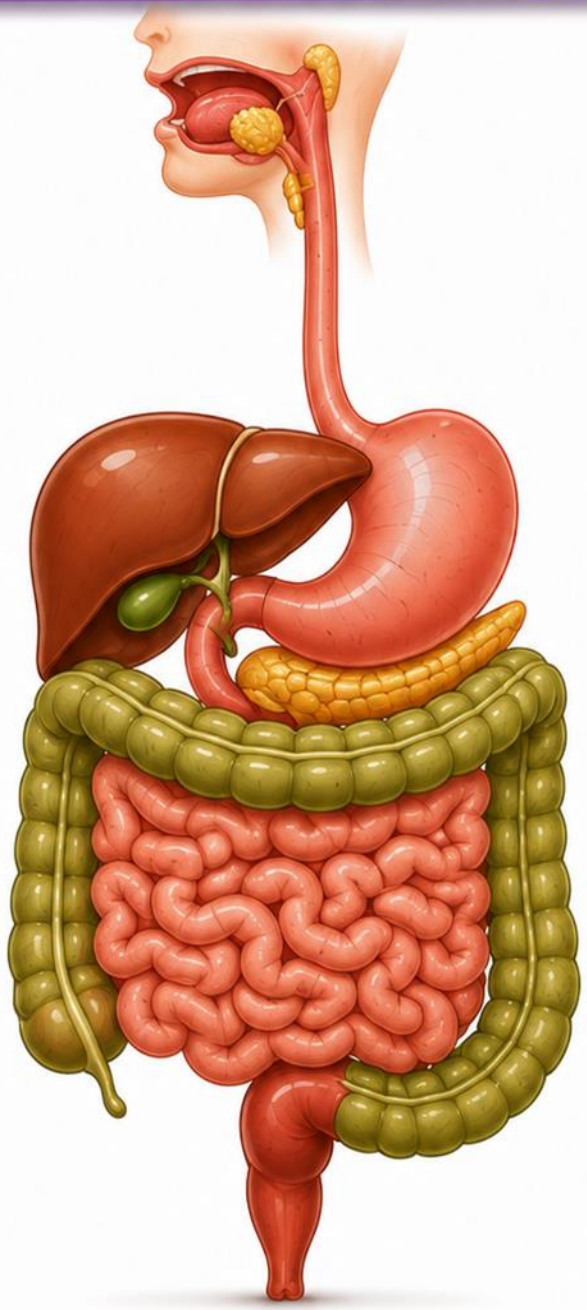
2 Breathing in Cats

Cats breathe air using their **lungs**.



1. Air (with **oxygen**) enters through the **nose** or **mouth**.
2. Breathing in opens the pathway and pulls air into the **lungs**.
3. The **oxygen** in the air goes into the blood, which then travels to all parts of the body.

Chapter 3
DIGESTION



WHAT GOES IN

Water

Food

WHAT COMES OUT

Waste

Lesson 3-1 :

Digestion and How Does It Work?

Digestion is the process of breaking down the food we eat into smaller pieces so our body can absorb the **nutrients** it needs for **energy**, **growth**, and **repair**.

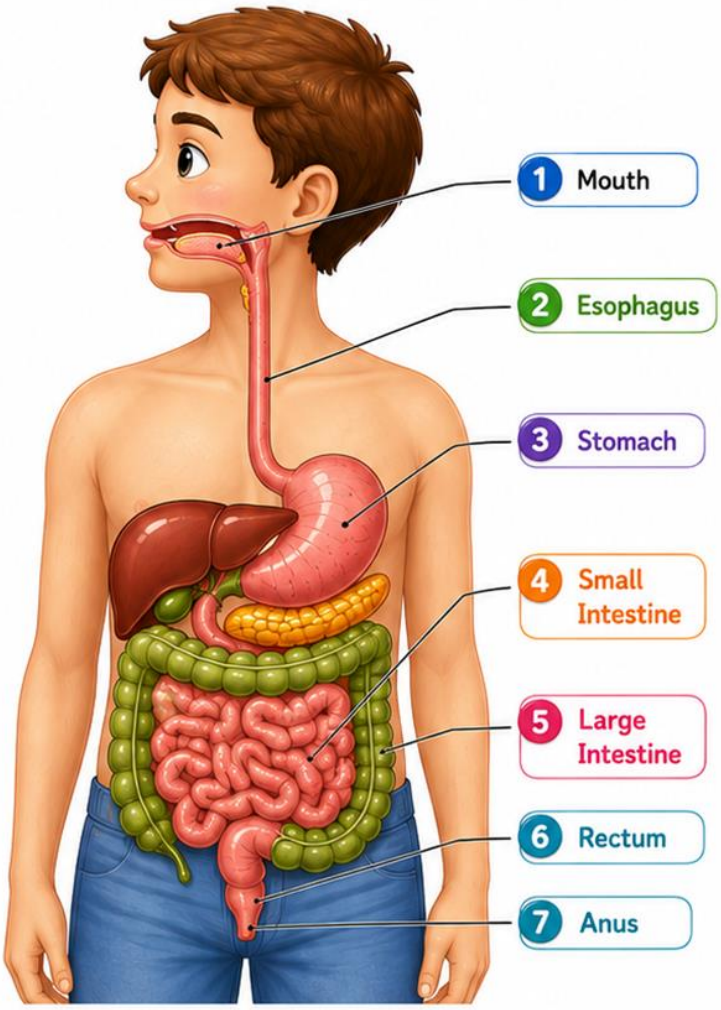
What is Digestion?

When we eat, our body changes the food into simple **nutrients**. These nutrients go into the **blood** and are used by all parts of the body.

The Main Parts of the Digestive System

1. **Mouth** – Food enters here. Teeth chew the food and saliva starts to break it down.
2. **Esophagus** – A tube that moves food from the mouth to the stomach.
3. **Stomach** – Stores food and mixes it with strong juices to break it down further.
4. **Small Intestine** – The place where most nutrients are absorbed into the blood.
5. **Large Intestine** – Absorbs water from the leftover food and forms waste.
6. **Rectum** – Stores the waste.
7. **Anus** – The waste leaves the body through this opening.

The Digestive System



Our body is amazing! It knows how to turn food into **energy**!

Eat **healthy food**, chew well, and drink plenty of **water** to help your digestive system stay strong!

Lesson 3-1

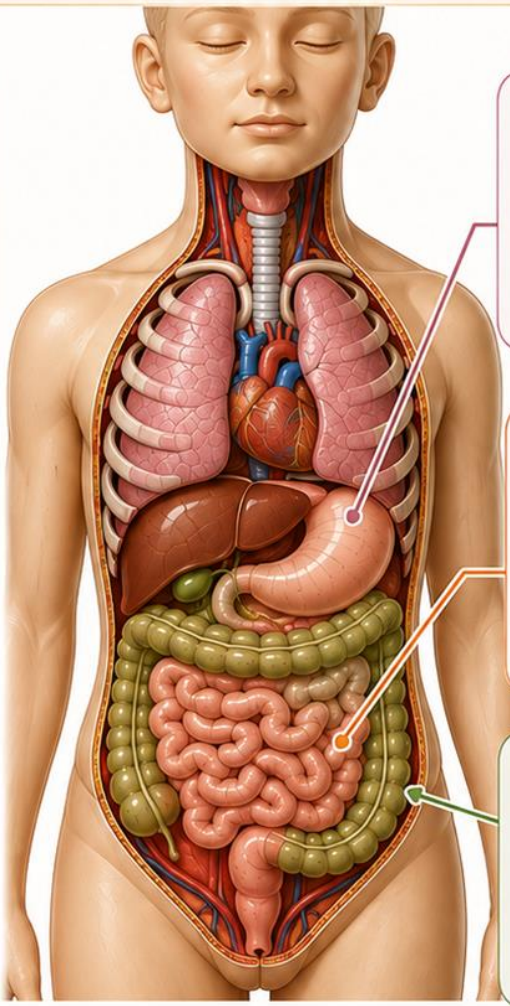
Where Are the Stomach and Intestines?

The stomach and intestines are important organs in the digestive system. They are inside your **abdomen** and work together to digest food and absorb nutrients.

Where in the Body?

These organs are located in the **abdomen**, which is the area between the chest and the hips.

1. **Chest (Thorax)**
   Where the heart and lungs are.
2. **Upper Abdomen**
   Just below the ribs.
3. **Middle Abdomen**
   The center part of the abdomen.
4. **Lower Abdomen**
   Near the hips. Where the lower intestines and rectum are.



STOMACH

The stomach is in the upper abdomen, mostly on the **left side**, just below the ribs.

SMALL INTESTINE

The small intestine is in the **middle** and **lower** abdomen. It is long and coiled below the stomach.

LARGE INTESTINE (COLON)

The large intestine **surrounds** the small intestine. It goes around the edges of the abdomen.

- The **stomach** is where food is mixed and broken down.
- The **small intestine** absorbs most of the nutrients.
- The **large intestine** absorbs water and removes waste.

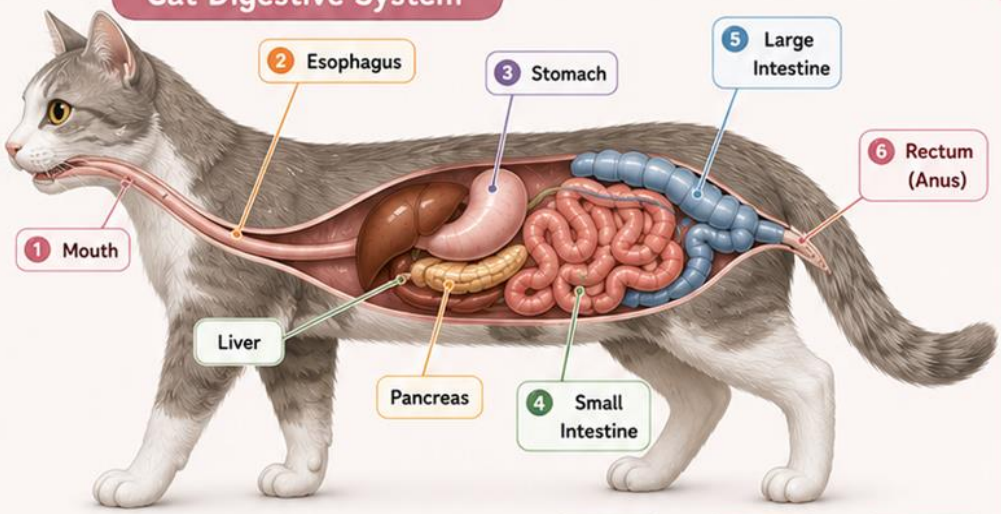
Remember: The stomach is mainly in the upper-left abdomen. The small intestine is in the middle and lower abdomen, surrounded by the large intestine.

Digestive Systems of a Cat and a Fish

All animals need food to live and grow.
Their digestive systems break down food and absorb nutrients.

Cat Digestive System

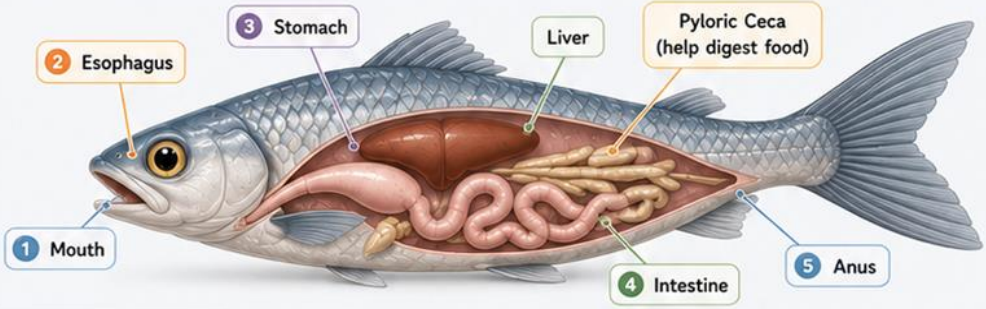
1. Food enters the mouth.
2. Food travels down the esophagus to the stomach.
3. The stomach mixes the food with juices and breaks it down.
4. The small intestine absorbs nutrients from the food.
5. The large intestine absorbs water and forms waste.
6. Waste is stored in the rectum and leaves the body through the anus.



Food Path: Mouth → Esophagus → Stomach → Small Intestine → Large Intestine → Rectum/ Anus

Fish Digestive System

1. Food enters the mouth.
2. Food moves down the esophagus to the stomach.
3. The stomach stores food and starts to break it down.
4. Nutrients are absorbed in the intestine.
5. Waste leaves the body through the anus.



Food Path: Mouth → Esophagus → Stomach → Intestine → Anus

Cats have a long small intestine and a large intestine that absorbs water and forms waste.

Both Animals Digest Food!

Cats and fish both have digestive systems to break down food, absorb nutrients, and get rid of waste. The organs are arranged differently, but the job is the same!

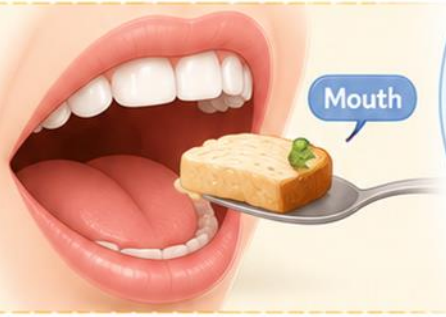
Fish have pyloric ceca to help digest food and a shorter intestine.

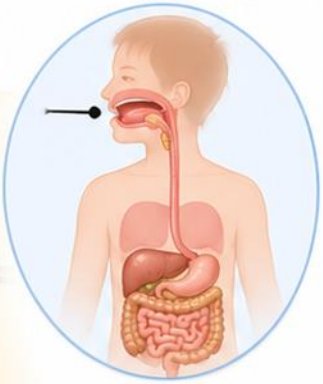
★ Both cats and fish digest food and absorb nutrients, but their digestive systems are arranged differently.

Lesson 3-2

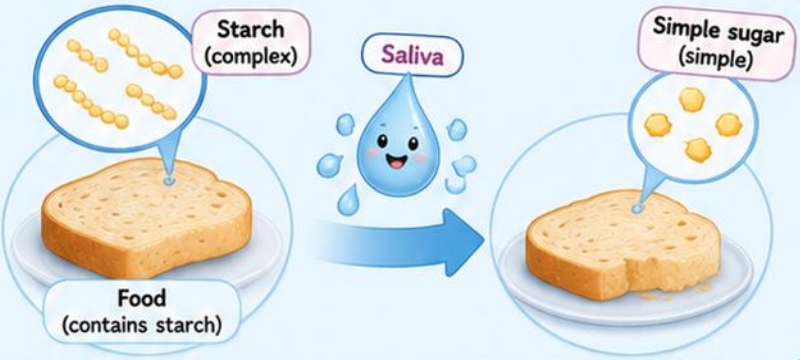
The Function of the Saliva

1 The **first step** of digestion occurs in the **mouth**.
Here, food begins to change from a complex form into a simple one so it can be easily absrobed by the body.





2 The mouth contains **saliva**.
Saliva is a **watery secretion** that helps break down **starch** in food and changes it into a simple sugar that can be easily absorbed.



3 A liquid that has the function of digesting food like saliva is called a **Digestive Juice**.

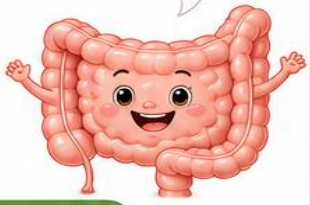
4 Digestive juices are not only produced in the **mouth**, but also in the **stomach** and **intestines**.

So, **saliva** starts the process of digestion in the mouth and helps change complex food into simple food for the body to use.

Lesson 3-3

How Does Absorption Work?

Let's learn how absorption works inside our body!

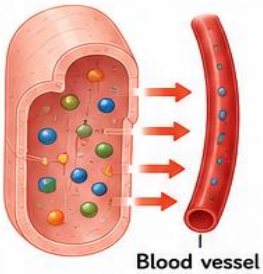


After food is digested into small, simple nutrients, the body absorbs these nutrients and uses them for energy, growth and repair.

1. What is meant by absorption?

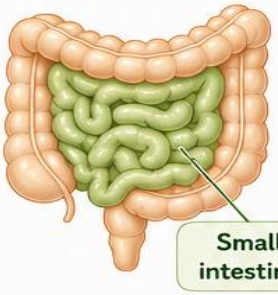
Absorption is the process where digested nutrients and water pass from the intestine into the blood.

- Nutrient
- Water
- Blood vessel



2. Where does absorption occur?

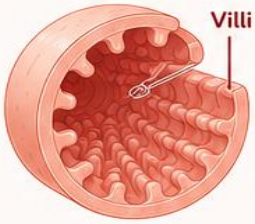
The nutrients from the digested food are mainly absorbed in the **small intestine** along with water.



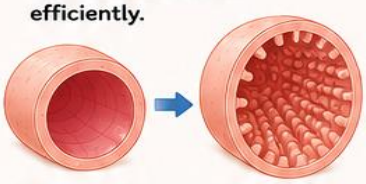
3. What is the structure of the small intestine?

Inside the small intestine there are tiny structures called **Villi.**

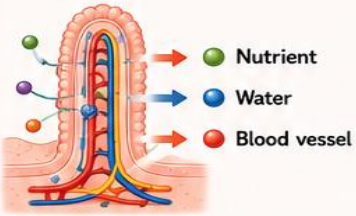
Villi are small bumps on the inside of the small intestine lined up closely.



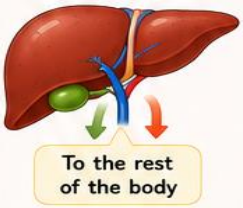
These bumps increase the inner surface area of the small intestine, so it can absorb nutrients and water efficiently.



Villi are rich in blood vessels which allow nutrients and water to be absorbed and carried by the blood.



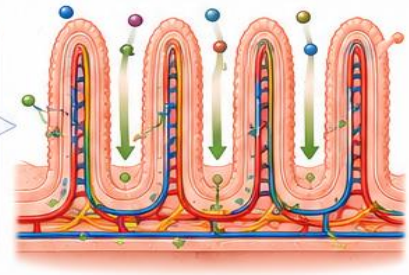
These nutrients pass through the liver which stores some of them and sends them out to the whole body when they are needed.



4. Why are villi important?

If you look at a surface of a towel, it has many loop-like threads to absorb water better than a flat cloth.

In the same way, villi increase the surface area inside the small intestine so it can absorb more nutrients and water.



This helps:

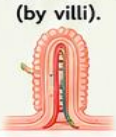
- ✔ Provide energy for the body
- ✔ Support growth and repair
- ✔ Keep the body healthy

In Summary – The Absorption Process

1. Food is digested.

2. Nutrients become small and simple.

3. Absorbed in the small intestine (by villi).



4. Carried by blood.

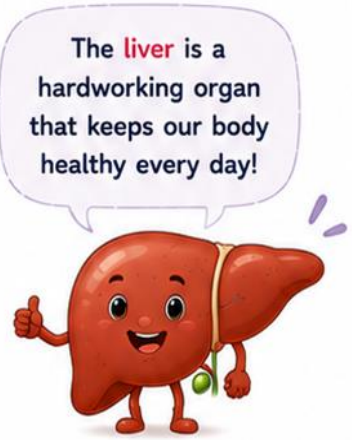
5. Go to the liver and then to all parts of the body when needed.

That is how absorption works!

Lesson 3-3

What is the Importance of Liver?

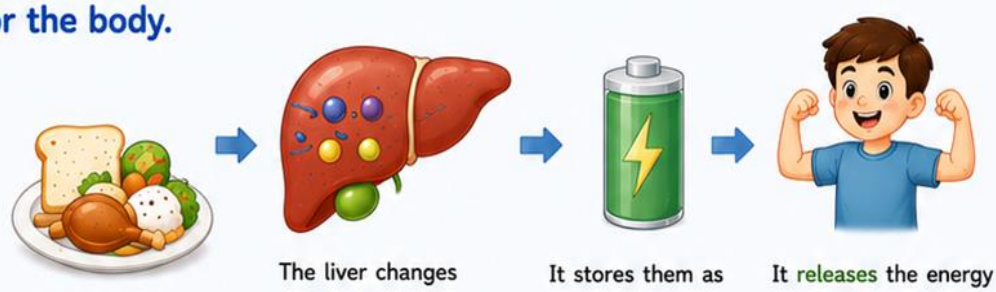
The liver has many important jobs that help our body work in the best way.



1 It stores energy for the body.

The liver converts some nutrients from the food into a simple form that the body can use easily.

Then it stores them as energy and releases them when the body needs them.



2 It cleans the body.

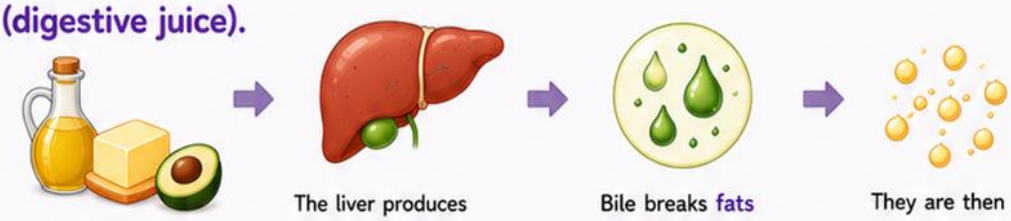
The liver breaks down harmful substances in the body and gets rid of them through urine, keeping the body safe and healthy.

Examples of harmful substances broken down by the liver:

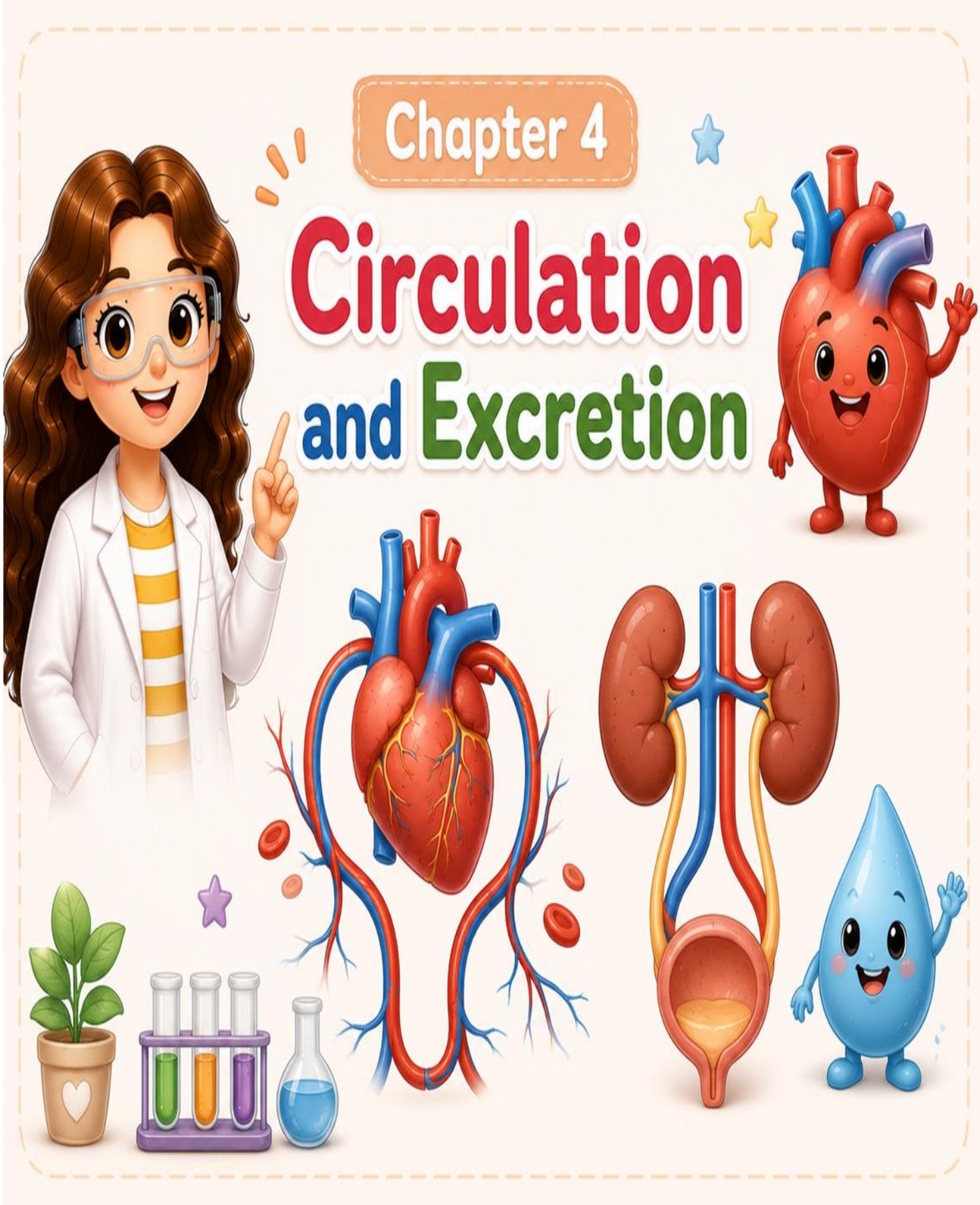
- Medicine and drug leftovers
- Alcohol
- Nicotine (from cigarettes)
- Toxins from pollution
- Certain poisons
- Chemicals from household products

3 It produces bile (digestive juice).

The liver produces a digestive juice called bile. Bile helps in the break down of fats into tiny drops so they can be digested and absorbed by the body.



Thanks to the liver, our body can get energy, stay clean, and digest food properly!


Chapter 4
Circulation
and Excretion

Lesson 4-1

Heart and Blood

The circulatory system is the body's transport system. It carries important substances to and from all parts of the body.

1 What is the importance of the circulatory system?

- Nutrients absorbed from the small intestine are carried by the blood to the body's cells.

- Oxygen taken in by the lungs is carried by the blood to all body parts.

- Carbon dioxide and other wastes produced by body cells are carried by the blood back to the organs that remove them from the body.

2 What are the main parts of the circulatory system?

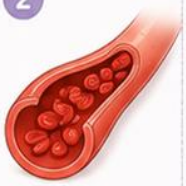
The circulatory system has three main parts:

1

Heart

A strong muscle that pumps the blood around the body.

2



Blood vessels

Tubes that carry blood to every part of the body.

3

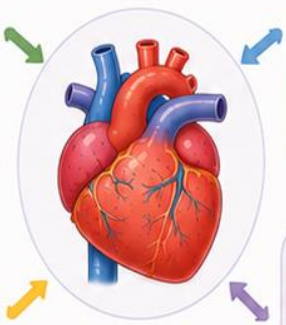
Blood

The liquid that carries oxygen, nutrients, carbon dioxide, and other substances.

3 What is the function of the heart?

The heart is a strong muscular organ that works like a pump.

It receives blood returning from the body.



It pumps blood to the lungs to pick up oxygen.

It pumps oxygen-rich blood to all parts of the body.

It keeps the blood moving continuously around the body.

Remember:

Heart = pump

It pumps the blood all the time.

Blood vessels = pathways

They are the roads that carry the blood.

Blood = transport system

It carries important substances to and from the body.

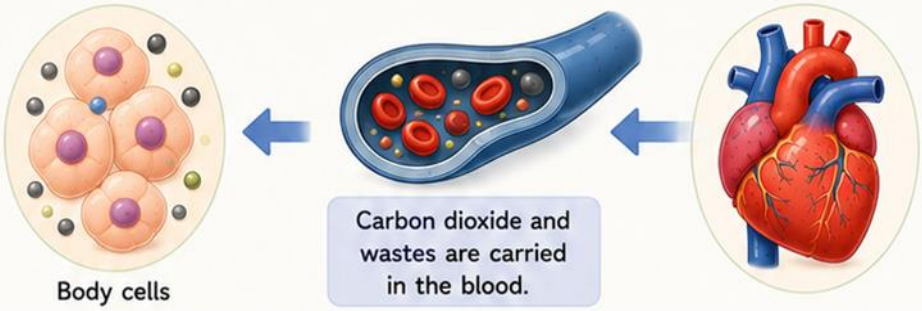
Lesson 4-1

Blood Circulation

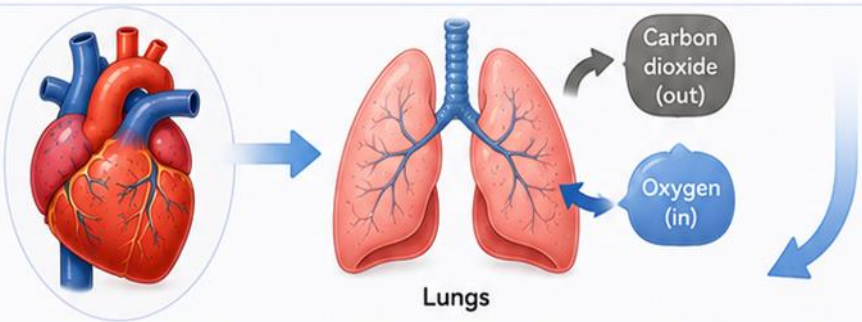
Blood flows through blood vessels that spread all over the body and circulates inside the body.

a The heart pumps blood that is rich in oxygen (oxygenated blood) and nutrients to all parts of the body.

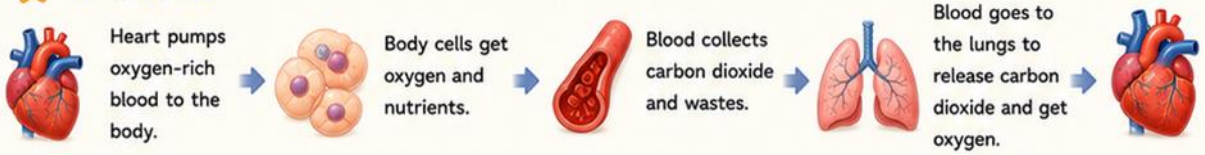
b Blood collects carbon dioxide (deoxygenated blood) and wastes produced by the cells, and returns to the heart again.



c The deoxygenated blood is sent to the lungs to release carbon dioxide and take in fresh oxygen.

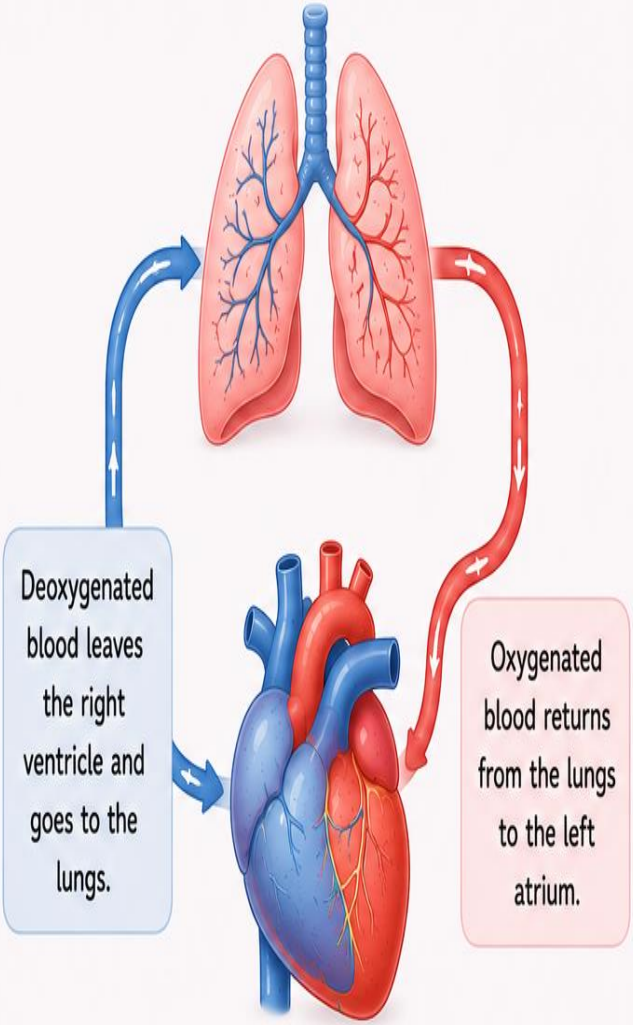


In short:



Minor Circulation
(Pulmonary Circulation)

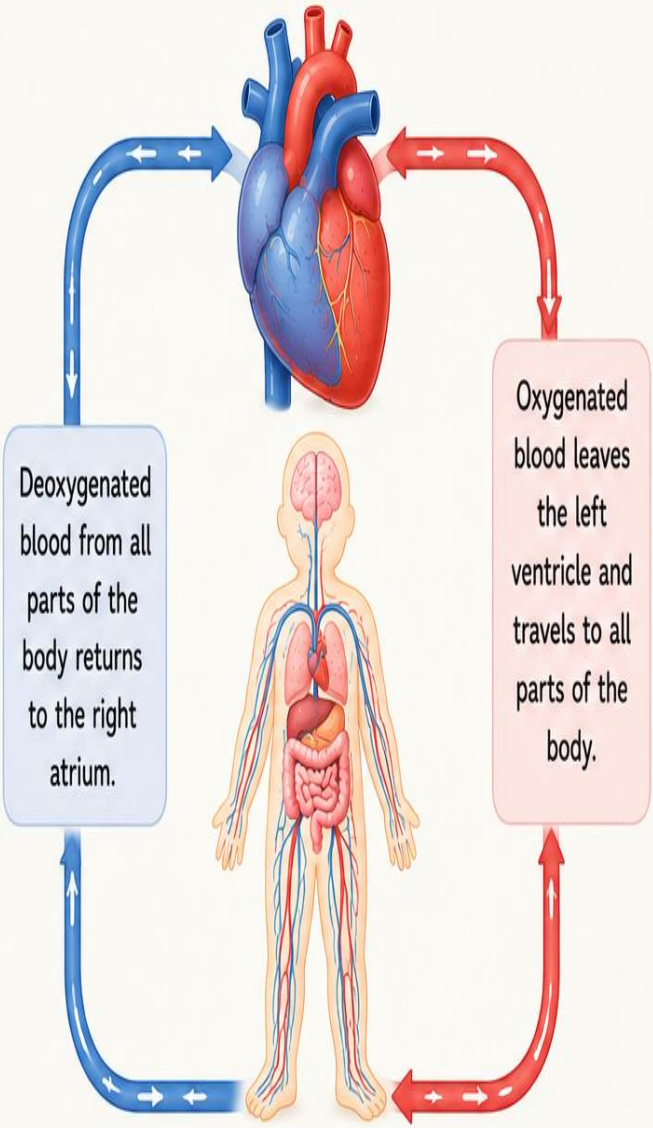
Blood travels between the heart and the lungs.



Deoxygenated blood
Oxygenated blood

Major Circulation
(Systemic Circulation)

Blood travels between the heart and the rest of the body.



Deoxygenated blood
Oxygenated blood

Heart Beats and Pulse

- The movement of the heart pumping out blood is called a heartbeat.
- When the heartbeat travels through the blood vessels, it can be felt at the wrist as a pulse.
- Pulse is the number of heart beats in one minute.

Activity: How Does the Pulse Change from Rest to Exercise?

Tools:

Stop watch

Steps:

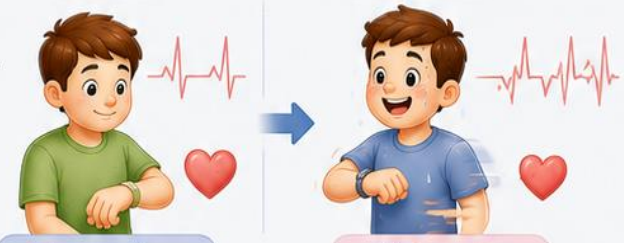
1. Put your fingers on your wrist and find the place where it is beating.

2. While sitting quietly, count how many beats you feel in one minute. Record your result.

3. After jumping lightly in place for one minute, measure your pulse again. Record your result.

Conclusion

When you exercise, your heart needs to send more blood to your muscles.
So, your heart beats faster and your pulse increases.
After you rest, your heart rate goes back down and your pulse becomes slower.

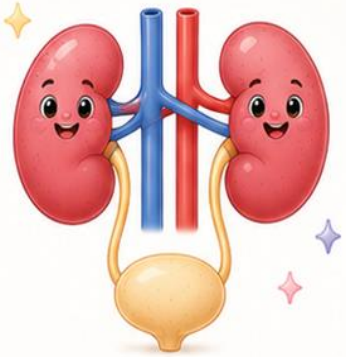


Remember!

- A healthy heart keeps our body strong and full of energy.
- Regular exercise strengthens the heart.
- Rest, healthy food, and enough sleep help keep the heart healthy.

Lesson 4-2

How do Kidneys Work?



Our body removes wastes to stay healthy. The kidneys are the main organs that help remove these wastes.

1 What is meant by excretion process?

Excretion is the process by which our body removes harmful and unwanted substances (wastes) from the body.

2 What is meant by waste products?

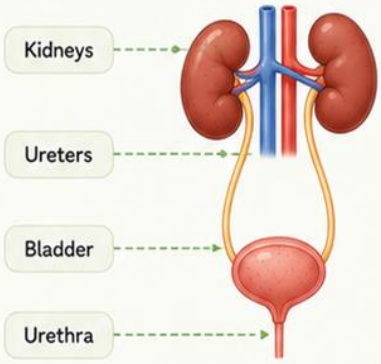
Waste products are unwanted substances produced in our body cells as a result of using food and oxygen for energy.

Examples of waste products:

- Carbon dioxide (from breathing out)
- Urea (from breaking down proteins)
- Excess water and salts

3 One of the systems responsible for excretion is the urinary system.

The urinary system removes waste products from the blood and gets rid of them from the body in the form of urine.



The parts of the urinary system and their jobs:

Kidneys: clean the blood and remove wastes.

Ureters: carry urine from the kidneys to the bladder.

Bladder: stores urine.

Urethra: carries urine out of the body.

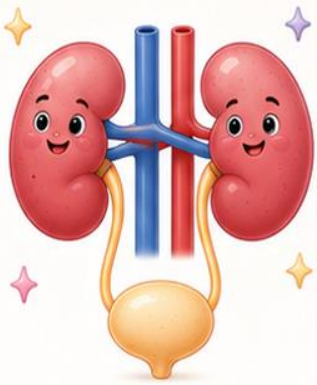
In short

Our body makes waste products. The kidneys take these wastes out of the blood and remove them from the body as urine.

We eat food. → Cells use food and oxygen. → Wastes are formed in the cells. → Kidneys remove wastes from the blood. → Wastes leave the body as urine.

Lesson 4-2

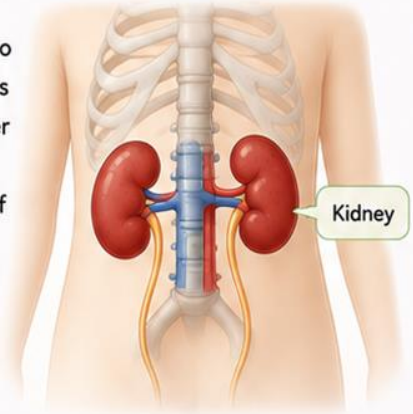
The Kidneys



The kidneys are important organs that help keep our body clean and healthy by removing wastes.

1 Where are the kidneys located?

The kidneys are two bean-shaped organs located in the lower part of the back, one on each side of the backbone.



2 What is the function of the kidneys?

The kidneys act like filters.

- They clean the blood by removing harmful wastes and extra water.
- They help balance the amount of water and salts in the body.
- They keep our body healthy.

Blood with wastes enters the kidney.

Kidney filters the blood.

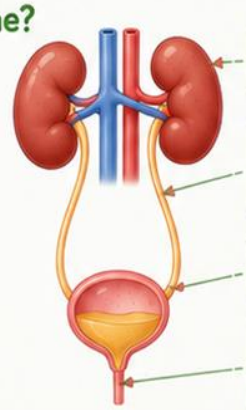
Clean blood returns to the body.

Wastes and extra water become urine.

3 What is urine?

Urine is a yellow liquid made of wastes and extra water.

It is made in the kidneys and stored in the bladder until it leaves the body.



4 What happens if the human body does not get rid of urine?

If the body does not get rid of urine, wastes and extra water will build up in the body.

This can make the person feel sick and can be dangerous.

In short

The kidneys are in the lower part of the back, one on each side.

They filter the blood and remove wastes and extra water.

Urine is the liquid waste that leaves the body.

Getting rid of urine keeps our body clean and healthy.

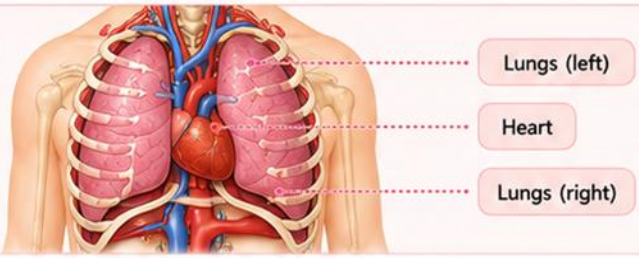
Taking care of our kidneys helps our whole body stay healthy!

Lesson 4-3:
Various Organs

In our body, many organs are lined up closely without gaps, and they work together each other.

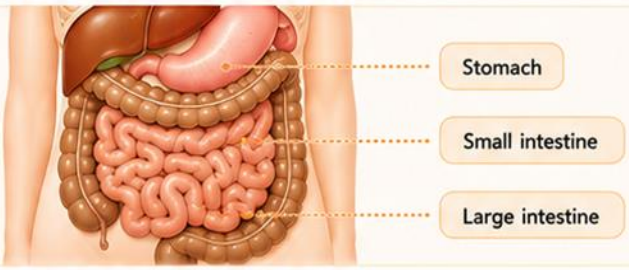
1 Organs in the chest:

There are left and right **lungs**, and the **heart** sandwiched between them. They are protected by the ribs.



2 Organs in the abdomen:

There is the **stomach** at the top, the long **small intestine** below it, and the **large intestine** around it.



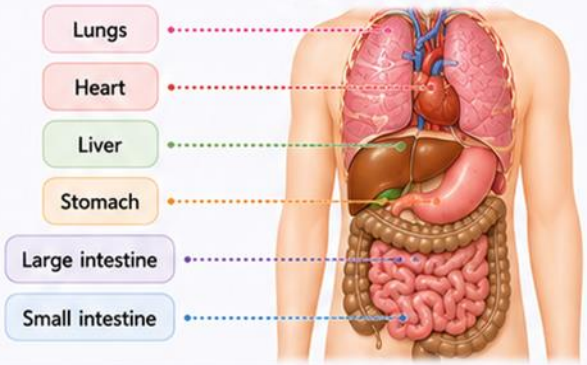
3 Organs in the lower back on the back side:

There is one **kidney** on each side.

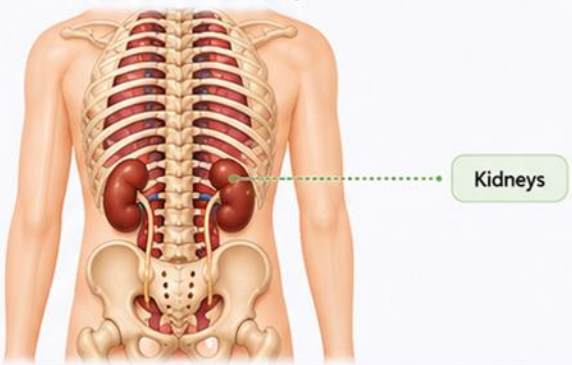
4

All organs are connected by **blood vessels**, and the whole body moves by the blood carrying nutrients and oxygen and collecting waste products.

▼ Front of the body

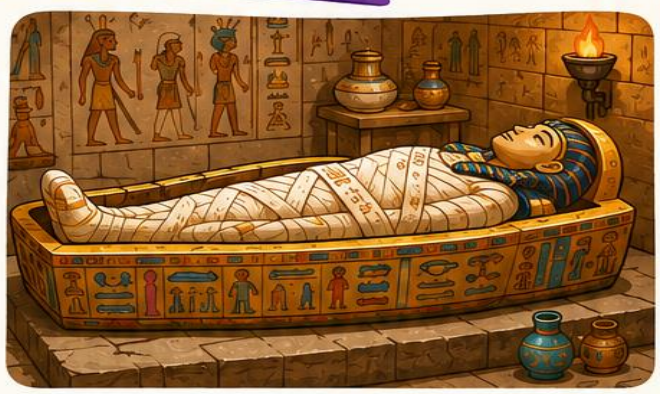


▼ Back of the body



Chapter 4:
The Mystery of the Human Body

Ancient Wisdom and Modern Science



1 What Mummies Taught Us About the Locations of Our Organs

In Chapters 2 through 4, we learned how the **lungs**, **stomach**, **heart**, and **kidneys** work.

In fact, **ancient Egyptians** were the first people in the world to know the exact locations and shapes of these organs, thanks to their mummy-making skills.

2 The Secret of Canopic Jars:

When ancient Egyptians made mummies, they kept four very important organs — the lungs, stomach, liver, and intestines — safe in special jars. They knew that these were important "engines" that supported human life.

3 The Heart is the "Place of Wisdom"

As we learned in Chapter 4, the heart is a pump that sends blood throughout the body. However, ancient people believed the heart was where the "mind" (or soul) resided, so they left it inside the body.

The Four Canopic Jars

Imsety
Has a human shape and was believed to be the god who protects the **liver**.

Hapi
Has a baboon shape and was believed to be the god who protects the **lungs**.

Duamutef
Has a jackal (wild dog) shape and was believed to be the god who protects the **stomach**.

Qebehsenuef
Has a falcon shape and was believed to be the god who protects the **intestines**.

Remember!
Mummies and canopic jars help us see how smart **ancient Egyptians** were. They used observation and careful study — just like **scientists** do today!

Did You Know?
Ancient Egyptians removed the brain through the nose because they did not think the brain was important. They believed the **heart** controlled the body and thoughts!

Chapter 4:
Keeping Our Body's Organs Healthy
Ancient Wisdom and Modern Science

2 In modern science, we learn that our **organs** do not work separately. They are all connected together by the **circulation of blood**.

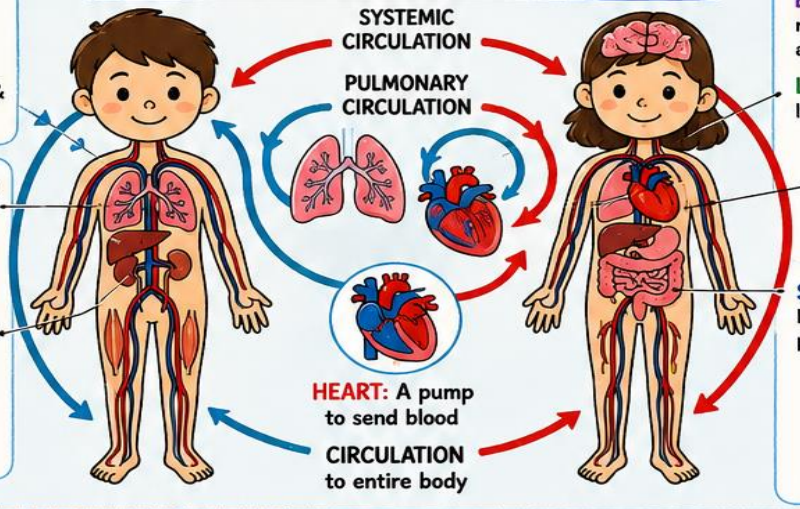
ORGANS CONNECTED BY BLOOD CIRCULATION

TYPES OF BLOOD
- **RED:** Oxygen & nutrient-rich blood
- **BLUE:** Carbon dioxide & waste-rich blood

LIVER: Stores nutrients, breaks down harmful substances

KIDNEY: Filters waste and extra water from blood to make urine

FILTRATION IN KIDNEY



BRAIN: Gives commands, requires many nutrients and oxygen

LUNG: Takes in oxygen, lets out carbon dioxide

GAS EXCHANGE IN ALVEOLI

STOMACH & INTESTINE: Digests and absorbs food, passes nutrients to blood

NUTRIENT ABSORPTION IN INTESTINE

1 The Transporters of Oxygen and Nutrients

The **oxygen** we breathe in through our lungs and the **nutrients** we absorb from our small intestine are all carried to cells all over our body by the **blood**, using the pumping power of the **heart**.

2 Egyptian Food for Good Health

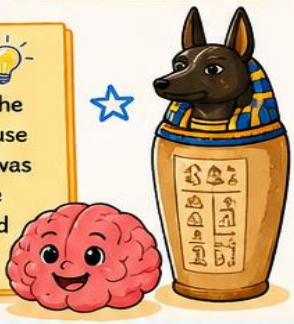
Our bodies are made from the food we eat. A traditional Egyptian food called **"Molokhia"** is full of nutrients that help keep our blood flowing smoothly and help us digest our food. Learning how our body works helps us understand why taking care of our daily meals is so important.

Remember!

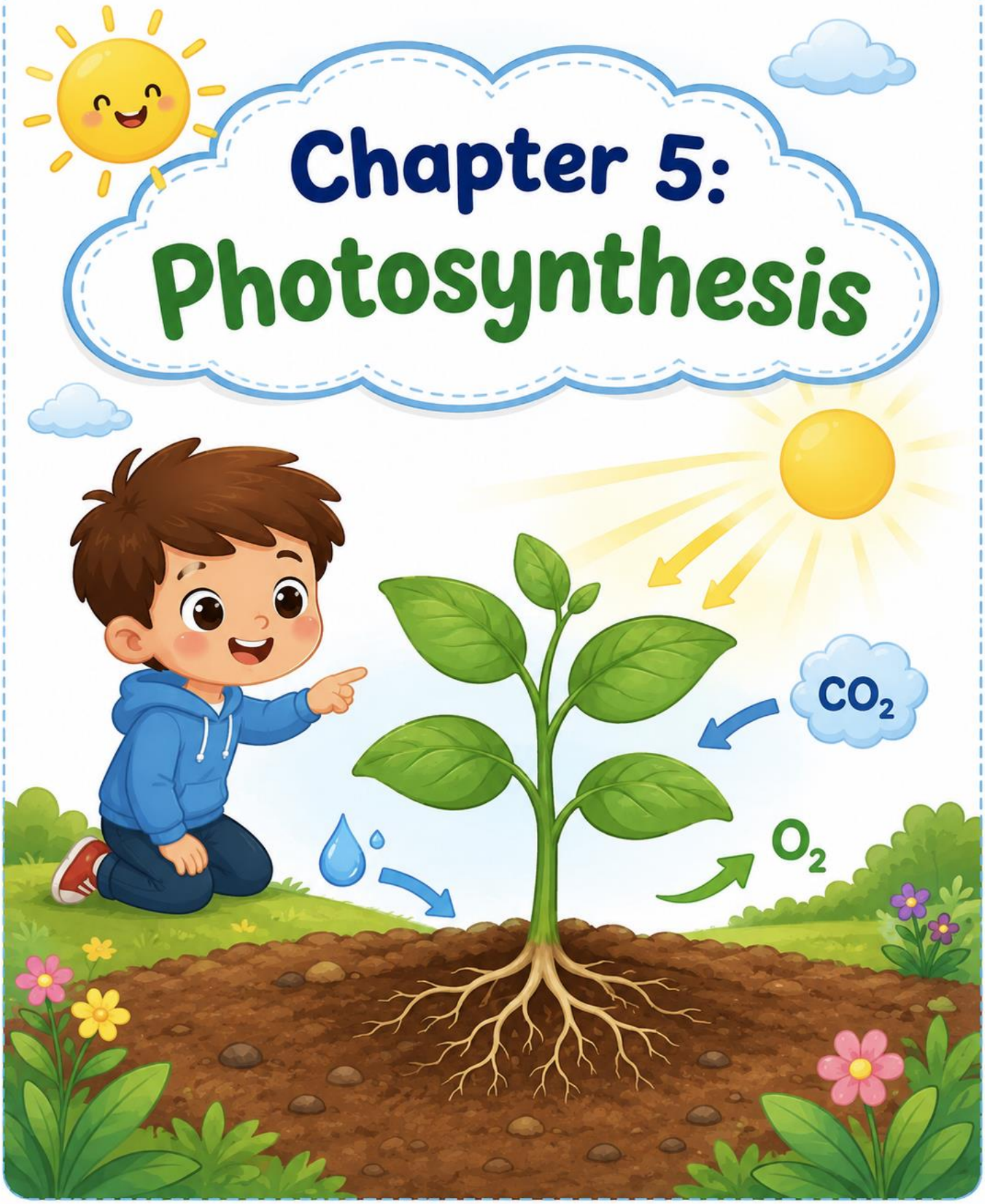
Our organs work together like a team! The **heart**, **lungs**, **brain**, **liver**, **kidneys**, **stomach**, and **intestines** all have special jobs. When we eat healthy food, drink enough water, exercise, and get enough rest, we help our organs stay strong and healthy!

Did You Know?

Ancient Egyptians removed the brain through the nose because they did not think the brain was important. They believed the **heart** controlled the body and thoughts!



Chapter 5: Photosynthesis



Lesson 5-1:
Sunlight and Starch

Plants are living things. They have needs and do important jobs to stay alive and grow.

1 What do plants need to grow?

Plants need certain things from their environment to grow well. These include:

Sunlight
from the sun

Water
from rain or the soil

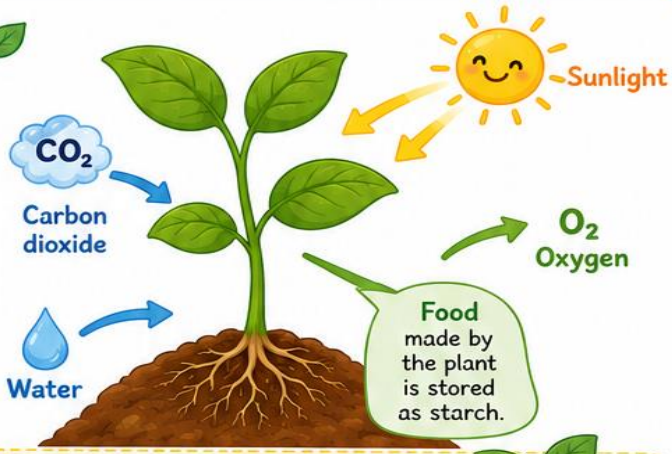
Carbon dioxide
from the air

Nutrients
from the soil

Air
all around us

2 What is photosynthesis?

Photosynthesis is how plants make their own food. They use sunlight, water, and carbon dioxide to make food for themselves.



In photosynthesis, plants make food and store it as starch. They also release oxygen, which we need to breathe!

Activity

Is Sunlight Needed to Make Starch?

What are we changing?
We will check whether the leaf **gets sunlight or not.**

Why cover the leaf before the experiment?
Leaves may already have starch from sunlight during the day.
So, cover them the day before so no sunlight can reach them.

How do we detect starch?
Add iodine solution to the leaf. If starch is present, the color will turn **blue-black**.

How to Test for Starch in Leaves

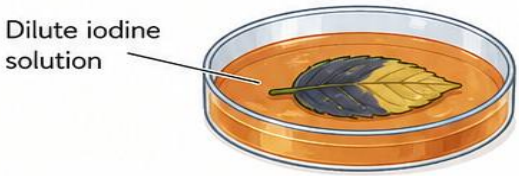
1 Soften the leaf.
Put the leaf in hot water for a few minutes to make it soft.

Be careful!
Do not touch the hot water. You could get burned!

2 Remove the green color.
Place the leaf from step 1 in ethanol in a beaker of hot water. This removes the green color from the leaf.

Be careful!
Ethanol is very flammable. Do not heat it over a flame!

3 Test with iodine solution.
Rinse the leaf with warm water, then place it in dilute iodine solution. Watch for a color change.



If starch is in the leaf, it will turn **blue-black**.
If there is no starch, the leaf will stay **yellow or brown**.

Activity

Sunlight and Starch in Leaves

Let's find out how sunlight affects starch in plant leaves.

1 What we need:

Plant

Ethanol

Aluminum foil

Beaker

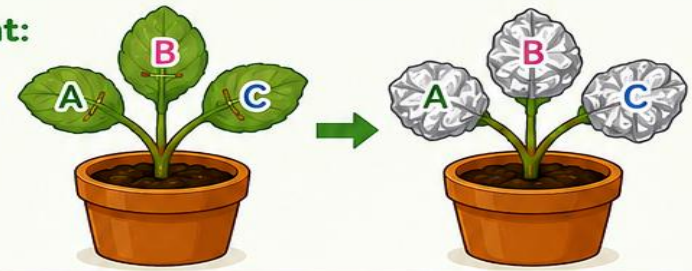
Dilute iodine solution

Petri dish

Hot water (about 80°C)

2 The day before the experiment:

- Make three small cuts on different parts of the leaves and label them **A**, **B**, and **C**.
- Cover all the leaves with aluminum foil so they do not get any sunlight.



Observation: All leaves are covered with foil, so no part of the leaf receives sunlight.

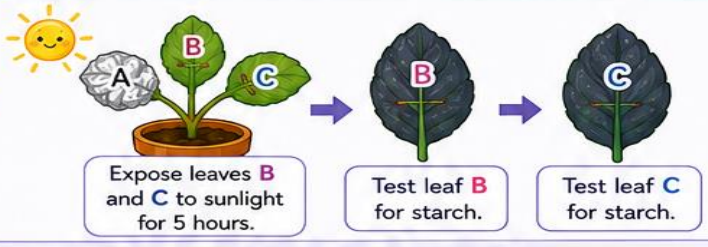
3 In the morning:

- Remove the foil from leaf **B**.
- Take leaf **A** and test it for starch.

Observation: Leaf **A** does not turn blue-black (no starch).
Leaf **B** may show a small change or no blue-black yet (starch may not have formed).

4 After 5 hours:

- Expose leaves **B** and **C** to sunlight for 5 hours.
- Then test both leaves for starch.



Blue-black color means starch is present in the leaf.

Observation: Leaves **B** and **C** turn blue-black (starch is present).
Leaf **A** stays yellow/brown (no starch).

Conclusion:

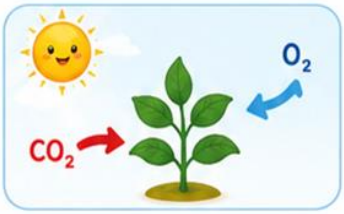
Starch is found in the parts of the leaf that received sunlight (**B** and **C**), but not in the part that stayed covered (**A**).

Therefore, sunlight is necessary for plants to make starch during photosynthesis.

Lesson 5-2:
Gas Exchange

Plants exchange gases with the air every day.

When sunlight shines on the leaves, plants take in **carbon dioxide** from the air and give out **oxygen**.

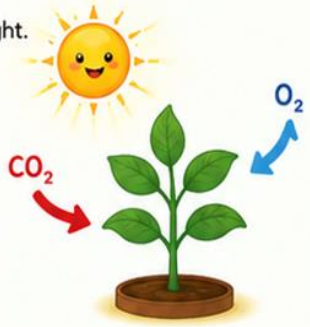


Plants also breathe to live and stay healthy. When they breathe, they take in **oxygen** and release **carbon dioxide**.

Because of these two processes, there is a difference between **daytime** and **nighttime** for plants.

In the Daytime (when there is light)

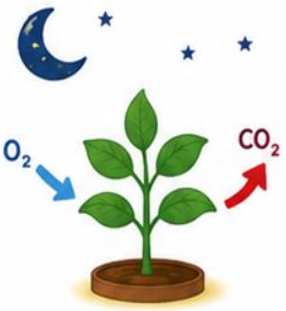
- Plants make food using sunlight.
- They take in **carbon dioxide** and release **oxygen**.
- This process is very active during the day.
- Although plants also breathe, making carbon dioxide, the amount of **oxygen** released is greater.



In short:
In the daytime, plants take in more **carbon dioxide** and release more **oxygen**.

At Nighttime (when there is no light)

- Plants cannot make food without light.
- They only breathe.
- They take in **oxygen** and release **carbon dioxide**.
- More carbon dioxide is released because no food is being made.



In short:
At nighttime, plants take in **oxygen** and release more **carbon dioxide**.

Conclusion

Plants exchange gases with the air all the time.
In the **daytime**, they take in **carbon dioxide** and release **oxygen**.
At **nighttime**, they take in **oxygen** and release **carbon dioxide**.
This difference happens because light is needed for plants to make food.

Activity: Gas Exchange in Plants

Let's find out how the air inside the bag will change after some time when a plant is sealed in a bag.

1 What we need:

Plant

Polyethylene bag

Straw

Twist tie

Gas detector tube for oxygen

Gas detector tube for carbon dioxide

Gas sampling pump

Scissors

2 What we do:

1 On a sunny day, place a bag over the plant. Tie the bottom tightly with a twist tie. Then, breathe into the bag through the straw.

Observation:
At the start (right after breathing in):
- Oxygen (O_2) percentage will be LOWER.
- Carbon dioxide (CO_2) percentage will be HIGHER.

2 Make a small hole in the bag with scissors. Insert the gas detector tube (use the pump) and measure the percentage of oxygen and carbon dioxide right away.

Observation:
First reading (immediately):
- Oxygen (O_2) percentage will be LOWER.
- Carbon dioxide (CO_2) percentage will be HIGHER.

3 Leave the plant in a place with sunlight for about 1 hour without opening the bag.

Observation:
After 1 hour in sunlight:
- Oxygen (O_2) percentage will be HIGHER than before.
- Carbon dioxide (CO_2) percentage will be LOWER than before.

4 After about 1 hour, insert the gas detector tube again (use the pump) and measure the percentage of oxygen and carbon dioxide one more time.

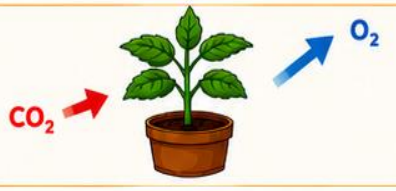
Observation:
Second reading (after 1 hour):
- Oxygen (O_2) percentage will be HIGHER than the first reading.
- Carbon dioxide (CO_2) percentage will be LOWER than the first reading.

3 What we record:

Write the percentages you measured in steps 2 and 4.

Gas	First Reading (Step 2) (at the start)	Second Reading (Step 4) (after 1 hour)	Change (Increase or Decrease)
Oxygen (O_2)			
Carbon Dioxide (CO_2)			

- Plants exchange gases with the air.
- In sunlight, the plant uses carbon dioxide and releases oxygen.
- After some time, the oxygen inside the bag increases, while the carbon dioxide decreases.
- This happens because the plant is making food using sunlight (photosynthesis).



Chapter 6:
Transpiration of water in plants

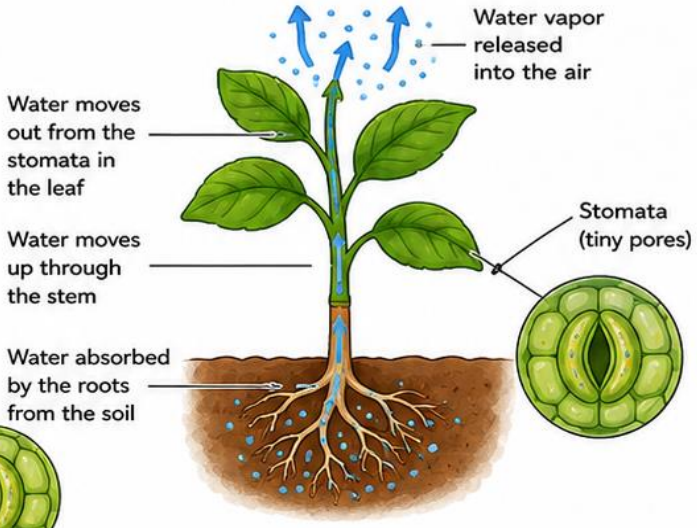
Lesson 6-1

Transpiration

Transpiration is the process by which water in plants changes into water vapor and is released into the air.

What happens?

Water is absorbed by the roots from the soil. It moves up through the stem to the leaves. Then, it changes into water vapor and goes out into the air through tiny pores in the leaves called stomata.



What are stomata?

Stomata are tiny pores mostly found on the underside of leaves. They open to let out water vapor (transpiration) and close to reduce water loss.

Let's Investigate!

1 Take two similar plants.

Put one plant in sunlight and another in a dark place.

2 Cover some leaves.

Cover a few leaves on each plant with a clear plastic bag and tie it gently.

3 Wait for some time.

Leave the plants for a few hours.

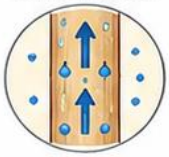
4 Observe.

You will see tiny water droplets inside the bag on the plant kept in sunlight, but very few or no droplets on the plant kept in the dark.

Why is transpiration important?

Helps in cooling the plant.

Helps in the upward movement of water and minerals.



Keeps the leaves fresh and turgid.

Helps in the distribution of water in the plant.

Quick Tip!

Stomata open in the daytime (when there is light) and close in the night.

Activity

Let's find the water drops in the bag!

Let's observe that water is coming out of the leaves.

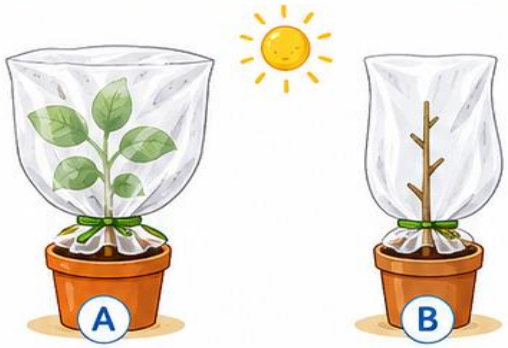
1 Preparation:

- Plants (Plant A with leaves and Plant B with all the leaves taken off)
- Plastic bags
- Twist ties or strings (to tie the bag)

2 Experiment:

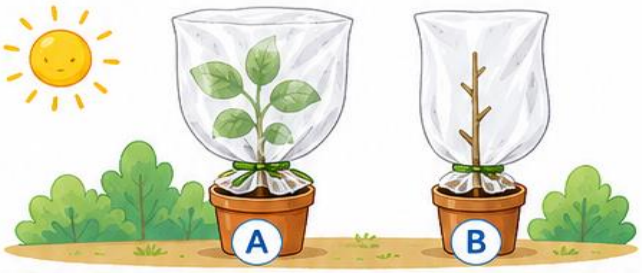
1. On a sunny day, put bags over Plant A (with leaves) and Plant B (with all leaves taken off), and tie the bases with twist ties or strings.

Observation:
Both bags are clear at first. No water drops inside the bags.



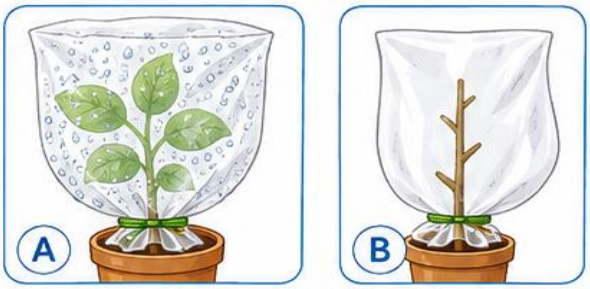
2. Leave them in a sunny place for a while.

Observation:
After some time, tiny water drops start to appear inside the bag on Plant A, but not on Plant B.



3. Check if water drops form inside the bags. This is the water that transpired from the leaves.

Observation:
Many water drops are inside the bag on Plant A. There are no drops in the bag on Plant B.



Conclusion

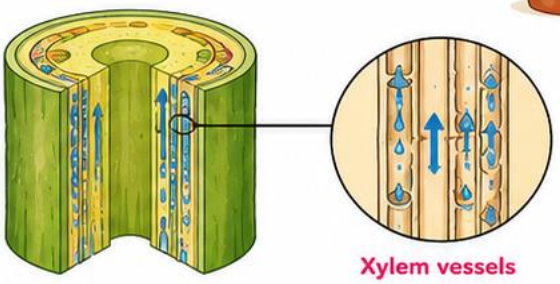
Water comes out of the leaves in the form of tiny drops (water vapor). This process is called transpiration. Plants with leaves (Plant A) release water vapor, but plants without leaves (Plant B) do not.

Lesson 6-2:

The Path of Water

1 Where does the water absorbed from the roots pass through to be carried to the whole plant?

The water absorbed from the roots passes through tiny tubes inside the plant called **xylem vessels**. These vessels carry the water up through the stem to the leaves and all other parts of the plant.



2 Does the plant come back to life after watering, why?

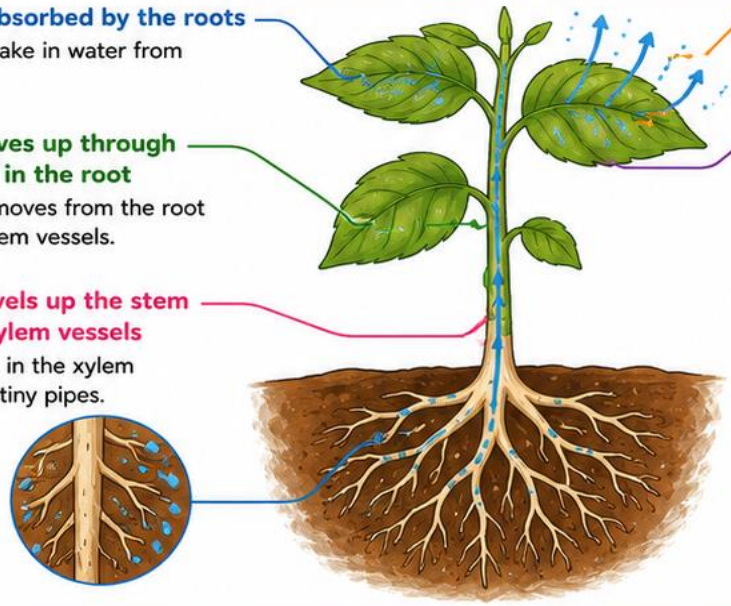
Yes, the plant comes back to life after watering. Water fills the cells, makes them firm again, and helps the plant to do its work such as photosynthesis. This also helps the leaves to stand up and stay green and healthy.

The Path of Water in a Plant

1 Water is absorbed by the roots
Root hairs take in water from the soil.

2 Water moves up through the xylem in the root
The water moves from the root into the xylem vessels.

3 Water travels up the stem through xylem vessels
It moves up in the xylem vessels like tiny pipes.



4 Water reaches the leaves
The water reaches the leaves through the xylem vessels.

5 Water is released from the leaves
Water changes into water vapor and goes out into the air through tiny openings called stomata (transpiration).

Inside the Xylem

- Xylem vessels are like tiny pipes.
- They carry water and minerals from the roots to all parts of the plant.
- The arrows show the direction of water.

Water has an important path in the plant. It starts from the roots, travels up through the stem in the xylem, reaches the leaves, and helps the plant stay alive and healthy.

Lesson 6-2:
The Path of Water

Let's use colored water to visualize where the water passes through.

1 Preparation:

Plant

Colored water

Flask

Absorbent cotton

Magnifying glass or microscope

Cutter knife

2 Experiment:

1 Dig up the plant with the soil, and wash the roots with water.

Observation:
The plant is clean and ready. The roots are white.

2 Put colored water in the flask and soak the roots in it.

Observation:
The roots absorb the colored water.

3 After a few hours, observe the color change in the plant.

Observation:
The leaves and stems change to red.

4 Cut the roots, stems, and leaves with a cutter knife, and observe the cut surfaces.

Observation:
The inside parts of the roots, stems, and leaves are red.

Conclusion

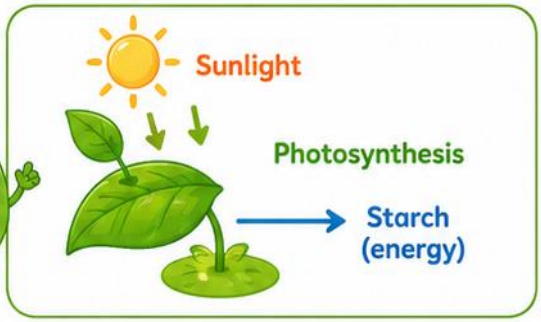
The water absorbed by the roots travels through tiny tubes inside the plant called **xylem vessels**.
These tubes carry water and minerals up through the stem to the leaves and all parts of the plant.

Chapter 6 Column

Blessings of the Nile: Plant Survival Strategies

1 The "Green Factory" That Turns Sunlight into Electricity

In Chapters 5 and 6, we learned about photosynthesis, where plants use sunlight to produce starch.

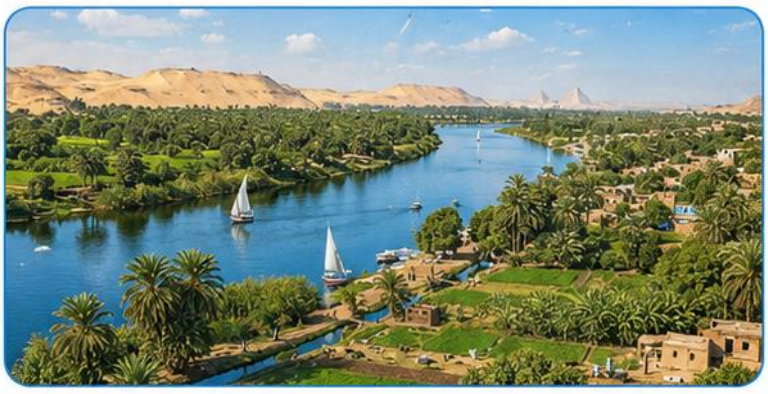


2 Solar Panels on the Water Lotus:

The leaves of the lotus (water lily) floating on the Nile River have a flat and large shape. This shape helps them catch as much of Egypt's strong sunlight as possible so they can produce nutrients (starch) efficiently.

3 Exchanging Gases:

When plants do photosynthesis, they take in the carbon dioxide that humans breathe out and release oxygen. The beautiful green plants along the Nile River act as a **"natural air purifier"** to keep the air in Egypt clean.



The Nile River and its plants are a great **blessing**.
They give us food, clean our air, and support life in many ways!

Blessings of the Nile-Plant Survival Strategies

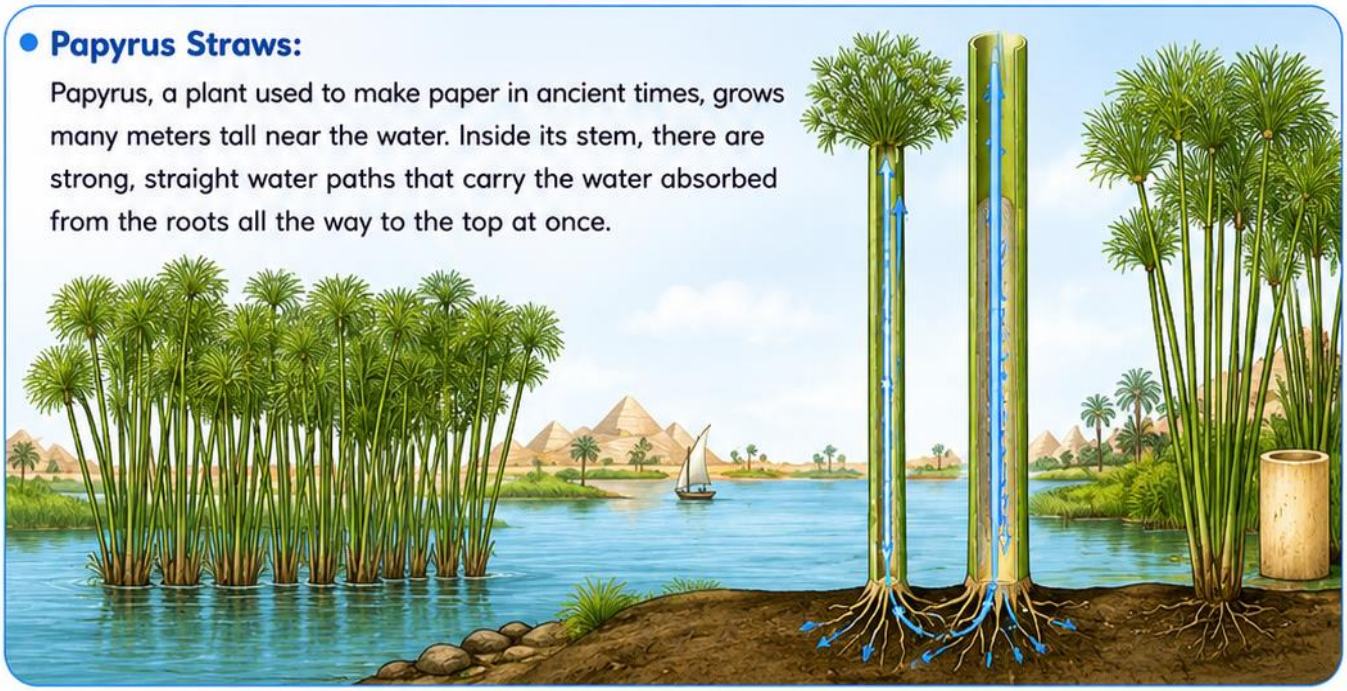
Chapter 6
Column

2 Water Paths and the Wisdom of Papyrus

In Chapter 6, we learned how water absorbed by the roots travels up to the leaves and escapes into the air through transpiration.

• Papyrus Straws:

Papyrus, a plant used to make paper in ancient times, grows many meters tall near the water. Inside its stem, there are strong, straight water paths that carry the water absorbed from the roots all the way to the top at once.

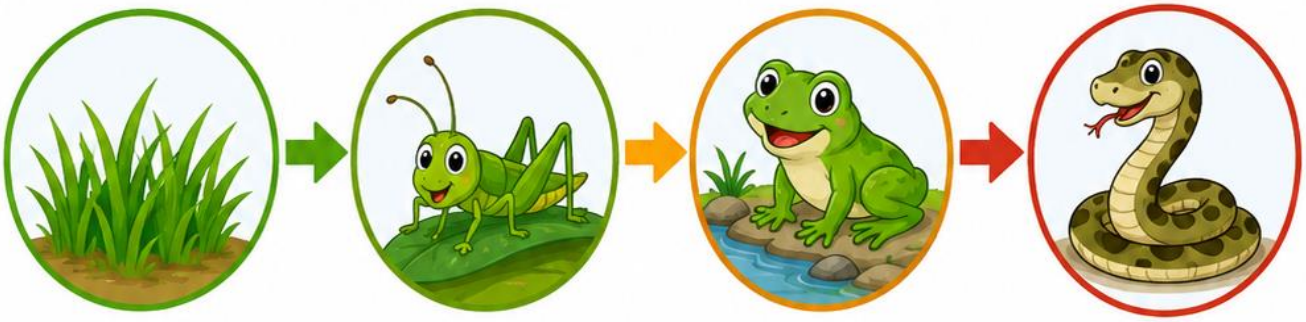


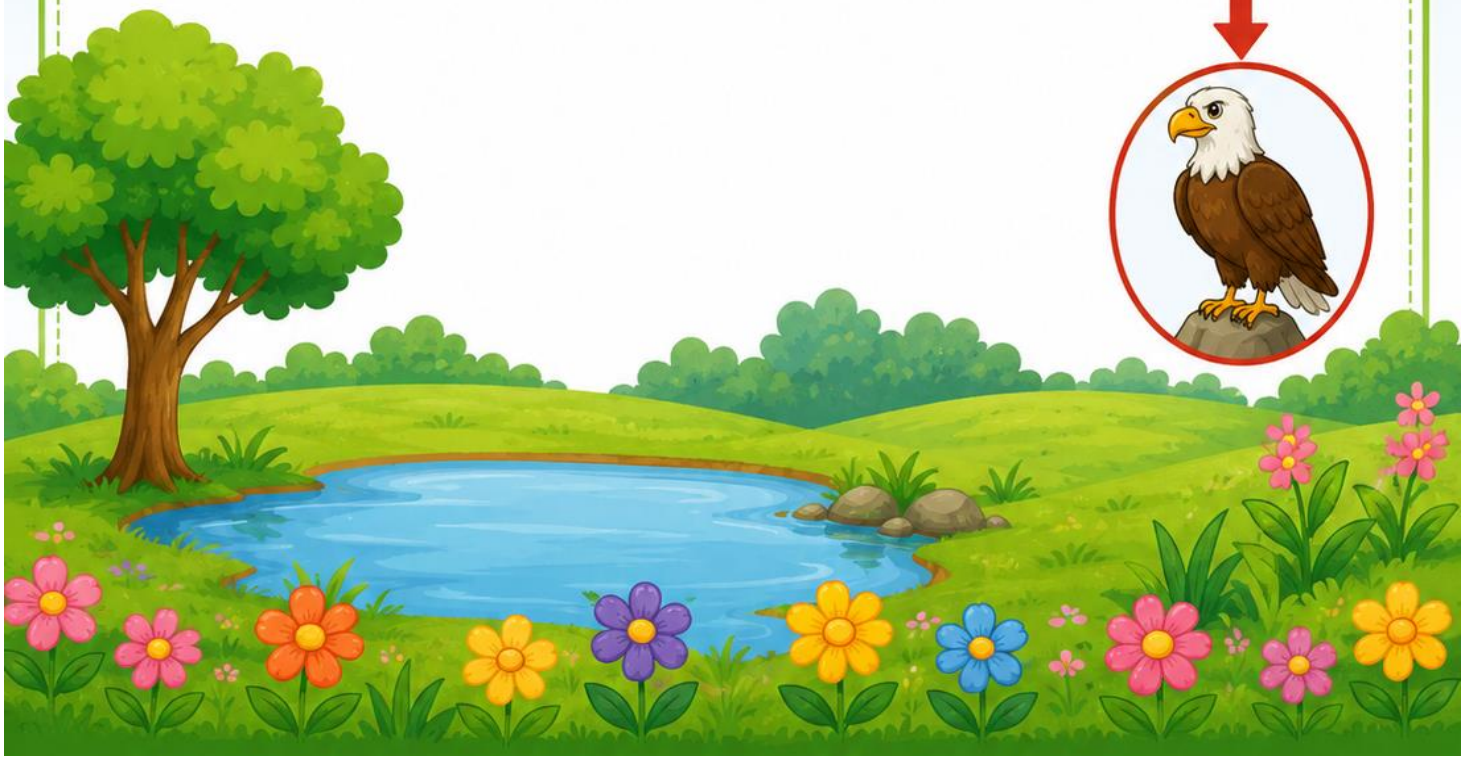
• The Science of Sprinkling Water:

Transpiration is the process where water vapor comes out of the leaves, and it works to lower the temperature around the plant. In the desert country of Egypt, the reason it feels cool when you get close to plants is because they act like a natural sprinkler to cool the air for us.

Chapter 7

Food Chain





7-1: Food Chain

How are living organisms connected by what they eat?

All living organisms need food to live and grow. They are connected to each other by what they eat. When one organism eats another, energy is passed from one to the next.

What is meant by food chain?

A food chain is a series of living things that are connected to each other by the eating and being eaten.
It shows how energy and nutrients move from one organism to another.

Producer (Plant) → Primary Consumer (Herbivore) → Secondary Consumer (Carnivore) → Tertiary Consumer (Carnivore) → Apex Predator

In a **food chain**, every living thing depends on another for food. If one part is missing, it can affect the whole chain!

How the Food Chain Works

Point! Living things are connected to one another through a feeding relationship. This "eat and be eaten" connection is called a **food chain**.

1 Plants (Producers):

Plants use sunlight, water, and air to make their own food.
They are the first part of the food chain.

2 Herbivores (Primary Consumers):

Herbivores are animals that eat plants.
They get energy from plants.

3 Carnivores (Secondary and Higher Consumers):

Carnivores eat other animals.
They get energy by eating herbivores or even other carnivores.

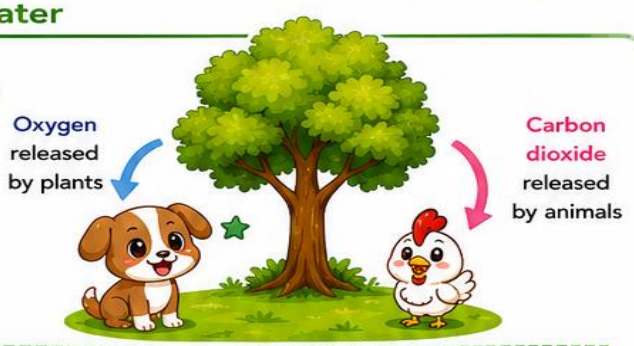
The arrows (→) show the direction in which energy is passed from one organism to another.
When one organism is eaten by another, the energy moves to the next living thing.

Lesson 7-2: Living Organisms, Air, and Water

Point! Circulation of Air and Water

Connection of living organisms through air

Animals breathe in **oxygen** released by plants, and plants take in **carbon dioxide** released by animals for photosynthesis.



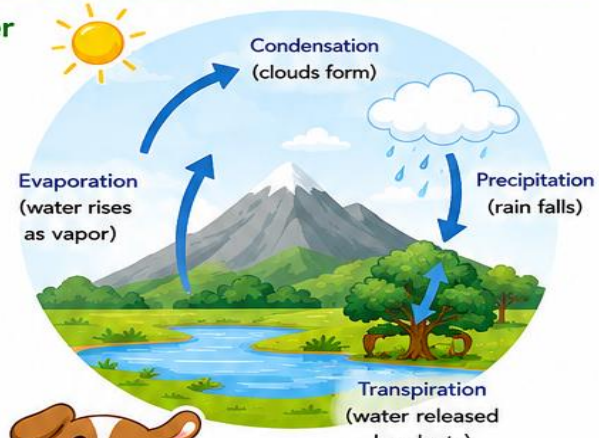
Connection of living organisms through water

Plants release water into the air through **transpiration**.

This water vapor joins the water cycle.

Water also evaporates from oceans, rivers, and lakes, rises into the atmosphere, forms clouds, and falls as rain.

Transpiration is an important part of the water cycle that keeps water moving between living organisms and the environment.



Our lives are constantly supported by the oxygen produced by plants and the water circulating around the Earth!

Activity Let's explore how air and water move between living organisms!

1 Check the path of breathing:

Be aware of your own breathing and imagine how "the breath you just breathed out is used for photosynthesis by the tree next to you".

2 Trace the journey of water:

Let's talk about the possibility that the water you drink today was once a cloud, or water that a tree in some country absorbed.

3 Think about the balance:

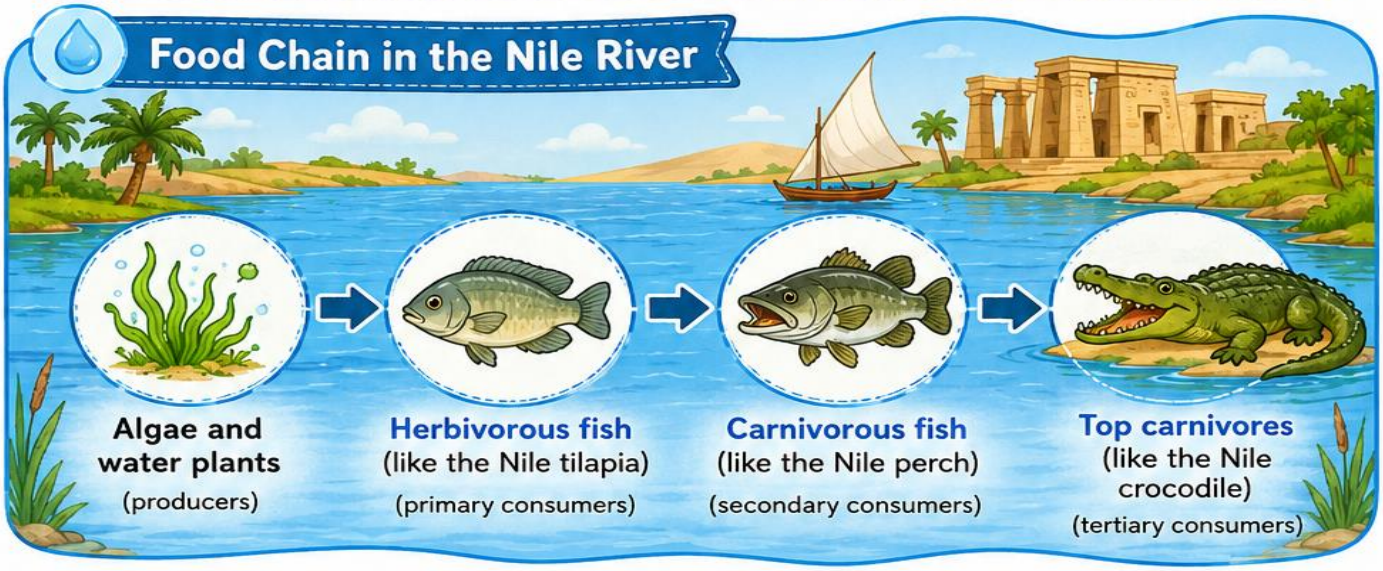
Let's predict what would happen to the oxygen and carbon dioxide in the air if plants completely disappeared.

Chapter 8: Egypt Special Part

Lesson 8-1: Desert and Nile Food Chains

Egypt has completely different food chains in **aquatic environments** (the Nile River) and **arid areas (deserts)**.

Food Chain in the Nile River



Algae and water plants (producers)

Herbivorous fish (like the Nile tilapia) (primary consumers)

Carnivorous fish (like the Nile perch) (secondary consumers)

Top carnivores (like the Nile crocodile) (tertiary consumers)

Food Chain in the Desert

Desert shrubs and grass (producers)

Herbivores (like grasshoppers and jerboas) (primary consumers)

Top carnivores (like fennec foxes and Egyptian vultures) (secondary consumers)

Even if the types of animals are different, the chain always starts with plants and ends with large carnivores.

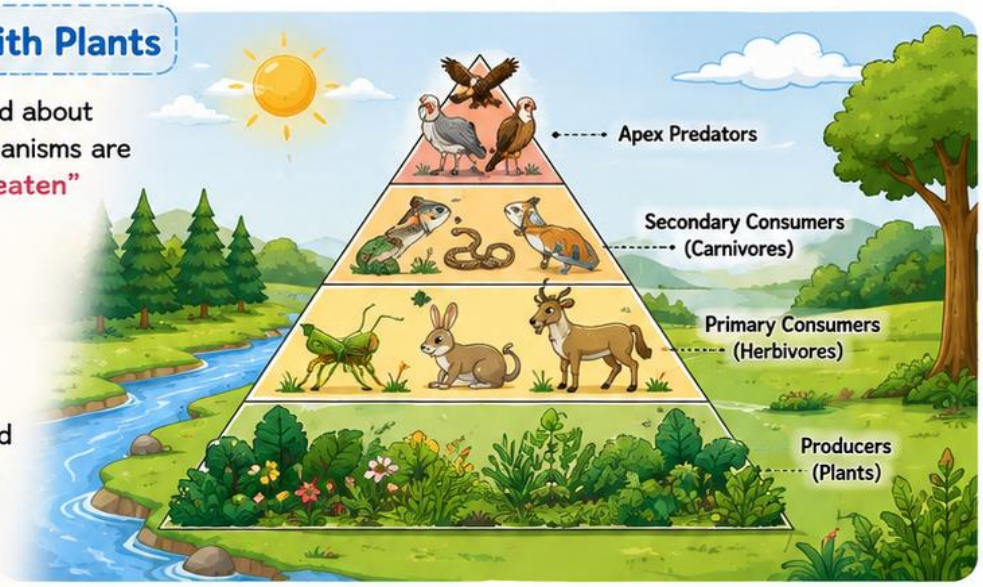
Chapter 8 Column:
It Always Starts with "Plants"

1 It Always Starts with Plants

In Chapters 7 and 8, we learned about the food chain, where living organisms are connected by an "eat and be eaten" relationship.

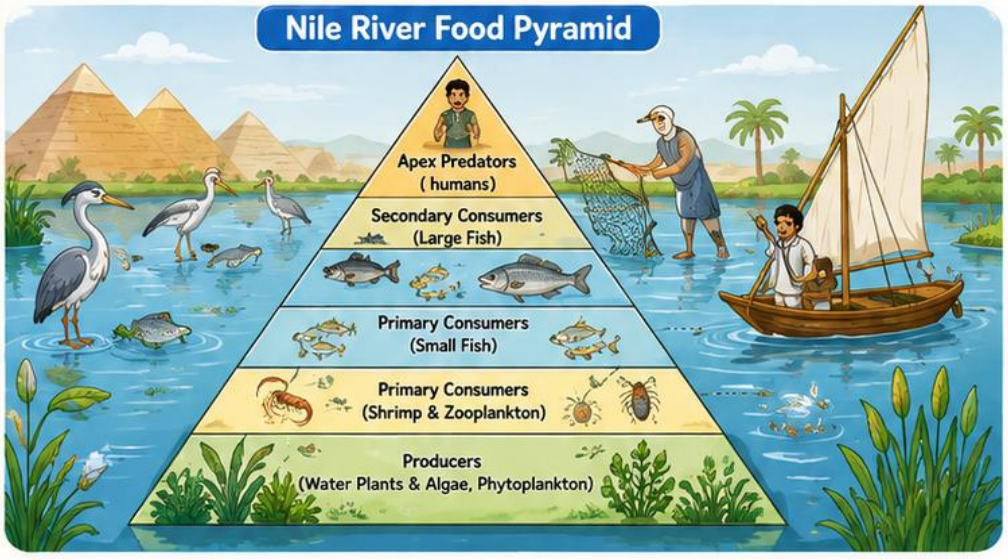
The Base of the Pyramid:

The diagram of a food chain is called a food pyramid, and the very bottom of this pyramid is always supported by plants (producers), which produce their own food from sunlight.



The Nile's Chain:

In the Nile River, there is a beautiful circle of life: small shrimp eat water plants, fish (like the Nile tilapia) eat the shrimp, and then water birds and humans eat the fish.



Even if the types of animals are different, the chain always starts with **plants** and ends with **large carnivores.**

Chapter 8 Column: It Always Starts with "Plants"

2 Surviving in the Harsh Desert

- Egypt's water is part of a global water cycle.
- This cycle connects Egypt with regions far to the south.

The Desert Hunter Fennec Fox:

- Desert plants are limited, so food can be scarce.
- Some animals feed on grass, seeds, and available plant material.
- Lizards and other small animals become food for desert predators.
- The fennec fox is an important consumer in the desert food chain.
- It helps keep the desert ecosystem balanced.

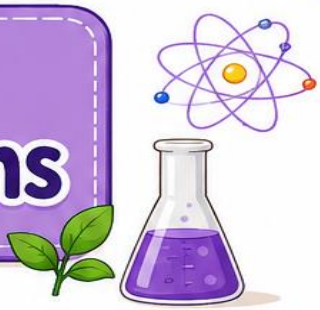
Memories of Wadi al-Hitan:

- Wadi al-Hitan is known as the "Valley of the Whales."
- It is a desert today, but millions of years ago it was an ocean.
- Fossils there show evidence of ancient marine life.
- These organisms formed a complete food chain in the ancient ocean.
- The fossils help us understand how ecosystems changed over time.

Even if the types of animals are different, the chain always starts with **plants** and ends with **large carnivores**.

Chapter 9:
Properties of Solutions

Lesson 9-1: Grouping Solutions



What is meant by a solution?

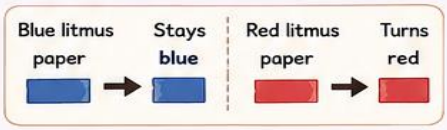
- A solution is a mixture in which one substance (the **solute**) is completely dissolved in another substance (the **solvent**).
- It forms a **uniform mixture** that looks the same throughout.

What are the different types of solutions?

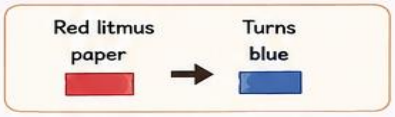
Solutions can be divided into **3** groups by how they change the **color** of litmus paper.

- **Acidic:** Turns **blue** litmus paper **red**.
  (Example: lemon juice, vinegar, hydrochloric acid)

- **Neutral:** Does **not change** the color of either litmus paper.
  (Example: salt water, sugar water, distilled water)

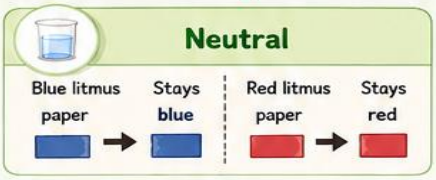


- **Alkaline:** Turns **red** litmus paper **blue**.
  (Example: soapy water, limewater, ammonia solution)



How can we differentiate between them?

- We can use **litmus paper** to test the solution.
- Each type of solution shows a **different change** in the color of litmus paper.



Different solutions show different **properties**. Understanding these properties helps us use solutions **safely and wisely** in our daily life!

Activity

Let's group solutions using litmus paper!

Let's use various solutions and observe the color changes.

Tools

Solutions (salt water, carbonated water, dilute hydrochloric acid, baking soda solution, dilute ammonia solution)

Beaker

Test tube

Test tube rack

Tweezers

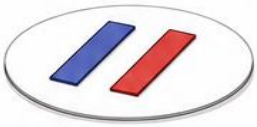
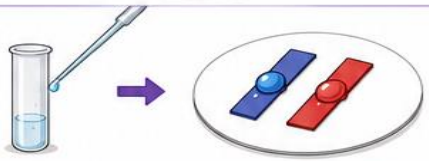
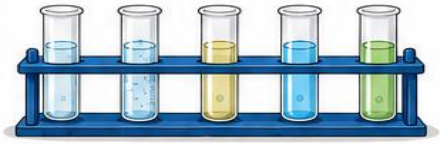
Litmus paper (blue and red)

Filter paper

Safety goggles

Glass rod

Steps

Step	What to do	Pictures	Observation
1	Put blue and red strips of litmus paper on the filter paper.		—
2	Use a glass rod to add one drop of the solution you want to test onto both litmus strips.		—
3	Check the color change of the litmus strips and group them into acidic, neutral, or alkaline.		The colors of the strips change depending on the type of solution.

Caution!

- When changing the solution, rinse the glass rod with water in a beaker and wipe it off every time.
- Make sure to ventilate the room. Also, wear safety goggles.

Summary: Let's summarize the results in a table.

Solution	Blue litmus paper (observation)	Red litmus paper (observation)	Group (acidic / neutral / alkaline)
Salt water	Stays blue	Stays red	Neutral
Carbonated water	Turns red	Stays red	Acidic
Dilute hydrochloric acid	Turns red	Stays red	Acidic
Baking soda solution	Stays blue	Turns blue	Alkaline
Dilute ammonia solution	Stays blue	Turns blue	Alkaline

Conclusion:

Solutions can be grouped into three types: acidic, neutral, and alkaline based on how they change the color of litmus paper.

Acidic

Neutral

Alkaline

Lesson 9-2: What is Dissolved in Solutions?

What happens when we evaporate a solution?

- When we **evaporate (heat)** a solution, the water changes into vapor and leaves. The **dissolved substance stays behind** in the dish.

Solids dissolved in water

- When you evaporate the water, the dissolved substances (such as salt or baking soda) will remain at the bottom of the dish.

Examples:

Gases dissolved in water

- Even if you evaporate the water, **nothing will remain** because the dissolved substance escapes into the air.
- Sparkling water has carbon dioxide gas dissolved in it.

Think and Answer!

1. What will remain after evaporating salt water? _____
2. What will remain after evaporating baking soda solution? _____
3. What will remain after evaporating sparkling water? _____

Summary

- By evaporating a solution, we can find out what was dissolved inside.
- **Solids** (like salt or baking soda) remain at the bottom of the dish.
- **Gases** (like carbon dioxide in sparkling water) do not remain because they escape into the air.

Activity

Let's heat and extract what is dissolved!

Let's extract and observe what became invisible by dissolving in water.

Tools

Solutions
(salt water, carbonated water, dilute hydrochloric acid, baking soda solution, dilute ammonia solution)

Beaker

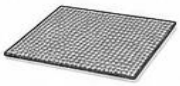
Test tube

Test tube rack

Pipette

Evaporating dish

Portable gas stove



Wire gauze

Safety goggles

Safety First!

- Wear safety goggles.
- Do not bring your face close to the evaporating dish to smell it or breathe it in directly.
- To avoid burns, do not touch heated objects until they cool down completely.

Steps

Step	What to do	Picture	Observation
1	Use a pipette to transfer the solution from the test tube into the evaporating dish.		The solution is transferred into the evaporating dish.
2	Heat it gently over low heat with a portable gas stove.		The water starts to evaporate. Steam (water vapor) is seen rising from the dish.
3	Turn off the heat while a little liquid is still left, and let the remaining heat dry it.		The remaining liquid disappears as the dish cools down.
4	Observe what is left.		A solid substance remains at the bottom of the dish.

Summary

- Let's summarize what was left in the evaporating dish in a table.
- Also, write down how the solutions look and smell.

Solution	What was left in the evaporating dish	How the solution looks	How the solution smells
Salt water	White crystals (salt)	Clear, colorless	No smell (odorless)
Carbonated water	No solid left	Clear, colorless	Slight fizzy smell (bubbles)
Dilute hydrochloric acid	No solid left	Clear, colorless	Pungent, acidic smell
Baking soda solution	White powder (baking soda)	Clear, colorless	No smell (odorless)
Dilute ammonia solution	No solid left	Clear, colorless	Strong ammonia smell

Conclusion

By evaporating the solutions, we discovered that:

- **Solids** (such as salt or baking soda) remain behind because they do not evaporate.
- **Gases** (such as carbon dioxide in sparkling water or ammonia gas in ammonia solution) escape into the air, so nothing remains.

1 The Secret of Solutions and the Pharaoh's Wisdom

Water solutions may look clear, but they hide mysterious properties — **acidic**, **neutral**, or **alkaline** — that can only be told apart using littmus paper. Ancient Egyptians also used chemical properties wisely thousands of years ago.

Natural Alkaline Powder "Natron"

Ancient Egyptians used a white powder called **natron**, found in desert salt lakes. It was strongly **alkaline** and had a powerful ability to absorb water from the body. This made it an essential material for creating mummies that kept their shape for thousands of years.

Natron alkaline absorbs water helped preserve mummies → → →

2 Acids and Alkalis in Our Daily Lives

Even today, we use these properties without realizing it — such as lemon juice for cooking and soapy water for cleaning.

Lemon Juice — Acidic

Lemon juice is acidic and is commonly used in cooking.

Soapy Water — Alkaline

Soapy water is alkaline and is commonly used for cleaning.

Remember

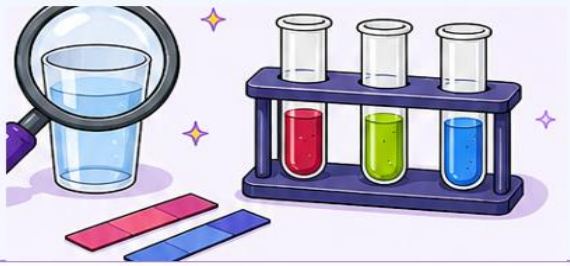
A clear solution can still be **acidic**, **neutral**, or **alkaline**. Litmus paper is our "**chemical eye**" for identifying these hidden properties.

The "Chemical Eye"
That Sees Invisible Forces

Chapter 9

2 Diagnosing the Earth and the Nile

- In modern Egypt, testing water solutions helps us understand the condition of the land and the Nile River.
- Litmus paper changes color to show the properties of a solution.
- These scientific messages help us protect our beautiful environment.



The Health of the Soil for Growing Crops

- Farmers check whether the soil is acidic or alkaline.
- They adjust the soil to provide the best conditions for plants.
- Cotton and wheat grow well when the soil has suitable properties.
- Good soil conditions help farmers achieve a large harvest.

Water Quality Management of the Nile River

- The quality of Nile River water is scientifically monitored.
- Scientists monitor wastewater from factories.
- They check that the water is not strongly acidic or alkaline.
- Monitoring water quality helps protect fish, birds, and the environment.

Clean Water for Life

Healthy River, Safe Animals

Science for a Better Future

Science helps us understand, protect, and preserve our wonderful world!

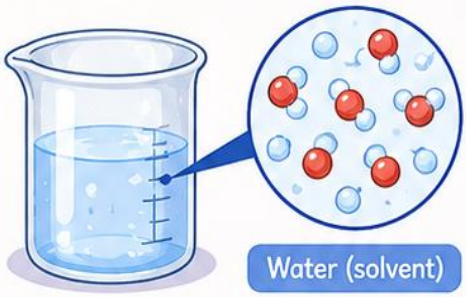
Chapter 10:
Reaction Between Solutions and Metals

Lesson 10-1:

Aqueous Solutions and Metals

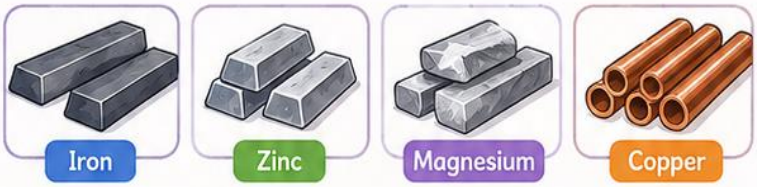
What does "aqueous" mean?

- Aqueous solution means a solution in which **water** is the solvent.

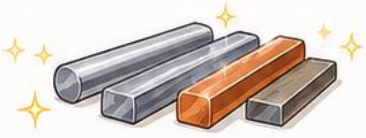


What are metals?

- Metals are materials such as **iron**, **zinc**, **magnesium** and **copper**.



- Many metals are **shiny** when freshly exposed.



- They are generally good conductors of **heat** and **electricity**.

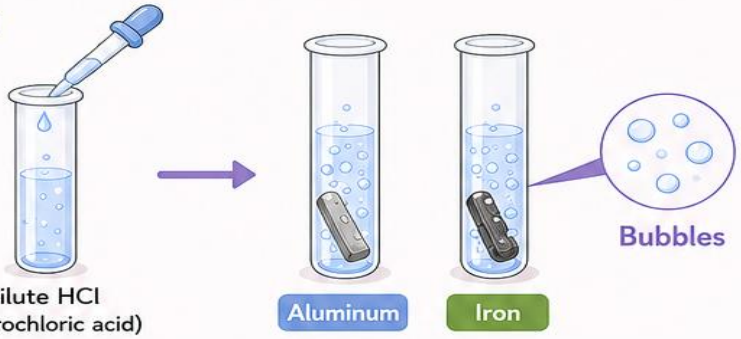
- Different metals have different **reactivities**.

Lesson 10-1:

Aqueous Solutions and Metals

Some aqueous solutions can react with metals.

- Hydrochloric acid (HCl) reacts with **aluminum** and **iron** when dilute HCl is poured on metals like aluminum and iron, they react and **produce bubbles.**

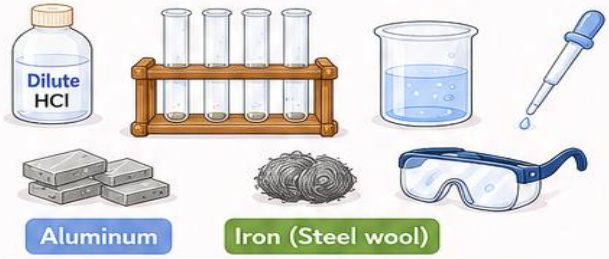


Activity

Let's investigate how hydrochloric acid reacts with metals.

1 Tools:

- Dilute hydrochloric acid (HCl)
- Test tubes (4)
- Test tube rack
- Aluminum (small pieces)
- Iron (steel wool)
- Pipette
- Test tube brush
- Water
- Beaker
- Safety goggles



2 Steps:

1. Put a small piece of aluminum in a test tube.
2. Put a small amount of steel wool (iron) in another test tube.
3. Use a pipette to add dilute HCl to the first test tube.
4. Use another pipette to add dilute HCl to the second test tube.
5. Observe the changes in both test tubes.

Observations

Test Tube Set	Metal	Liquid Added	What We Observed
1	Aluminum	Dilute HCl	✓ Bubbles were produced. ✓ The aluminum piece became hot. ✓ The aluminum surface started to change. ✓ The bubbles are a gas.
2	Iron (Steel wool)	Dilute HCl	✓ Bubbles were produced. ✓ The steel wool became hot. ✓ The steel wool surface started to change. ✓ The bubbles are a gas.

Conclusion

From our activity, we can conclude that dilute hydrochloric acid (HCl) reacts with metals such as **aluminum** and **iron**. When HCl is poured on these metals, they **react and produce bubbles (a gas).**

Lesson 10-2: What is Left After Evaporation?

Point! Changes in Metals

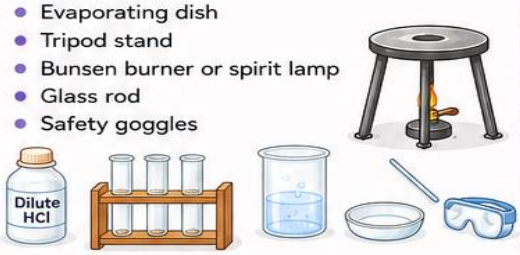
- When you evaporate the liquid of aluminum that reacted with dilute hydrochloric acid, a **white solid** appears.
- Differences from the original metal:
  - Appearance: It is not shiny.
  - Reaction: Even if you put the remaining solid into dilute hydrochloric acid again, it will not react and produce bubbles like the original metal.

→ We can say that the metal changed into a **substance with completely different properties** by reacting with the aqueous solution.

Activity Let's find out what is left after evaporation!

1 Tools:

- Aluminum (small pieces)
- Dilute hydrochloric acid (HCl)
- Beaker
- Test tubes (2)
- Test tube rack
- Evaporating dish
- Tripod stand
- Bunsen burner or spirit lamp
- Glass rod
- Safety goggles



2 Steps:

1. Put small pieces of aluminum into a beaker.
2. Add dilute hydrochloric acid to the beaker.
3. Let the reaction occur until no more bubbles are seen.
4. Filter the mixture (if needed) to get a clear solution.
5. Pour the clear solution into an evaporating dish.
6. Heat the evaporating dish gently to evaporate the liquid until it is completely dry.
7. Observe the solid (white powder) left in the dish.
8. Add dilute hydrochloric acid to the produced powder in another test tube.
9. Observe the result.

Observations

Step	What We Do	What We Observe
1 – 3	React aluminum with dilute HCl.	Bubbles are produced. The aluminum pieces get smaller and some may disappear.
4 – 5	Evaporate the clear solution until it is dry.	The liquid disappears. A white solid (powder) remains in the evaporating dish.
8 – 9	Add dilute HCl to the produced powder.	No bubbles are produced. The solid does not react with dilute HCl.

Conclusion

When the liquid is evaporated, a **white solid** remains.
This solid looks different from the original metal and does not react with dilute hydrochloric acid.
The metal has changed into a **substance with completely different properties**.

Chapter 10 Column: The Science of Change – How Metals are Reborn

1 The Mystery of Gold That Shines Forever

- Strong, hard metals react with an acid and produce **bubbles**.
- This gives us an important clue about **Egypt's** proud golden history.
- Figuring out the true nature of these violent changes in the experiment helps us solve the mystery of why the **Pharaoh's mask** continues to shine even after thousands of years.

Why the Pharaoh Chose Gold:

- The reason Tutankhamun's **golden** mask is still beautiful today is because **gold** has an **extremely stable** property.
- It **does not react** with strong solutions like hydrochloric acid.

The Energy Released When Metals React:

- The bubbles that are produced when iron or aluminum reacts with dilute hydrochloric acid are actually **hydrogen gas.**
- When a metal changes into another substance, a large amount of **chemical energy** is at work to create a new gas.

In Short!

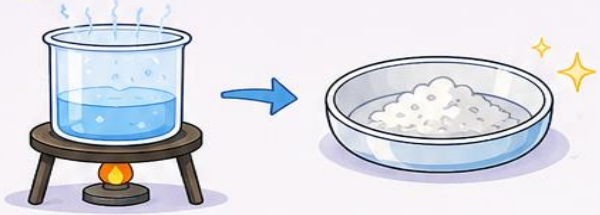
- Some metals react with **acids** and produce **bubbles.**
- The remaining solid can have completely **different properties.**
- **Gold** is special because it does not react with strong solutions.
- Chemical reactions involve **energy** and can create new gases.

Chapter 10 Column:
The Science of Change – How Metals are Reborn

2 The Power of Recycling to Reuse Resources

- When a metal reacts with a solution, it changes into something else.
- After the liquid is evaporated, we find a new solid left behind.
- This shows the metal has not disappeared. It has only changed form.
- Understanding these **"invisible changes"** helps us in factories and in recycling.
- This helps us **save Earth's resources** again and again.

The Technology of Evaporating to Recover



- We heat the solution to **evaporate** the water.
- The water turns to steam and disappears.
- The solid left behind is collected.
- This solid can be used to make very **pure materials**.

Egypt's Future and Recycling

- Instead of throwing away metal things, we can use the power of **chemistry** to turn them back into **clean materials**.
- What we learn about change helps us **reduce trash**.
- This makes Egypt's future **cleaner, greener and stronger**.

- ★ Even if a metal reacts and we can't see it anymore, its parts are **still there**.
- ★ Our job is to find them, **take them out and use them again**.
- ★ **Future scientists** will use these ideas to help our world!

Lab Safety and Professional Tools

In 6th-grade experiments, we handle fire and strong chemicals (like hydrochloric acid). This is a guide to help you do experiments safely and get the correct results.

1 Handling Chemicals

- **Wear safety goggles:**

Always wear them to keep solutions like hydrochloric acid from getting into your eyes.

- **How to smell chemicals:**

Waft the smell toward you with your hand. Do not put the chemical too close to your nose.

- **If you spill a chemical:**

Wash it off right away with lots of water and tell your teacher.

Be safe, be smart, and have fun in the lab!

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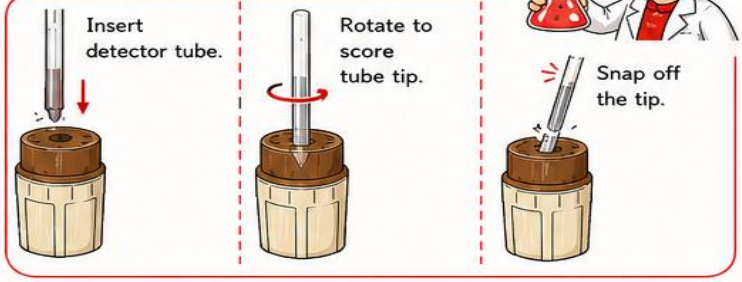
2 How to Use a Gas Detector Tube

1 Snap off both ends:

Use the special tip holder to snap off both ends of the detector tube.

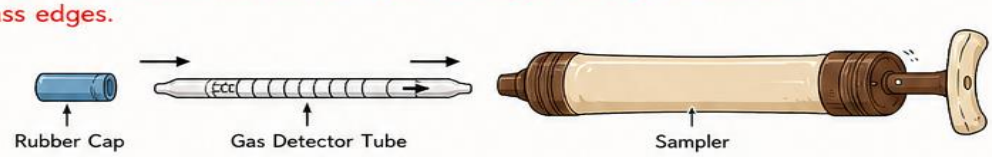


2 Check the arrow and attach the tube:

Push the detector tube firmly into the opening of the gas sampler so the arrow (⟶) points toward the sampler.

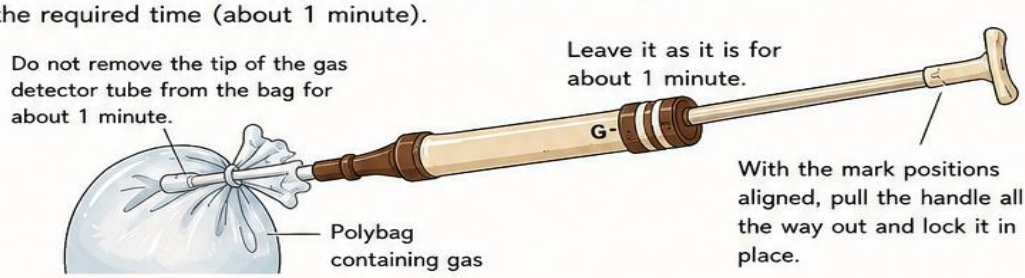
[Caution!]

- Always put on the rubber cap so you do not get hurt by the broken glass edges.



3 Draw in the gas and wait:

Pull the handle all the way out at once and lock it in place. Wait for the required time (about 1 minute).

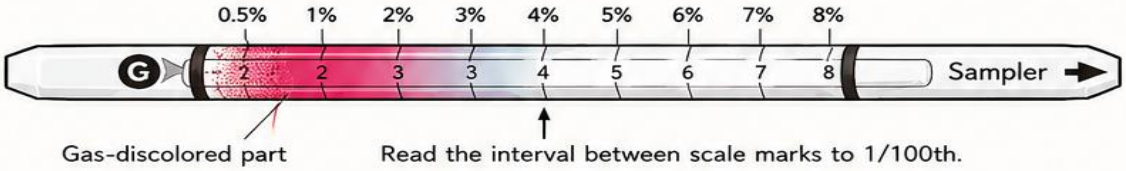


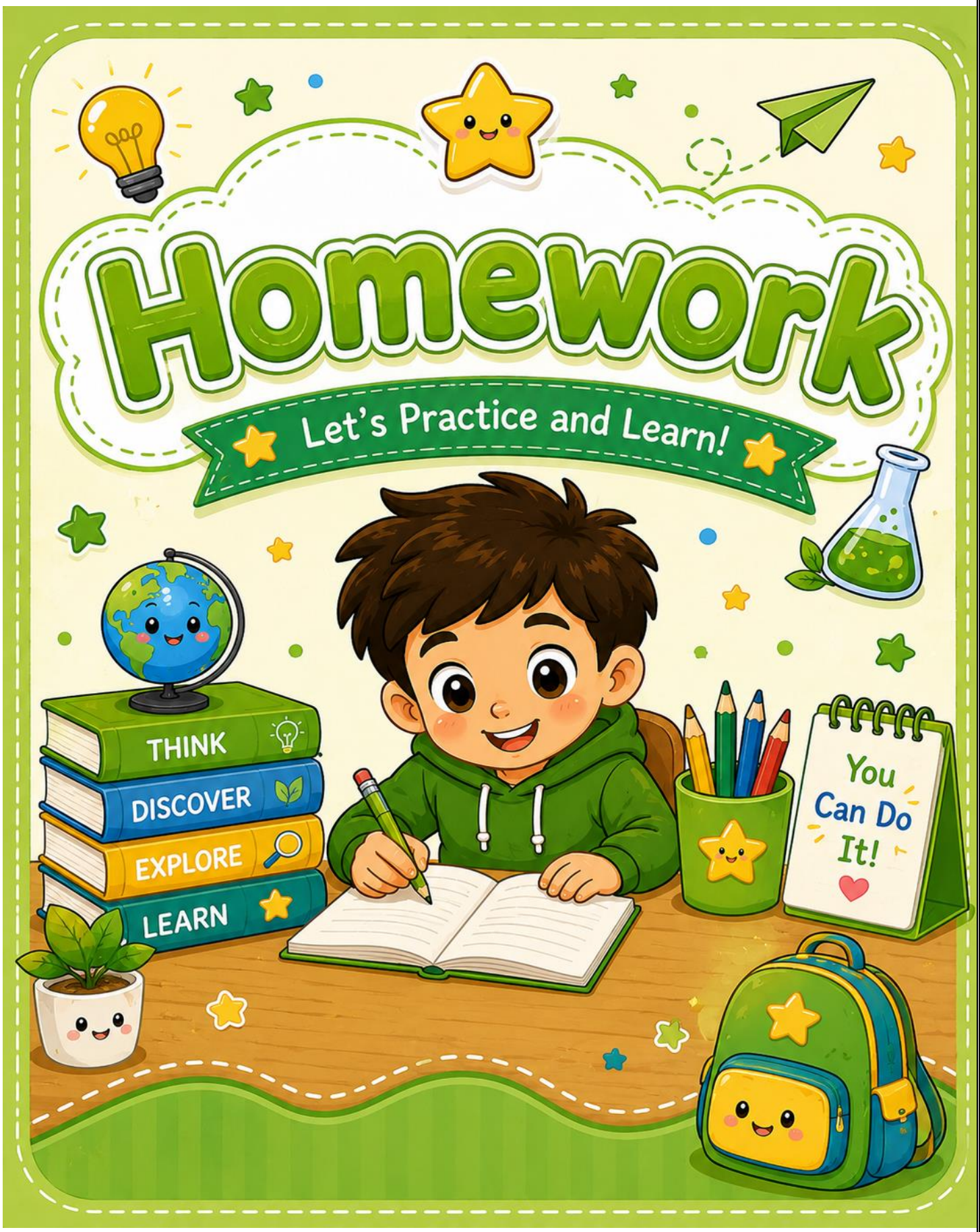
4 Read the scale:

Read the scale where the color has changed.

[Caution!]

- Oxygen detector tubes get hot, so do not touch them right away.
- Throw away used detector tubes exactly as your teacher tells you.




Homework
Let's Practice and Learn!
THINK
DISCOVER
EXPLORE
LEARN
You Can Do It!

Lesson 1-1: Combustion and Air Worksheet

Answer all the questions.

A. Choose the correct answer.

1. Which condition allows a candle to keep burning?
   a) No air supply b) Continuous fresh air
   c) A closed lid d) Less oxygen

2. In the experiment, which setup provides the least fresh air to the candle?
   a) Jar without a lid b) Jar with a small gap
   c) Completely closed jar d) Open candle

3. Making a small gap at the bottom of the jar causes _____.
   a) some fresh air to enter b) all the air to leave
   c) the candle to go out immediately
   d) the oxygen to disappear

4. Which flame would you expect to be the strongest?
   a) Flame in a completely closed jar
   b) Flame in a jar with a small gap
   c) Flame in a completely open jar
   d) Flame with no oxygen

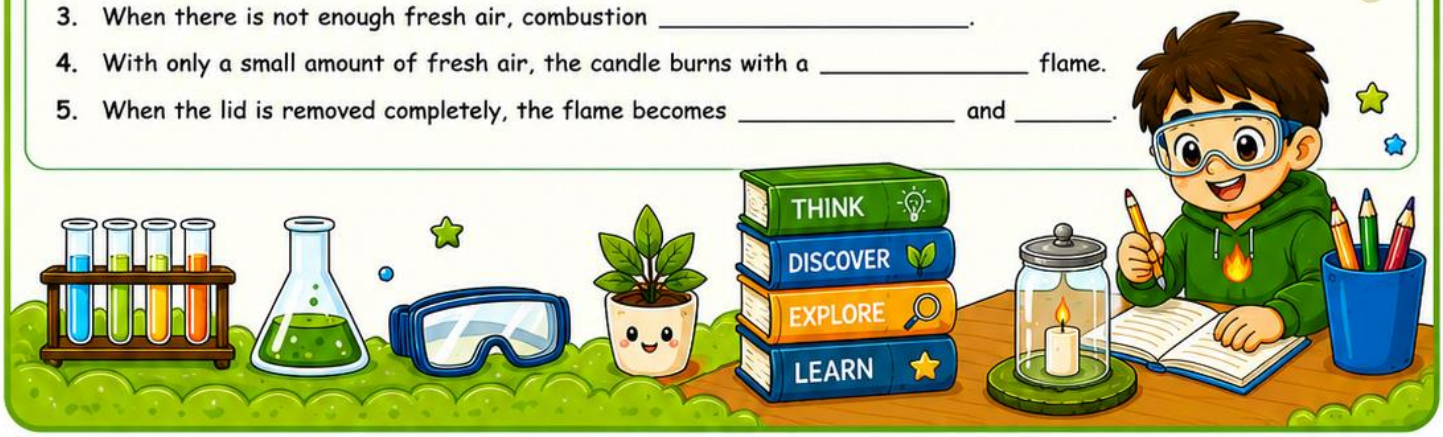
5. Which of the following best describes the relationship between air supply and combustion?
   a) More fresh air supports stronger combustion.
   b) Less fresh air makes combustion stronger.
   c) Air has no effect on combustion.
   d) Combustion removes the need for oxygen.

B. Write True or False.

1. A candle can keep burning without oxygen. ( )
2. A small opening allows some fresh air to reach the candle. ( )
3. The amount of air available can affect the strength of a candle flame. ( )
4. The flame is strongest when the jar is completely sealed. ( )
5. During combustion, oxygen in the surrounding air is consumed.

C. Complete the following.

1. The process in which substances burn is called _____.
2. During combustion, _____ in the air is gradually consumed.
3. When there is not enough fresh air, combustion _____.
4. With only a small amount of fresh air, the candle burns with a _____ flame.
5. When the lid is removed completely, the flame becomes _____ and _____.



D. What happens when...?

1. A candle is lit and left where fresh air can reach it freely?

_____

2. A burning candle is covered by a jar and the jar is completely closed?

_____

3. The oxygen trapped inside a closed jar is gradually used up?

_____

4. A small opening is made at the bottom of the jar while the top remains covered?

_____

5. The lid is removed completely after the candle has been burning with only a small air gap?

_____

E. Give reasons for the following

1. Oxygen is necessary for combustion.

_____

2. Fresh air must continuously flow in and out for combustion to continue.

_____

3. The flame in the completely closed jar goes out after a while.

_____

4. The candle can still burn when there is a small gap at the bottom of the jar.

_____

5. The flame becomes brighter and stronger when the lid is completely removed.

_____

Lesson 1-2 Role of Oxygen

Try

1 (1) In the experiment of burning a standing candle inside a jar, what happens to the flame if you put a lid on the jar to stop air from coming in and out?

---

(2) Why did the flame go out inside the jar with the lid? Choose the correct answer from below.

A: Because there was no water vapor in the jar.

B: Because it got dark inside the jar.

C: Because fresh air stopped coming in.

2 (1) What gas is included in the air and has the power to help things burn?

---

(2) What happens to the burning when you put a lit incense stick into a jar of oxygen? Choose the correct answer from below.

A: It becomes weaker.

B: It burns brightly.

C: The flame goes out.

(3) What happens to the burning when you put a lit incense stick into a jar of carbon dioxide? Choose the correct answer from below.

A: It keeps burning as it is.

B: It burns brightly.

C: The flame goes out.

Exercise

1 (1) In the experiment of burning a candle inside a jar, what happens to the flame if you leave gaps at the top and bottom of the jar? Choose the correct answer from below.

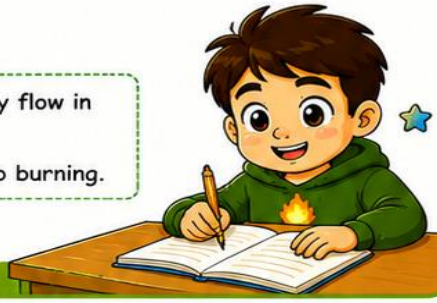
A: It keeps burning.

B: It goes out immediately.

(2) Fill in the blanks with the correct words.

For things to keep burning, fresh (❶ _____) must constantly flow in and out.

In a place where air is not replaced, things (❷ _____) keep burning.



Lesson 1-2: The Role of Oxygen

A. Choose the correct answer

1 Air is best described as:

a) one pure gas b) a mixture of gases
c) only oxygen d) only nitrogen

2 Which gas represents about one-fifth of the volume of air?

a) Nitrogen b) Oxygen
c) Carbon dioxide d) Water vapor

3 Which pair of gases does NOT support combustion?

a) Oxygen and nitrogen
b) Nitrogen and carbon dioxide
c) Oxygen and carbon dioxide
d) Oxygen and water vapor

4 Which tool carries gas from a gas cylinder to the collecting jar?

a) Water basin
b) Candle holder
c) Bent tube
d) Lid

5 About how much of the jar should be filled with gas before it is removed from the water basin?

a) 10%
b) 21%
c) 50%
d) 70%

B. Complete the following

1 Air surrounds the _____ and is made up of several different gases.

2 Oxygen supports _____, while nitrogen and carbon dioxide do not.

3 To collect a gas by water displacement, the gas jar is first completely filled with _____.

4 As gas enters the inverted jar, it _____ the water inside the jar.

5 During water displacement, the gas jar is placed _____ in the water basin.

C. Write True or False

1 Water vapor is one of the gases that may be found in air. ( )

2 Nitrogen is present in a smaller amount than oxygen in air. ( )

3 Equal amounts of gases should be used when comparing their effects on a candle. ( )

4 A collecting jar is placed upright in the basin during water displacement. ( )

5 The water level inside the collecting jar falls as gas enters it. ( )

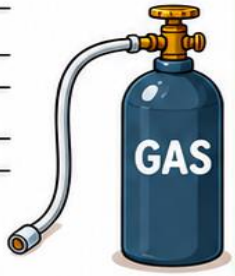
Lesson 1-2: The Role of Oxygen

D. Write the scientific term

1. The process in which a substance burns. _____
2. The gas that makes up the greatest percentage of air. _____
3. The gas needed for combustion and for breathing: _____
4. The container used to collect a gas during the experiment. _____
5. The method in which a gas is collected by pushing water out of an inverted container. _____

E. What happens when...?

1. A lighted candle is placed in a jar containing oxygen? _____
2. A lighted candle is placed in a jar containing nitrogen? _____
3. A lighted candle is placed in a jar containing carbon dioxide? _____
4. Gas begins to enter a completely water-filled inverted jar through the bent tube? _____
5. A burning substance is supplied with more oxygen than before? _____



F. Give reasons for the following

1. Air is considered a mixture rather than a single substance. _____
2. Equal amounts of oxygen, nitrogen, and carbon dioxide are prepared when comparing their effects on burning. _____
3. The gas jar must be inverted when gas is collected by water displacement. _____
4. A lighted candle is useful for identifying which of the three gases supports combustion. _____
5. Oxygen is described as essential to living organisms as well as important in combustion. _____

Lesson 1-3: Air After Burning

Try

(1) Answer the following questions about the air after things burn.

① What gas has increased? _____

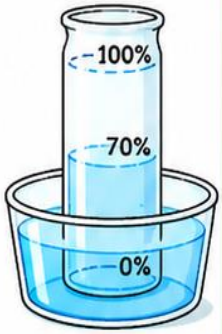
② What gas has decreased? _____

(2) What is the colorless and clear liquid used to check if there is carbon dioxide?

_____

Exercise

1. What happens when you put limewater in a jar with increased carbon dioxide and shake it? Choose the correct answer from below.



A: It turns blue.

B: It turns yellow.

C: It turns milky white.

2. How does the amount of oxygen change after burning?

_____

3. How does the amount of carbon dioxide change after burning?

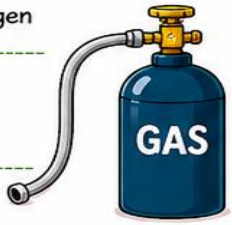
_____

Lesson 1-3: Air After Burning

A. Choose the Correct Answer

1 Which gas increases in the air after burning?

a) Oxygen b) Carbon dioxide c) Nitrogen d) Hydrogen



2 Which substance is used to test for carbon dioxide gas?

a) Distilled water b) Limewater c) Salt water d) Oil

3 What happens to limewater when carbon dioxide passes through it?

a) It turns blue. b) It turns green. c) It turns milky white. d) It remains colorless.

4 Which tool contains chemicals that react with certain gases and change color?

a) Gas detection tube b) Beaker c) Thermometer d) Combustion spoon

5 What is the main function of a sampling pump?

a) To heat a gas b) To draw a gas sample through a detection tube

c) To produce oxygen d) To measure temperature

6 Which device displays the percentage of a gas electronically?

a) Gas monitoring equipment b) Combustion spoon d) Safety goggles

7 Ancient Egyptians used bellows mainly to:

a) remove oxygen from a furnace. b) push more air into a furnace.

c) decrease the temperature of a fire. d) produce carbon dioxide.

8 Which of the following is a renewable source of energy?

a) Coal b) Natural gas c) Sunlight d) Petroleum

9 Increasing carbon dioxide in the atmosphere contributes to:

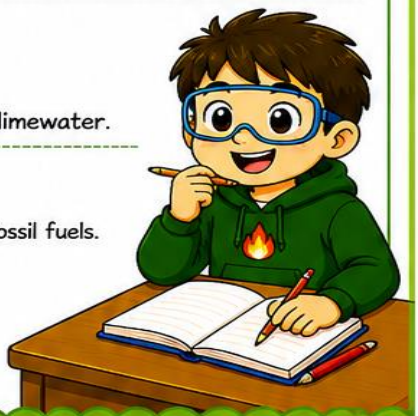
a) the greenhouse effect. b) less combustion.

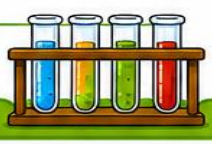
c) lower oxygen use. d) the production of limewater.

10 A solar power plant helps protect the environment because it:

a) burns more fuel. b) produces electricity without burning fossil fuels.

c) increases carbon dioxide production. d) uses oxygen to produce electricity.





Lesson 1-3: Air After Burning

B. Write True or False

1. The amount of oxygen in air increases after burning. ( )
2. Limewater can be used as an indicator for the presence of carbon dioxide. ( )
3. Limewater remains completely clear when carbon dioxide is passed through it. ( )
4. Different gas-detection tubes may be designed to detect different gases. ( )
5. A gas-monitoring device can show gas concentration as a numerical percentage. ( )
6. Gas-monitoring equipment must always be broken after one use. ( )
7. Ancient Egyptians used combustion in some of their daily activities. ( )
8. Increasing the oxygen supply can make combustion occur faster. ( )
9. Renewable energy depends only on fossil fuels. ( )
10. Solar power can reduce some of the negative effects associated with carbon dioxide emissions. ( )

C. Write the Scientific Term

1. A process in which substances react with oxygen and burn. _____
2. The gas that is produced during burning and causes limewater to turn milky white. _____
3. A colorless, clear liquid used to test for the presence of carbon dioxide gas. _____
4. Small glass tubes containing chemicals that react with particular gases. _____
5. A device used to draw a sample of gas through a gas-detection tube. _____
6. An electronic device with a sensor that measures the percentage of gas in the surrounding air. _____
7. A blowing tool used to force additional air into a fire or furnace. _____
8. The rise in Earth's temperature caused by increasing carbon dioxide in the atmosphere. _____
9. Energy obtained from natural sources that do not run out. _____
10. A power plant that uses sunlight to generate electricity. _____

Lesson 1-3: Air After Burning

E. What Happens When...?

1. A burning candle is sealed inside a jar and left for some time?
2. A burning material receives a larger supply of fresh air? The oxygen available to a burning substance decreases?
3. More oxygen is supplied to a burning fire?
4. A gas sample reacts with the chemical inside a suitable gas-detection tube?
5. A sampling pump is operated while connected correctly to a gas-detection tube?
6. Bellows are used to push more air into a hot furnace?
7. The percentage of carbon dioxide in the atmosphere increases greatly?
8. Sunlight falls on the solar panels of a solar power plant?
9. Electricity is generated using solar energy instead of burning fossil fuels?

Give Reasons for the Following

1. Air after combustion is different from air before combustion.
2. Carbon dioxide increases in the air after some substances burn.
3. Limewater is useful when investigating the gas produced during burning.
4. Two jars—one with a burning candle and one without—can be used in the limewater experiment.
5. Gas-detection tubes are useful for identifying gases in air.
6. A color chart is supplied with gas-detection tubes.
7. Gas-monitoring equipment is useful in specialized hospitals and intensive care units.
8. Ancient Egyptians used bellows when working with furnaces.
9. Increasing atmospheric carbon dioxide can raise Earth's temperature.
10. Using solar power can help reduce the negative effects of carbon dioxide.

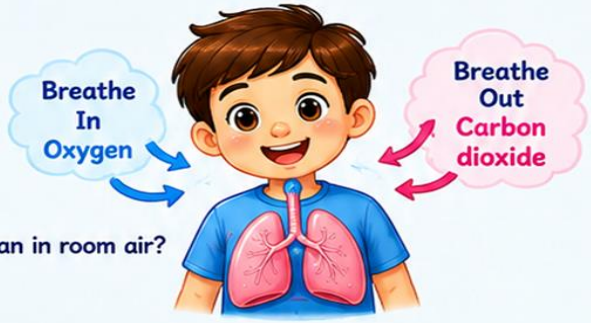
Chapter 2:
Lesson 2-1:
How Breathing Works

Breathing helps us take in oxygen and release carbon dioxide.

A. Choose the Correct Answer

Choose the correct answer.

1. **Which statement best describes breathing?**
   a) Taking in carbon dioxide only
   b) Taking in oxygen and releasing carbon dioxide
   c) Taking in food and releasing oxygen
   d) Releasing oxygen and taking in carbon dioxide



2. **Which gas is present in a larger amount in exhaled air than in room air?**
   a) Oxygen b) Nitrogen
   c) Carbon dioxide d) Hydrogen

3. **Which material is used in the activity to compare the carbon dioxide in exhaled air and room air?**
   a) Salt water b) Limewater c) Distilled water d) Oil

4. **A student wants to compare the amounts of oxygen and carbon dioxide in two air samples. Which tool should be used?**
   a) Scissors b) Gas-detection tubes c) Safety goggles d) Plastic bags only

5. **Which observation provides evidence that exhaled air contains more carbon dioxide than room air?**
   a) Limewater turns milky more quickly with exhaled air.
   b) Exhaled air becomes blue.
   c) Room air turns limewater red.
   d) Exhaled air does not react with limewater.

B. Write True or False

Think Like a Scientist!

Write True or False.

1. Respiration is one of the important life processes carried out by living organisms. ( )
2. During inhalation, air containing oxygen enters our lungs. ( )
3. During exhalation, carbon dioxide leaves our lungs. ( )
4. The two plastic bags in the experiment contain air collected from exactly the same source. ( )
5. Safety goggles should be worn when carrying out the limewater experiment. ( )

Small Scientists Make a Big Difference!

Healthy Lungs Brighter Tomorrows!

Explore Learn Grow!

Chapter 2:
Lesson 2-1:
How Breathing Works

Breathing helps us take in **oxygen** and release **carbon dioxide**.

Healthy Lungs Happy You!

Small Steps Big Scientists!

C. Write the Scientific Term

Use what you have learned!

1. The process of taking air into the lungs. _____
2. The process of pushing air out of the lungs. _____
3. The gas needed by the body to release energy from food. _____
4. The gas released from the body during breathing. _____
5. Small tubes containing chemicals that can be used to test the amounts of particular gases. _____

Breathe Learn Be Healthy!

D. Give Reasons for the Following

Think Like a Scientist!

1. Our body needs a continuous supply of oxygen.
2. Two plastic bags are used in the breathing experiment instead of only one.
3. One plastic bag is filled with exhaled air while the other is filled with normal room air.
4. The two bags should be shaken after adding the same amount of limewater.
5. Exhaled air and room air give different results when tested with gas-detection tubes.

E. What Happens When...?

Predict Think Explain!

1. A person breathes in?
2. A person breathes out?
3. Limewater is added to a bag of exhaled air and the bag is shaken?
4. The oxygen gas-detection tube is used to compare exhaled air with room air?
5. The carbon dioxide gas-detection tube is used to compare exhaled air with room air?

Clean Air Healthy Lungs Brighter Futures!

Chapter 2 Respiration

Lesson 2-2: How Do Lungs Work?

Breathe Learn Grow!

Try

1 (1) What gas do we take into our body when we breathe in?

_____

(2) When comparing the air we breathe in and the air we breathe out, which gas increases in the air we breathe out? Choose the correct answer from below.

A. Oxygen
B. Carbon dioxide
C. Neither increases

Clean Air Healthy Lungs Happier You!

2 (1) What is the organ in our chest called that exchanges oxygen and carbon dioxide? _____

(2) Fill in the blanks with the correct words to complete the sentence.

Through the (① _____ around the lungs, (② _____ in the air is taken into the blood, and (③ _____ in the blood is let out into the air.

Exercise

1 (1) What happens to limewater when you blow breathed-out air into it? Choose the correct answer from below.

A. It turns milky white.
B. It turns blue.
C. It disappears.

Science in Everyday Life!

(2) Which statements correctly explain breathing? Choose all correct answers from below.

A. The air we breathe out has no oxygen at all.
B. The air we breathe out also has oxygen.
C. The air we breathe out has more oxygen than the air we breathe in.
D. The air we breathe out has more carbon dioxide than the air we breathe in.

Breathe Well Live Well Be Amazing!

2 (1) Which statement correctly explains gas exchange in the lungs? Choose the correct answer from below.

A. It lets out oxygen from the blood into the air.
B. It takes in oxygen from the air into the blood.
C. It takes in carbon dioxide from the air into the blood.

(2) What carries the oxygen taken in by the lungs to the whole body? Choose the correct answer from below.

A. Blood
B. Muscles
C. Bones

Explore Learn Grow!

Chapter 2

Lesson 2-2:

How Do Lungs Work?

Our lungs help us take in oxygen and remove carbon dioxide.

Healthy Lungs Happy You!

A. Choose the Correct Answer

Choose the correct answer.

1. Which structure is the main entrance for air into the respiratory system?
   a) Alveoli b) Nose c) Diaphragm d) Bronchioles

2. Which sequence correctly shows the path of air through the respiratory system?
   a) Nose → bronchi → trachea → alveoli
   b) Nose → trachea → bronchi → bronchioles → alveoli
   c) Nose → alveoli → bronchi → trachea
   d) Trachea → nose → bronchioles → alveoli

3. Where does gas exchange between the air and blood take place?
   a) Trachea b) Nose c) Alveoli d) Diaphragm

4. During inhalation, which change helps air enter the lungs?
   a) The chest cavity becomes smaller.
   b) The diaphragm moves upward.
   c) The chest cavity becomes larger.
   d) The lungs become smaller.

5. A fish obtains the oxygen it needs mainly through its:
   a) lungs. b) gills. c) nose. d) bronchi.

Breathe In Oxygen Breathe Out Carbon Dioxide

Your Lungs Work Hard for You!

B. Write True or False

Think Check Be a Scientist!

Write True or False.

1. The nose helps filter, warm, and moisten the air we breathe in. ( )
2. The trachea carries air from the nose or mouth toward the lungs. ( )
3. Bronchioles are larger tubes than the bronchi. ( )
4. Cats breathe air using lungs. ( )
5. Oxygen absorbed by a cat's lungs can be transported by the blood to different parts of its body. ( )

Healthy Lungs Brighter Futures!

Explore
Learn
Discover

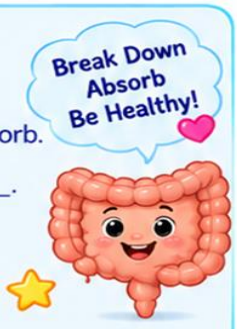
Chapter 3 Digestion

Lesson 3-1:

Digestion and How Does It Work?

A. Complete the Following

Fill in the blanks with the correct words.



1. Digestion changes the food we eat into simple _____ that the body can absorb.
2. Food moves from the mouth to the stomach through a tube called the _____.
3. The stomach is located mainly in the _____ abdomen.
4. In a cat, waste is stored in the _____ before leaving the body.
5. In a fish, food passes from the mouth to the esophagus and then reaches the _____.

B. Write True or False

Write True (T) or False (F).

1. Teeth start breaking down food in the mouth. ( )
2. The small intestine is shorter than the stomach. ( )
3. The large intestine surrounds the small intestine in the human digestive system. ( )
4. Cats and fish both have digestive systems, although their digestive organs are arranged differently. ( )
5. All animals need food to live and grow. ( )

C. Cross Out the Odd One

In each group, one word does not belong. Cross it out.

1. Mouth – Esophagus – Stomach – Trachea
2. Stomach – Small intestine – Large intestine – Lungs
3. Chest – Upper abdomen – Middle abdomen – Alveoli
4. Cat – Fish – Human – Solar panel
5. Digestion – Absorption – Removal of waste – Inhalation

Amazing Bodies Brighter Futures!

Chapter 3 Digestion

Learn Explore Grow!

Lesson 3-1:

Digestion and How Does It Work?

Healthy Food Healthy You!

Think Recall Write!

D. Write the Scientific Term

Write the correct scientific term for each description.

1. The process of breaking down food into smaller substances that the body can absorb. _____
2. The organ that stores food and mixes it with strong digestive juices. _____
3. The organ where most nutrients from digested food are absorbed into the blood. _____
4. The organ that absorbs water from leftover food and helps form waste. _____
5. The opening through which solid waste leaves the digestive system. _____

E. What Happens When...?

Write what happens in each situation.

Think Explain Be a Scientist!

1. Food enters the mouth? _____
2. Food reaches the stomach? _____
3. Digested food reaches the small intestine? _____
4. Leftover food enters the large intestine? _____
5. Food passes through the digestive system of a fish and reaches the end of its intestine? _____

F. Give Reasons for the Following

Write the reason for each statement.

Think Deeper Give a Reason You Can Do It!

1. Food must be digested before the body can make full use of its nutrients. _____
2. The stomach and intestines are important parts of the digestive system. _____
3. The small intestine is long and coiled inside the abdomen. _____
4. The large intestine is important even though most nutrient absorption occurs in the small intestine. _____
5. A cat and a fish can digest food even though their digestive systems are not arranged in exactly the same way. _____

Good Food Bright Minds Brighter Futures!

Chapter 3 Digestion

Lesson 3-2:

The Function of the Saliva

Try

Follow the Right Path for a Healthy You!

1 (1) What do we call the process of changing food into a form that is easy to absorb into the body?

_____

(2) Choose the correct words from below to fill in the blanks.

Food that enters from the mouth is carried in the order of Mouth ➡ (① _____) ➡ (② _____) ➡ (③ _____) ➡ (④ _____), and comes out from the anus.

Large intestine / Stomach / Food pipe / Lungs / Small intestine

Exercise

Good Digestion Brighter Days!

1 (1) What is the correct order that food passes through? Choose the correct answer from below.

- A: Mouth ➡ Stomach ➡ Food pipe ➡ Small intestine ➡ Large intestine
- B: Mouth ➡ Food pipe ➡ Stomach ➡ Large intestine ➡ Small intestine
- C: Mouth ➡ Food pipe ➡ Stomach ➡ Small intestine ➡ Large intestine
- D: Mouth ➡ Food pipe ➡ Large intestine ➡ Small intestine ➡ Stomach

(2) What is the part of food from the mouth to the anus called?

_____

2 (1) In the experiment to check the function of saliva, why do you warm the test tubes in hot water of about 40 degrees Celsius? Choose the correct answer from below.

- A: To disinfect.
- B: To evaporate the water.
- C: To make it about the same temperature as inside the body.

(2) After mixing saliva and starch and warming it, did the color change when you added the iodine solution? Choose the correct answer from below.

It turned blue-black. / It did not change.

Healthy Habits Happier You!

Chapter 3 Digestion

Lesson 3-2:

The Function of the Saliva

A. Complete the Following

Fill in the blanks with the correct words.

1. The first step of digestion takes place in the _____.
2. Saliva is a _____ secretion found in the mouth.
3. Saliva helps break down _____ found in food.
4. Starch is changed into a simple _____ that can be easily absorbed by the body.
5. A liquid that helps digest food is called a digestive _____.

B. Write True or False

Write True (T) or False (F).

1. Digestion begins only after food reaches the stomach. ( )
2. Food begins to change from a complex form into a simpler form in the mouth. ( )
3. Saliva helps the body digest some of the food we eat. ( )
4. Digestive juices are produced only in the mouth. ( )
5. The stomach and intestines also produce digestive juices. ( )

Good Food
Bright Minds
Brighter Futures!

Chapter 3 Digestion

Explore Learn Grow!

3-3 How Absorption Works

Try

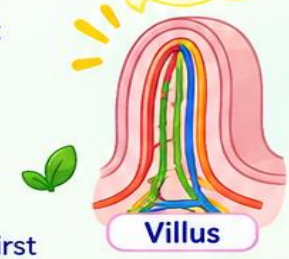
Tiny Bumps Big Help! Nutrients for a Stronger You!

1 (1) What do we call the small bumps on the inner wall of the small intestine that absorb nutrients?

_____

2 (2) What carries the absorbed nutrients to the body? Choose the correct answer from below.

A: Blood vessels
B: Nerves
C: Muscles



3 (3) Where do the nutrients absorbed in the small intestine pass through first to be carried to the whole body? Choose the correct answer from below.

A: Stomach
B: Large intestine
C: Liver

Healthy Food Better Absorption Happier You!

Exercise

A Larger Surface Area A Healthier You!

1 (1) What is the advantage of having villi on the wall of the small intestine? Choose the correct answer from below.

A: It makes it easier to swallow food.
B: The surface area of the wall of the small intestine becomes larger, and nutrients can be absorbed efficiently.
C: It makes it easier for nutrients to pass through.

2 (2) Fill in the blanks with the correct words.

The nutrients of digested food, along with water, are mainly absorbed in the (① _____). The nutrients absorbed in the small intestine are carried by the (② _____) through the (③ _____) to the whole body.

Good Digestion Brighter Minds Brighter Futures!

Chapter 3 Digestion

Lesson 3-3

How Absorption Works

Explore Learn Grow!

Our body absorbs digested nutrients and uses them for energy, growth and repair.

A. Choose the Correct Answer

Choose the correct answer.

1. Which structure increases the surface area of the small intestine for absorption?
   a) Stomach b) Villi c) Liver d) Esophagus
2. After nutrients are absorbed through the wall of the small intestine, they enter the:
   a) blood vessels. b) stomach. c) large intestine. d) mouth.
3. Where do absorbed nutrients pass through before being carried to the rest of the body?
   a) Lungs b) Liver c) Stomach d) Esophagus
4. Which substance produced by the liver helps break fats into tiny droplets?
   a) Saliva b) Water c) Bile d) Blood
5. Which statement best describes one way the liver helps clean the body?
   a) It absorbs oxygen from the air. b) It breaks down harmful substances.
   c) It stores solid waste. d) It pumps blood around the body.

B. Write True or False

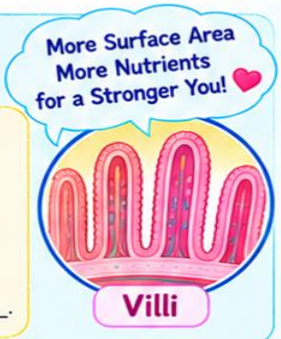
Write True (T) or False (F).

1. Absorption is the movement of digested nutrients and water from the intestine into the blood. ( )
2. Villi make the inner surface area of the small intestine smaller. ( )
3. The nutrients in food must become small and simple before they can be absorbed efficiently. ( )
4. The liver can store energy and release it when the body needs it. ( )
5. Blood vessels inside the villi carry absorbed nutrients away from the small intestine. ( )

C. Complete the Following

Fill in the blanks with the correct words.

1. Most nutrients from digested food are absorbed in the _____ intestine.
2. The tiny bumps lining the inner wall of the small intestine are called _____.
3. Villi contain many _____ vessels that carry absorbed nutrients.
4. The liver changes some nutrients from food into a simple form and stores them as _____.
5. Harmful substances broken down by the liver can be removed from the body through _____.



Good Nutrition
Brighter Minds
Brighter Futures!

Chapter 3 Digestion

Lesson 3-3

How Absorption Works

Explore Learn Grow!

Small Intestine Absorbs Nutrients!

Our body absorbs digested nutrients and uses them for energy, growth and repair.

D. What Happens When...?

Write what happens in each of the following situations.

Think Like a Scientist!

1. Digested nutrients reach the villi of the small intestine?
2. The inner wall of the small intestine has many villi instead of being smooth?
3. Absorbed nutrients enter the blood vessels inside the villi?
4. The body needs some of the energy that has been stored by the liver?
5. The villi in the small intestine are damaged or reduced in number?

Observe Think Explain Learn!

E. Give Reasons for the Following

Write the reason for each of the following.

Good Reasons Show Great Thinking!

1. The small intestine is an important organ for absorption.
2. Villi are useful for absorbing nutrients efficiently.
3. The liver is important after nutrients have been absorbed into the blood.
4. The liver helps keep the body safe and healthy.
5. Breaking fats into tiny droplets helps the process of digestion.

Healthy Digestion
Brighter Minds
Brighter Futures!

Chapter 4 Circulation and Excretion

Lesson 4-1

Heart and Blood

A healthy heart keeps your body strong!

Try

1 (1) What is the organ that works like a pump to send blood to the whole body?

2 (2) What are the important things that blood carries to the whole body? Choose all correct answers from below.

A: Oxygen
B: Carbon dioxide
C: Nutrients
D: Muscles

3 (3) Why does your pulse become faster after exercising? Choose the correct answer from below.

A: To let the blood rest slowly.
B: To carry more oxygen and nutrients to the whole body quickly.
C: Because there are not enough nutrients.

Exercise

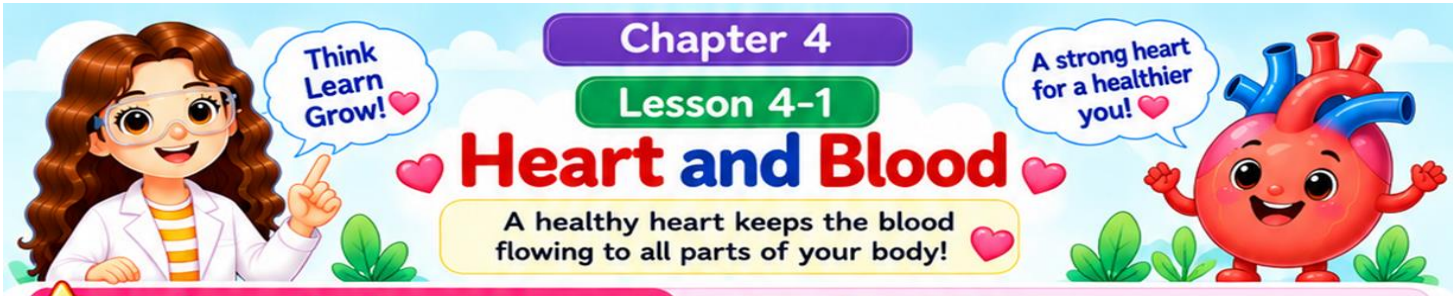
1 (1) Which statement correctly explains the heart? Choose the correct answer from below.

A: The heart plays a role in digesting the food we eat.
B: The heart is constantly beating to pump out blood.
C: The heart plays a role in storing nutrients.

2 (2) What do we call the movement of the heart pumping out blood?

3 (3) Fill in the blanks with the correct words.

The blood pumped out from the heart carries (① _____) and nutrients to the whole body, collects (② _____) and waste produced in the body, and returns to the heart again. The blood that has returned to the heart is sent to the (③ _____) to exchange (① _____) and (② _____).



Chapter 4

Lesson 4-1

Heart and Blood

A strong heart for a healthier you!

A healthy heart keeps the blood flowing to all parts of your body!

A. Choose the Correct Answer

Choose the best answer.

1. Which organ works like a pump to send blood around the whole body?

a) Lungs
b) Heart
c) Stomach
d) Small intestine

2. Which substances are carried by the blood to the body cells?

a) Oxygen and nutrients
b) Carbon dioxide and waste
c) Saliva and water
d) Carbon dioxide and nutrients

3. Where is blood sent to release carbon dioxide and take in fresh oxygen?

a) Stomach
b) Liver
c) Lungs
d) Small intestine

4. Which part of the circulatory system carries blood from one part of the body to another?

a) Blood vessels
b) Alveoli
c) Bronchi
d) Intestines

5. After delivering oxygen and nutrients to the body cells, blood collects:

a) oxygen and nutrients.
b) carbon dioxide and wastes.
c) food and water.
d) saliva and oxygen.

B. Write True or False

Write True (T) or False (F).

1. The heart is a strong muscular organ. ( )

2. Blood carries nutrients absorbed from the small intestine to body cells. ( )

3. Blood stays in the heart and does not travel around the body. ( )

4. The circulatory system includes the heart, blood vessels, and blood. ( )

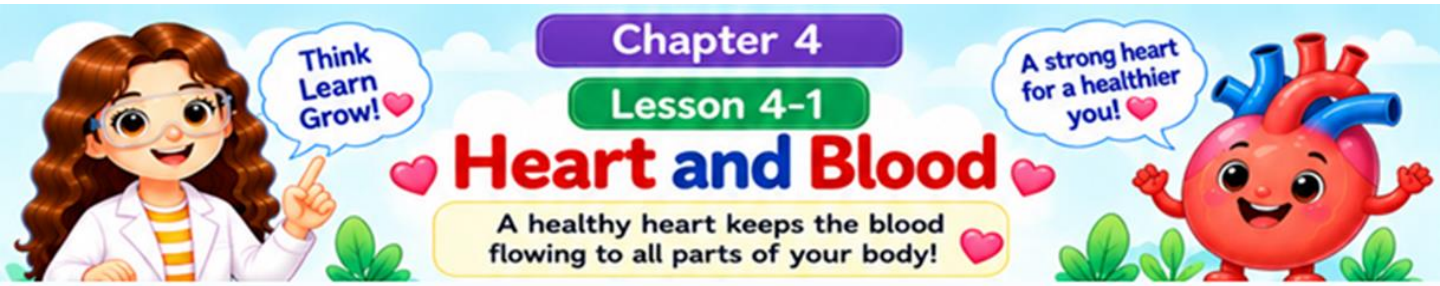
5. Oxygen-rich blood is pumped from the heart to different parts of the body. ( )

C. Complete the Following

Fill in the blanks with the correct words.

1. The heart works like a _____ that keeps blood moving around the body.
2. Blood carries oxygen and _____ to the body cells.
3. Carbon dioxide produced by body cells is collected by the _____.
4. Blood flows around the body through tubes called blood _____.
5. Blood returning from the body is sent to the _____ to exchange carbon dioxide for oxygen.

Healthy Hearts · Brighter Lives!



D. What Happens When...?

Write what happens.

1. You exercise and your body needs more oxygen?

2. Oxygen-rich blood leaves the heart and travels to the body cells?

3. Blood reaches the body cells after carrying oxygen and nutrients?

4. Blood containing carbon dioxide returns to the heart?

5. Blood reaches the lungs?

E. Give Reasons for the Following

Write the reason.

1. The heart beats continuously throughout a person's life.

2. The heart beats faster during exercise.

3. Blood must circulate to all parts of the body.

4. Blood is sent from the heart to the lungs.

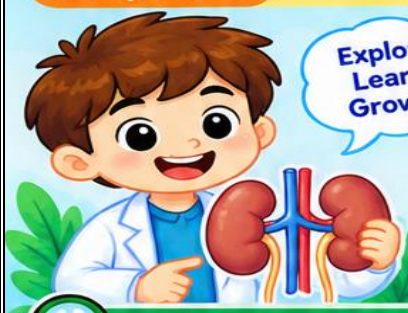
5. Blood returns to the heart after passing through the body.

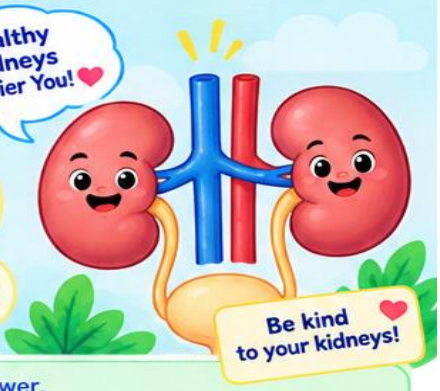
Chapter 4 Circulation and Excretion

Lesson 4-2

How do Kidneys Work?

Healthy kidneys keep your body clean and healthy!





A. Choose the Correct Answer

Choose the best answer.

1 Which organs are mainly responsible for removing wastes from the blood?

a) Lungs
b) Kidneys
c) Stomach
d) Heart

2 What is the main function of the kidneys?

a) To pump blood around the body
b) To filter the blood and remove wastes
c) To store urine
d) To carry urine out of the body

3 Where are the kidneys located in the human body?

a) In the chest
b) In the lower part of the back
c) Inside the head
d) In the neck

4 Which organ stores urine until it is ready to leave the body?

a) Kidney
b) Ureter
c) Bladder
d) Urethra

5 A person's kidneys are working properly. Which result would you expect?

a) Wastes remain in the blood.
b) Cleaned blood returns to the body.
c) Urine remains permanently in the kidneys.
d) The body stops producing wastes.

B. Write True or False

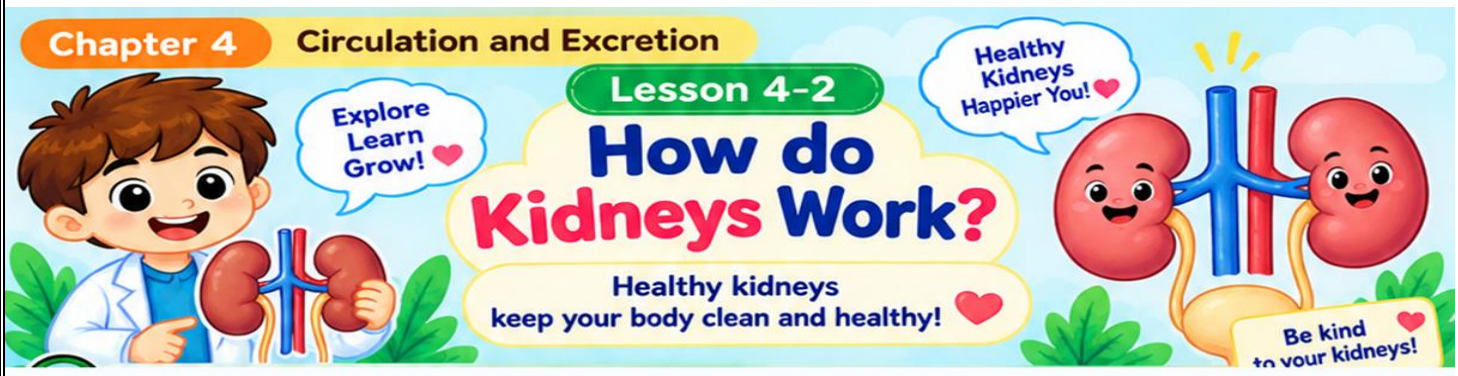
Write True (T) or False (F).

1. Excretion is the process by which the body removes harmful and unwanted substances. ( )
2. The kidneys help control the amount of water and salts in the body. ( )
3. The bladder produces urine by filtering the blood. ( )
4. The urinary system includes the kidneys, ureters, bladder, and urethra. ( )
5. Urine contains wastes and extra water removed from the blood. ( )

C. Complete the Following

Fill in the blanks with the correct words.

1. Unwanted substances produced by cells are called _____ products.
2. The process by which the body removes harmful and unwanted substances is called _____.
3. The urinary system removes waste products from the _____.
4. The kidneys help balance the amount of water and _____ in the body.
5. The kidneys remove wastes from the blood and get rid of them from the body in the form of _____



D. What Happens When...?

Think about the result, then answer.

1. Blood containing wastes enters the kidneys?
2. The kidneys remove wastes and extra water from the blood?
3. Urine leaves the kidneys and enters the ureters?
4. The bladder fills with urine?
5. The body fails to get rid of urine properly?

E. Give Reasons for the Following

Explain why each statement is true.

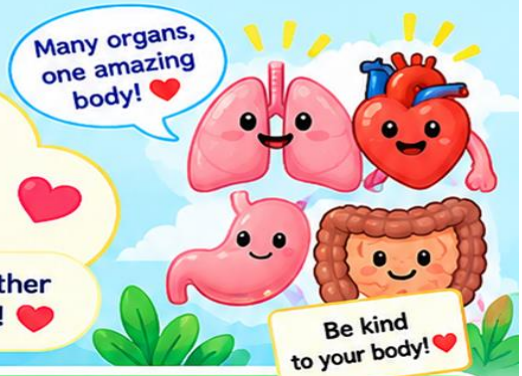
1. The kidneys are described as filters of the body.
2. The urinary system is important for excretion.
3. The kidneys help keep the body healthy.
4. The bladder is an important part of the urinary system.
5. Wastes produced by body cells must be removed from the body.

Chapter 4 Circulation and Excretion

Lesson 4-3

Various Organs

Different organs work together to keep us alive and healthy!



Try

Answer the following questions.

1 (1) What is the organ called that filters the blood and filters out waste products?

(2) What is the liquid waste produced in (1) called?

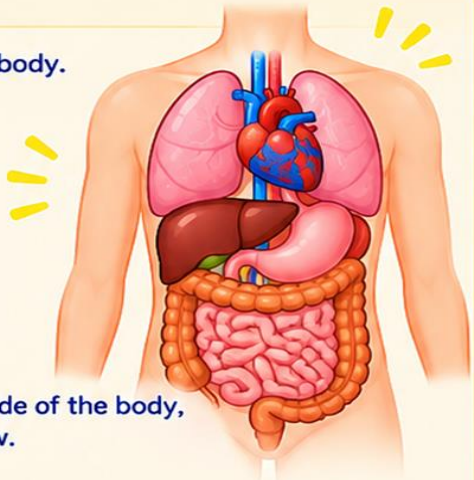
(3) How many of (1) are there in the body?

Exercise

Choose the correct answer.

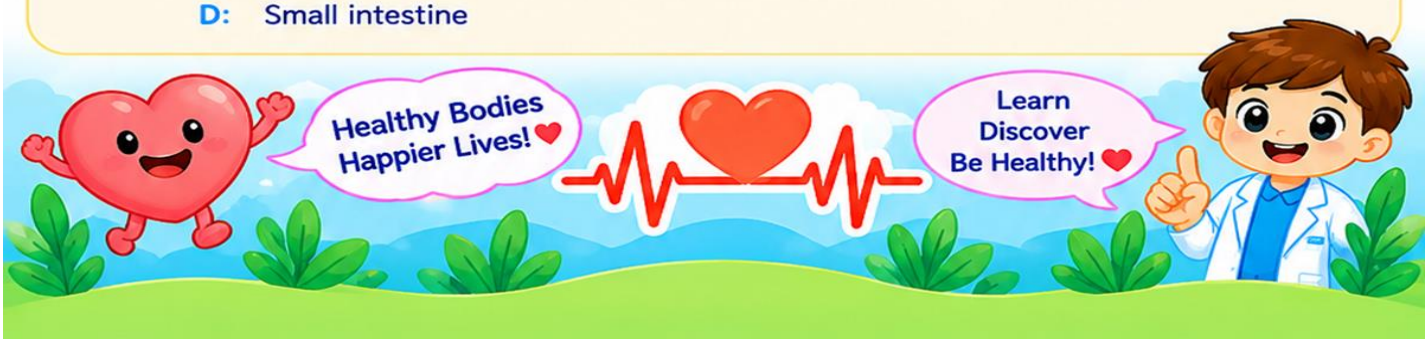
2 (1) Choose 2 organs that are inside the chest of the human body.

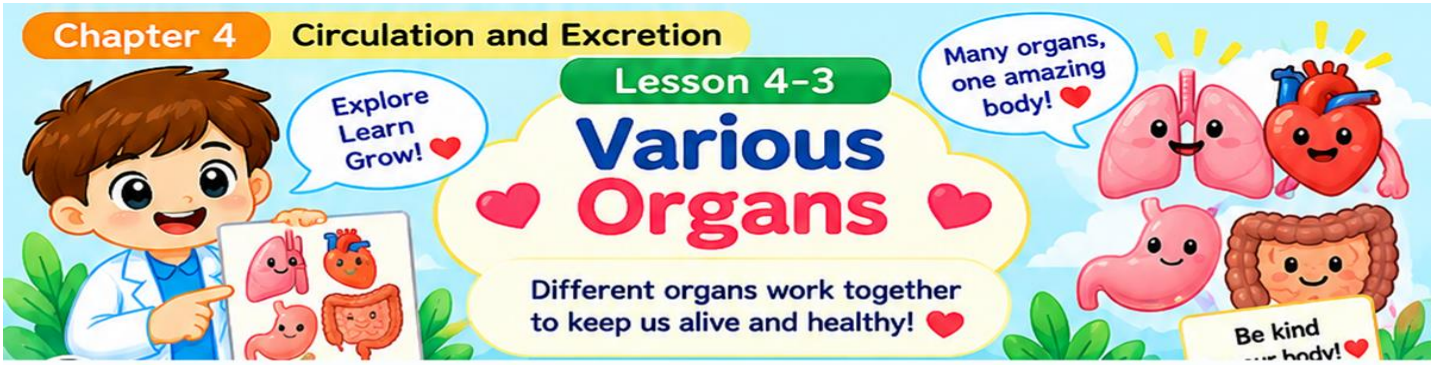
A: Stomach
B: Heart
C: Lungs
D: Large intestine
E: Small intestine



3 (2) Among the following organs, which one has one on each side of the body, for a total of two? Choose the correct answer from below.

A: Lungs
B: Stomach
C: Heart
D: Small intestine





Exercise

Continue the lesson questions.

1 Which statement correctly explains how the kidneys work? Choose the correct answer.

A: It absorbs oxygen from the blood.

B: It pumps blood to the whole body.

C: It takes out waste products from the blood.

Where is the urine produced in the kidneys temporarily stored until it is discharged out of the body?

2 Which statement correctly describes where the organs are located? Choose the correct answer.

A: The small intestine is outside the large intestine and surrounds the large intestine.

B: The stomach is connected to the food pipe.

C: The stomach is on the lower side of the abdomen.

D: There are two kidneys on the abdomen side.

Fill in the blank with the correct word.

Each organ is not moving separately, but all are connected by (_____) and work together.

Chapter 5 Photosynthesis

Lesson 5-1

Sunlight and Starch

Sunlight helps plants make their own food!

Try

Answer the following questions.

1 (1) What nutrient is produced when sunlight hits the leaves of plants?

2 (2) What color does a leaf that was in the sunlight turn when soaked in iodine solution? Choose the correct answer from below.

A: Does not change
B: Red
C: Blue-black
D: Yellow

3 (3) In the experiment to check if sunlight is needed to produce starch, what is the condition you change?

Exercise

Answer the following questions.

1 (1) If you hide a part of a leaf with aluminum foil to block the sunlight, how will that part react to the iodine solution reaction?

2 (2) Which of the following explanations is correct? Choose the correct answer from below.

A: Plants produce starch by receiving sunlight.
B: Plants produce starch by blocking sunlight.

Chapter 5 Photosynthesis

Lesson 5-1

Sunlight and Starch

Sunlight helps plants make their own food!

A Choose the Correct Answer

Circle the correct answer.

1. What food substance can be stored in a plant leaf after it receives sunlight?
   a) Starch b) Iodine c) Ethanol d) Salt
2. Which material is best for preventing light from reaching part of a leaf?
   a) Clear plastic b) Aluminum foil c) Tissue paper d) Water
3. Which observation is evidence that a tested leaf contains starch?
   a) It becomes soft. b) It turns blue-black. c) It becomes green. d) It produces bubbles.
4. In the sunlight experiment, which factor should be different between the two parts of the leaf?
   a) Amount of sunlight received b) Type of plant
   c) Amount of water d) Type of soil
5. Which result best supports the conclusion that sunlight is needed to make starch?
   a) Both parts of the leaf turn blue-black.
   b) Neither part changes color.
   c) Only the part exposed to sunlight turns blue-black.
   d) Only the covered part turns blue-black.

B True or False

Write True (T) or False (F).

1. Leaves can make food using energy from sunlight. ( )
2. The leaf should be tested for starch immediately without any preparation. ( )
3. The covered and uncovered parts of the same leaf can be compared in this experiment. ( )
4. A yellow or brown color after adding iodine indicates that starch is absent. ( )
5. The starch test can provide evidence about whether a leaf has made food. ( )

C Complete the Following

Fill in the blanks with the correct word.

1. Green plants make their own food by a process called _____ .
2. The leaf is first placed in _____ water to make it soft.
3. The green pigment found in leaves is called _____ .
4. During the starch test, _____ is used to remove the green color from the leaf.
5. A leaf that does not contain starch remains _____ or brown after iodine is added.

Chapter 5 Photosynthesis

Lesson 5-1

Sunlight and Starch

Sunlight helps plants make their own food!

Explore Learn Grow!

Plants use sunlight to make food!

Be kind to plants!

D What Happens When...?

Write what happens in each situation.

1. A healthy green plant is kept in darkness for a long time before being tested for starch?

2. The aluminum foil is removed from the leaf after it has been exposed to sunlight?

3. The green color is removed from the leaf during the experiment?

4. The uncovered part of the leaf is tested with iodine after receiving sunlight?

5. The covered and uncovered parts of the leaf are compared after the iodine test?

E Give Reasons for the Following

Write the reason for each statement.

1. The leaf is placed in hot water at the beginning of the starch test.

2. Aluminum foil should completely cover the selected part of the leaf.

3. The plant is placed in sunlight after part of its leaf has been covered.

4. The leaf must be handled carefully after it is heated in ethanol.

5. Iodine solution is added to the leaf at the end of the experiment.

Healthy Plants Happier Lives!

Learn Discover Be Healthy!

Chapter 5 Photosynthesis

Lesson 5-2

Gas Exchange

Plants take in and release gases to stay alive!

Try

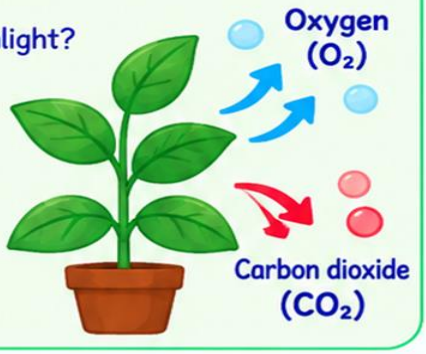
Answer the following questions.

1 What gas do plants release into the air when they are in the sunlight?

A: Oxygen B: Carbon dioxide C: Nitrogen

2 What gas do plants take in from the air when they breathe? Choose the correct answer from below.

A: Oxygen B: Carbon dioxide C: Nitrogen



Exercise

Choose the correct answer.

1 During the night, when there is no light, what kind of gas exchange do plants do? Choose the correct answer from below.

A: Take in oxygen and release carbon dioxide (breathing).

B: Take in carbon dioxide and release oxygen.

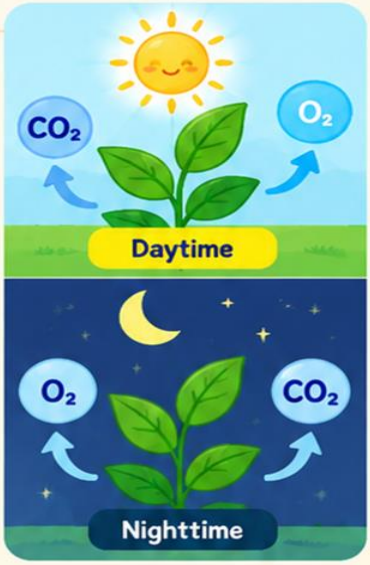
C: Do not take in or release any gases.

2 Which statement correctly explains plants and air? Choose the correct answer from below.

A: Plants keep releasing only oxygen all day long.

B: During the daytime, they take in more carbon dioxide than the carbon dioxide they release by breathing.

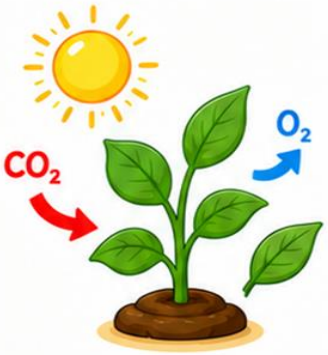
C: While the sunlight is hitting them, plants do not breathe.



Lesson 5-2
Gas Exchange in Plants

1 Complete the Following

1. During photosynthesis, plants take in _____ from the air and release _____ .
2. At night, plants take in _____ during respiration.
3. Plants exchange gases with the surrounding air during both _____ and _____ .
4. In sunlight, the amount of oxygen inside the sealed bag _____ after one hour.
5. The process by which green plants use light to make their food is called _____ .



2 Write True or False

1. Plants stop exchanging gases completely when there is no sunlight. ( )
2. During the daytime, a plant may carry out both photosynthesis and respiration. ( )
3. The percentage of carbon dioxide inside the bag becomes lower after the plant stays in sunlight for some time. ( )
4. At night, plants release more oxygen than carbon dioxide. ( )
5. A gas detector tube can be used to measure the percentage of a gas in the air. ( )

3 Write the Scientific Term

1. The process in which plants use light energy to make their own food. _____
2. The gas taken in by plants to make food in the presence of light. _____
3. The gas released in a greater amount by plants during photosynthesis. _____
4. The process by which plants take in oxygen and release carbon dioxide to obtain energy. _____
5. A device used in the experiment to draw a sample of gas from inside the bag. _____

Lesson 5-2
Gas Exchange in Plants

D What Happens When...?

4 Predict what will happen in each situation.

1. A green plant is kept in sunlight inside a sealed transparent bag for one hour?

   ---

2. A plant is kept in complete darkness for several hours?

   ---

3. The oxygen percentage inside the bag is measured after the plant has been exposed to sunlight?

   ---

4. The carbon dioxide percentage is compared before and after the plant stays in sunlight?

   ---

5. The light reaching a green plant is removed while gas exchange continues?

E Give Reasons for the Following

5 Explain why each statement is true.

1. The oxygen percentage inside the bag increases after the plant is left in sunlight.

   ---

2. Plants release more carbon dioxide at night than during the daytime.

   ---

3. The bag used in the experiment should be sealed tightly.

   ---

4. Two gas readings are taken during the experiment—one at the start and another after one hour.

   ---

5. The plant in the experiment is placed in sunlight before taking the second gas reading.

Lesson 6-1 Transpiration

A Choose the Correct Answer

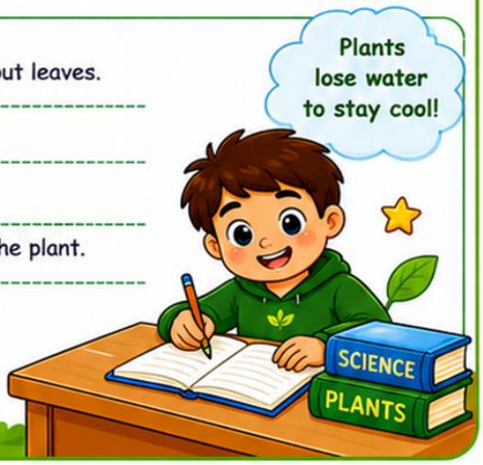
1. Which part of the plant absorbs water from the soil?
   A. Flowers B. Roots C. Leaves D. Fruits
2. Water vapor leaves the plant mainly through tiny pores in the leaves called _____.
   A. roots B. veins C. stomata D. stems
3. Which of the following describes transpiration?
   A. The absorption of sunlight by leaves
   B. The release of water vapor from the leaves
   C. The production of food by plants
   D. The absorption of gases by the roots
4. In the experiment, which plant will have more water droplets inside its plastic bag after being left in sunlight?
   A. Plant A with leaves B. Plant B without leaves
   C. Both plants equally D. Neither plant
5. Which of the following is an important role of transpiration?
   A. It prevents water from entering the roots.
   B. It stops minerals from moving through the plant.
   C. It helps cool the plant.
   D. It prevents the plant from releasing water vapor.

B What Happens When...?

1. A plant with leaves is covered with a clear plastic bag and left in sunlight for some time?
   _____
2. Water is absorbed from the soil by the roots?
   _____
3. Water reaches the leaves and moves through the stomata?
   _____
4. A plant loses water through transpiration on a warm sunny day?
   _____
5. The stomata close during the night?
   _____

C Give Reasons for the Following

1. Plant A with leaves has more water droplets inside its bag than Plant B without leaves.
   _____
2. Two similar plants are used in the transpiration experiment.
   _____
3. The plastic bags are tied tightly around the plants during the experiment.
   _____
4. Transpiration helps in the upward movement of water and minerals through the plant.
   _____
5. Stomata are important for reducing water loss from the plant.



Chapter 6
Transportation of Water in Plants
6-2 The Path of Water

Try

1 (1) What is it called when the water inside a plant changes into water vapor and is released from the leaves?

---

(2) In the experiment to check that water comes out of the leaves, what is the reason why water drops formed inside the plastic bag? Choose the correct answer from below.

A. Because it rained from outside.

B. Because water vapor came out of the leaves and changed into water.

C. Because the plant breathed and released carbon dioxide.

Exercise

1 (1) What are the small pores called, which are mostly on the underside of leaves, that act as exits for water vapor?

---

(2) Which statement correctly explains transpiration? Choose the correct answer from below.

A. Transpiration actively happens from the stems, not the leaves.

B. Transpiration is when plants release oxygen.

C. Transpiration happens through the small pores on the underside of the leaves, which act as exits for water vapor.

2 (1) Which statement correctly explains the path of water in plants? Choose the correct answer from below.

A. Water flows freely everywhere inside the stem.

B. Tubes are connected from the roots to the tips of the leaves to carry water.

C. The water absorbed from the roots passes through the stems, but does not go to the leaves.

(2) When a plant is wilted, what should you do to recover it?

Chapter 6

Transportation of Water in Plants

6-2 The Path of Water

A What Happens When...?

Answer the following questions.

1 The roots of a plant are placed in colored water for a few hours?

2 The colored water absorbed by the roots moves upward through the xylem vessels?

3 The roots, stem, and leaves are cut after the plant has been standing in colored water for several hours?

B Give Reasons for the Following

Write the reason for each statement.

1 The roots of the plant are washed before they are placed in the colored water.

2 Colored water is used instead of clear water in the experiment.

3 The cut surfaces of the roots, stem, and leaves show colored parts at the end of the experiment.

Lesson 7-1: Food Chain

Let's see how living things are connected by what they eat!

A Complete the Following

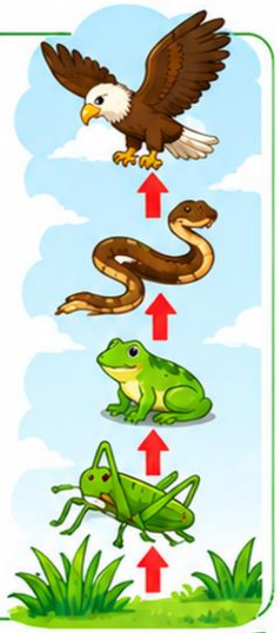
Fill in the blanks with the correct words.

1. A series of living organisms connected to one another by eating and being eaten is called a _____.
2. Green plants are called _____ because they make their own food.
3. Animals that eat plants are known as _____.
4. In a food chain, _____ passes from one organism to the next.
5. Animal at the top of a food chain that has no natural predator in that chain is called _____ predator.

B Write True or False

Write T for true or F for false.

1. All living organisms in a food chain depend directly or indirectly on other organisms for food. ( )
2. Carnivores get their energy by eating only plants. ( )
3. A primary consumer comes after a producer in a food chain. ( )
4. The arrows in a food chain point toward the organism receiving the energy. ( )
5. A frog that eats a grasshopper can act as a secondary consumer. ( )



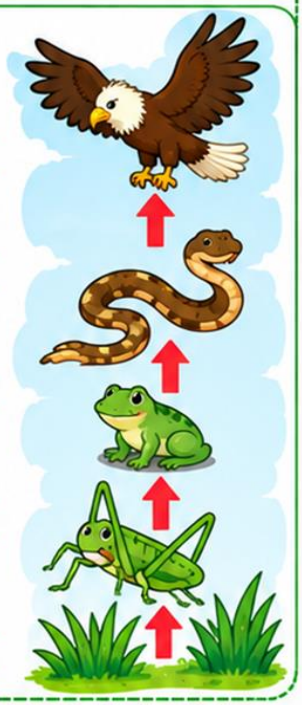
Lesson 7-1: Food Chain

Let's think more about food chains!

Living Things Are Connected!

C What Happens When...?

Write what happens in each situation.



1. A rabbit eats a green plant in a food chain?
2. A snake eats a frog that has eaten a grasshopper?
3. The number of producers in a food chain decreases greatly?
4. One organism disappears completely from a food chain?
5. A top predator is removed from a food chain?

D Give Reasons for the Following

Write the reason for each statement.

Every organism has a role!

1. Plants form the first part of most food chains.
2. A grasshopper is considered a primary consumer when it feeds on grass.
3. An eagle can be considered a top predator in a food chain.
4. Arrows are used between organisms when drawing a food chain.
5. Removing one organism from a food chain can affect several other organisms.

Healthy Ecosystems Need All Organisms!

Explore Inquire Discover!

Chapter 7 Food Chain

7-2 Living Organisms, Air, and Water

1 Try

1 (1) What do we call the feeding relationship between living organisms?

.....

(2) What kind of living organisms are the beginning of the food chain? Choose the correct answer from below.

A: Strong carnivores

B: Herbivores like insects

C: Plants that produce nutrients using sunlight

2 Exercise

1 (1) Arrange the following living organisms in the correct order of the food chain and connect them with arrows.

[Grasshopper / Wheat / Eagle / Frog]

(2) Which statement correctly explains the food chain? Choose the correct answer from below.

A: Arrows are drawn from the one that eats to the one that is eaten.

B: If carnivores disappear, plants will grow healthier.

C: Plants use sunlight to produce nutrients inside their bodies.

2 (1) Which statement correctly explains the relationship between living organisms and air? Choose the correct answer from below.

A: Both animals and plants only decrease oxygen and increase carbon dioxide.

B: The environment where animals can breathe is maintained because plants release oxygen.

(2) About the connection of living organisms through water, choose the correct word from below to fill in the blank.

The water on Earth is changing its shape and (_____), and it is used repeatedly by many living organisms.

circulating / disappearing / increasing

Lesson 7-2: Living Organisms, Air, and Water

Let's check what we have learned!

A Complete the Following

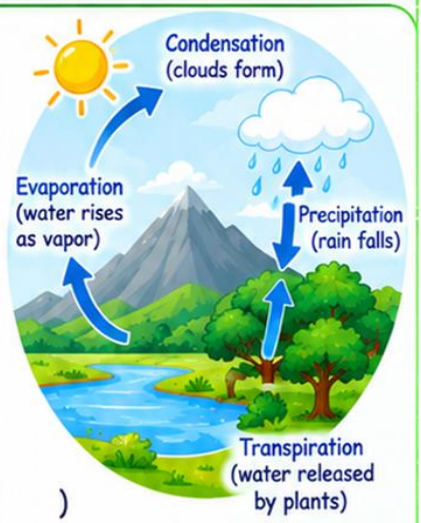
Fill in the blanks with the correct words.

1. Animals breathe in _____ released by plants.
2. Plants take in _____ released by animals for photosynthesis.
3. Plants release water vapor into the air through a process called _____.
4. The process in which water vapor changes into clouds is called _____.
5. Water falling from clouds to the Earth as rain is called _____.

B Write True or False

Write T for true or F for false.

1. Animals and plants help exchange oxygen and carbon dioxide in the environment. ( )
2. Transpiration adds water vapor to the atmosphere. ( )
3. Water disappears permanently after it evaporates from rivers and lakes. ( )
4. Plants are connected to both the air and the water cycle. ( )
5. Condensation occurs when water vapor forms clouds in the atmosphere. ( )



Lesson 7-2: Living Organisms, Air, and Water

Let's think more about air and water!

Caring for Air, Water, and Living Things!

C What Happens When...?

Write what happens in each situation.

1. An animal breathes out carbon dioxide near a green plant?
   _____
   _____
2. Water evaporates from rivers, lakes, and oceans and rises into the atmosphere?
   _____
   _____
3. Water vapor in the atmosphere cools and condenses?
   _____
   _____
4. Rainwater reaches the soil around a plant?
   _____
   _____
5. Plants completely disappear from an environment?
   _____
   _____

D Give Reasons for the Following

Write the reason for each statement.

1. Animals depend on plants for oxygen.
   _____
   _____
2. Plants and animals are connected through the gases in the air.
   _____
   _____
3. Transpiration is considered an important part of the water cycle.
   _____
   _____
4. Water on Earth is continuously circulated rather than being used only once.
   _____
   _____
5. The disappearance of plants can disturb the balance of oxygen and carbon dioxide in the air.
   _____
   _____

Chapter 8
Egypt Special Part 1

8-1 Desert and Nile Food Chains

Try

1 Give one example of a top carnivore at the top of the food chain in the Nile River.

_____

2 If all the plants in the desert died out, what do you think would happen to the carnivores? Write your own thoughts.

_____

_____

Exercise

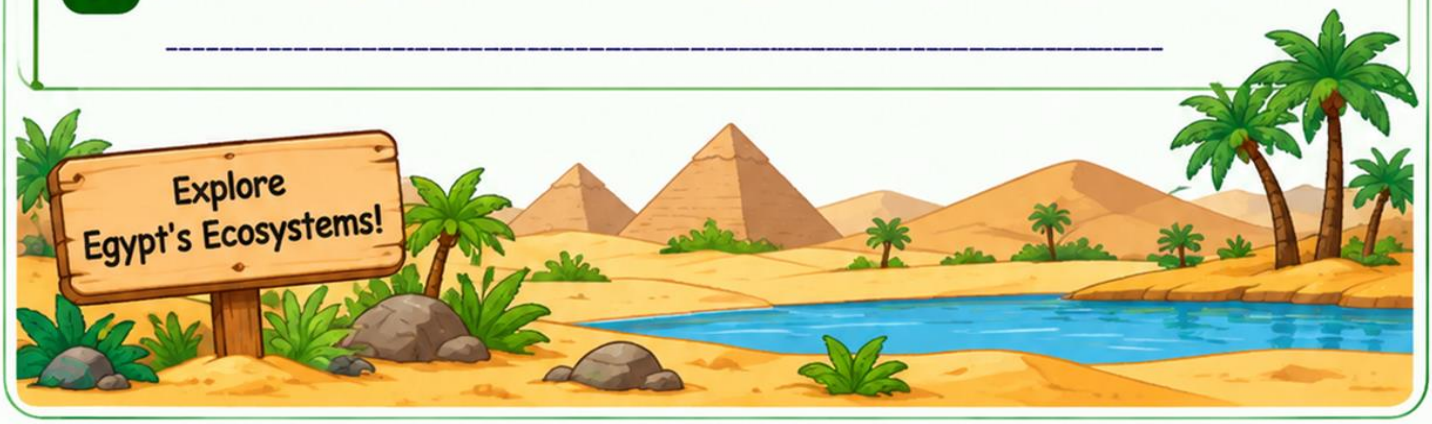
1 Which statement correctly explains the food chains in Egypt? Choose the correct answer from below.

A: In the Nile River, crocodiles eat algae directly.

B: In the desert, fennec foxes eat small animals like jerboas.

2 Give one example of a herbivore that eats plants in the desert food chain.

_____

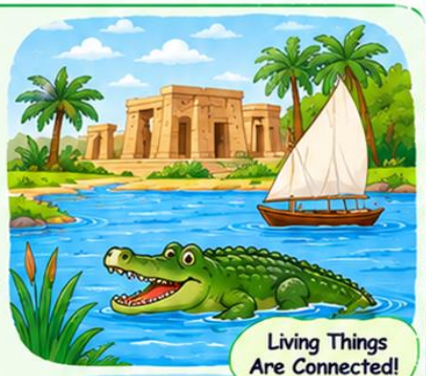


Lesson 8-1: Desert and Nile Food Chains

A Choose the Correct Answer

Read each question and choose the best answer (A, B, C, or D).

1. Which organism is a producer in the Nile River food chain?
   A. Nile perch B. Crocodile C. Algae D. Nile tilapia
2. Which animal can be a top carnivore in a desert food chain?
   A. Grasshopper B. Fennec fox C. Jerboa D. Desert grass
3. In a Nile River food chain, which organism may feed on a herbivorous fish?
   A. Carnivorous fish B. Algae C. Water plants D. Phytoplankton
4. Why are plants found at the base of a food pyramid?
   A. They eat other organisms.
   B. They make their own food using sunlight.
   C. They are top carnivores.
   D. They get energy from animals.
5. What would most likely happen to fennec foxes if the number of small desert animals decreased greatly?
   A. They would have less food available.
   B. They would become producers.
   C. They would make their own food.
   D. Their food supply would increase.



B Write True or False

Write T for true or F for false.

1. Desert shrubs and grasses can act as producers in a desert food chain. ( )
2. A crocodile gets its energy directly from algae in the Nile River food chain. ( )
3. Herbivores get energy by feeding on producers. ( )
4. The organisms at different levels of a food pyramid depend on one another for energy. ( )
5. Fossils at Wadi al-Hitan provide evidence that the area was once an ancient ocean. ( )

C Complete the Following

Fill in the blanks with the correct words.

1. A diagram showing the feeding levels in a food chain is called a food _____.
2. In the Nile River, fish such as Nile tilapia can act as _____ consumers when they feed on producers.
3. Animals such as fennec foxes that feed on other animals are called _____.
4. The food chain always begins with _____ that make their own food.
5. Wadi al-Hitan is also known as the Valley of the _____.

Even if the types of animals are different,
the chain always starts with plants and ends with large carnivores.

Chapter 9
Lesson 9-1: Properties of Solutions

Explore Investigate Learn Grow!

Different solutions, different properties!

A Choose the Correct Answer

Read each question and choose the best answer (A, B, C, or D).

1 Which statement best describes a solution?
- A. A mixture in which the substances can always be seen separately
- B. A mixture in which a solute dissolves completely in a solvent
- C. A substance made only of water
- D. A mixture that must contain a solid

2 Which of the following is an acidic solution?
- A. Salt water
- B. Baking soda solution
- C. Dilute hydrochloric acid
- D. Dilute ammonia solution

3 A solution turns red litmus paper blue. How should it be classified?
- A. Acidic
- B. Neutral
- C. Alkaline
- D. Pure water only

4 Which pair of litmus results indicates a neutral solution?
- A. Blue turns red; red stays red
- B. Blue stays blue; red stays red
- C. Blue stays blue; red turns blue
- D. Both papers turn blue

5 A student tests an unknown solution. The blue litmus paper stays blue and the red litmus paper turns blue. Which solution could it be?
- A. Lemon juice
- B. Vinegar
- C. Salt water
- D. Baking soda solution

Small Tests Big Discoveries!

B Write True or False

Write T for true or F for false.

1 The substance that is dissolved in a solution is called the solute. ( )

2 Carbonated water is an example of an acidic solution. ( )

3 Salt water is an example of a neutral solution. ( )

4 Dilute ammonia solution can be classified as alkaline using litmus paper. ( )

5 A solution usually looks uniform throughout when the solute has completely dissolved. ( )

C Complete the Following

Fill in the blanks with the correct words.

1 A mixture that looks the same throughout is called a _____ mixture.

2 In salt water, the salt is the _____.

3 Lemon juice and vinegar belong to the _____ group of solutions.

4 Solutions can be divided into _____ groups according to how they affect litmus paper.

5 In the activity, a _____ is used to place one drop of solution onto the litmus strips.

Understanding the properties of solutions helps us use them safely and wisely in everyday life!

Chapter 9
Lesson 9-1: Properties of Solutions

Different solutions, different properties!

D What Happens When...?

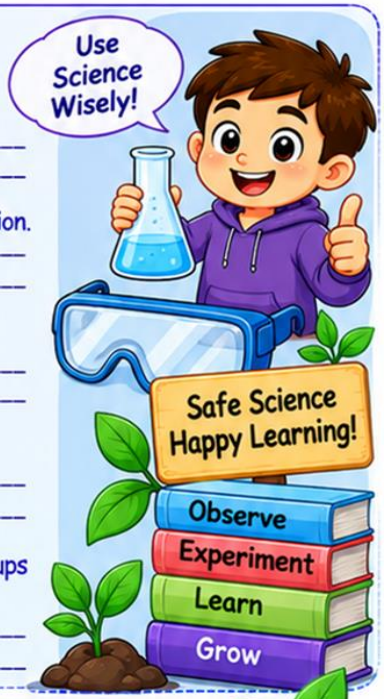
Write what happens in each situation.

1. Blue litmus paper is dipped into dilute hydrochloric acid?
   _____
   _____
2. Red litmus paper is tested with a baking soda solution?
   _____
   _____
3. Both red and blue litmus papers are tested with salt water?
   _____
   _____
4. Salt is added to water and stirred until it completely dissolves?
   _____
   _____
5. An unknown solution changes blue litmus paper to red but does not change red litmus paper?
   _____
   _____

E Give Reasons for the Following

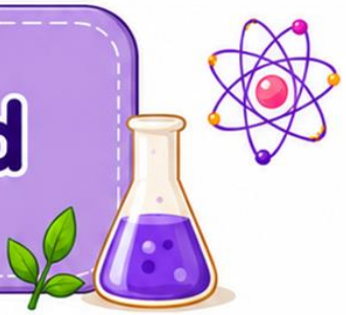
Write the reason for each statement.

1. Litmus paper can be used to group different solutions.
   _____
   _____
2. A glass rod should be rinsed with water and wiped before testing another solution.
   _____
   _____
3. Safety goggles should be worn while testing different solutions.
   _____
   _____
4. Salt water is classified as a neutral solution.
   _____
   _____
5. Dilute hydrochloric acid and dilute ammonia solution are placed in different groups even though both are solutions.
   _____
   _____



Exploring solutions today for a brighter tomorrow!

Lesson 9-2: What is Dissolved in Solutions?



Try

1 (1) Which is the correct combination for the litmus paper reaction? Choose the correct answer from below.

- **A:** An alkaline solution turns red litmus paper blue.
- **B:** An acidic solution turns red litmus paper blue.
- **C:** A neutral solution changes the color of both litmus papers.

2 (2) What are some neutral solutions? Answer with one example.

_____

Exercise

1 (1) What are the white solid left at the bottom of the dish when salt water was evaporated?

_____

2 (2) Why is nothing left at the bottom of the dish when dilute hydrochloric acid was evaporated? Choose the correct answer from below.

- **A:** Because nothing was dissolved in the dilute hydrochloric acid.
- **B:** Because what was dissolved escaped as a gas.
- **C:** Because it melted from the heat.

Remember!

- Some dissolved substances remain after evaporation (solids).
- Some dissolved substances do not remain after evaporation (gases).

Chapter 9
Lesson 9-2: What is Dissolved in Solutions?

Evaporation helps us discover what is dissolved!

A Choose the Correct Answer

Read each question and choose the best answer (A, B, C, or D).

1 Which solution contains a gas dissolved in water?

A. Salt water B. Baking soda solution
C. Carbonated water D. Sugar solution

2 What remains after salt water is completely evaporated?

A. Carbon dioxide B. Salt crystals
C. Water vapor D. Nothing

3 A white solid remains after an unknown solution is evaporated. What can you conclude?

A. The dissolved substance was a solid. B. Nothing was dissolved.
C. The dissolved substance was only a gas. D. The water did not evaporate.

4 Which equipment is used to heat the evaporating dish?

A. Test tube rack B. Portable gas stove
C. Pipette D. Safety goggles

5 Why may nothing remain after evaporating a solution containing a dissolved gas?

A. The gas changes into a solid. B. The gas stays in the dish.
C. The gas escapes into the air. D. The gas changes into water.

B Write True or False

Write T for true or F for false.

1 Evaporation can be used to investigate what is dissolved in some solutions. ( )

2 Baking soda remains as a solid after the water evaporates. ( )

3 All dissolved substances remain after evaporation. ( )

4 The evaporating dish should be heated gently during the activity. ( )

5 Safety goggles help protect the eyes while heating solutions. ( )

C Complete the Following

Fill in the blanks with the correct words.

1 During evaporation, liquid water changes into water _____.

2 The tool used to transfer a small amount of solution is a _____.

3 A shallow dish used for heating a solution is an _____ dish.

4 A dissolved solid does not _____ with the water during evaporation.

5 After evaporating salt water, the substance left behind is _____.

Explore solutions • Observe carefully • Discover with science!

Chapter 9

Lesson 9-2: What is Dissolved in Solutions?

Solids and gases behave differently during evaporation!

E What Happens When...?

Write what happens in each situation.

1. Salt water is heated gently in an evaporating dish until all the water has evaporated?
2. Carbonated water is heated until the water completely evaporates?
3. A baking soda solution is evaporated completely?
4. The water in a solution begins to boil during the heating activity?
5. A student touches the evaporating dish immediately after heating it?

Observe
Think
Predict
Explain!

Heat Gently
Observe Carefully
Stay Safe!

F Give Reasons for the Following

Write the reason for each statement.

1. An evaporating dish is used when investigating what is dissolved in a solution.
2. The solution should be heated gently during the evaporation experiment.
3. A white solid remains after salt water is evaporated.
4. Nothing remains after some solutions containing dissolved gases are evaporated.
5. Students should not bring their faces close to an evaporating dish while a solution is being heated.

Use
Science
Wisely!

Safe Science
Happy Learning!

Evaporate • Observe • Discover what was dissolved!

Lesson 10-1:

Aqueous Solutions and Metals

Activity Let's answer the following questions!

A Choose the correct answer.

1. In an aqueous solution, the solvent is:
   a) Iron b) Water c) Aluminum d) Gas

2. Which observation shows that a reaction occurs when dilute HCl is added to aluminum?
   a) Ice forms b) Bubbles are produced
   c) The liquid disappears immediatelly
   d) The metal turns into water

3. Which liquid is added to aluminum and iron in the activity?
   a) Salt water b) Distilled water
   c) Dilute hydrochloric acid d) Carbonated water

4. Which tool is used to add a small amount of dilute HCl to a test tube?
   a) Pipette b) Test tube rack
   c) Beaker d) Test tube brush

5. Which of the following is a property of many metals?
   a) They are good conductors of heat and electricity.
   b) They are always gases.
   c) They dissolve completely in water.
   d) They cannot conduct heat.

B Write the scientific term.

1. A solution in which water is the solvent.
   _____

2. A laboratory container used to hold small amounts of substances during an experiment.
   _____

3. A tool used to transfer small amounts of liquid.
   _____

4. A material that is generally a good conductor of heat and electricity.
   _____

5. The acid represented by the chemical formula HCl.
   _____

Lesson 10-1:

Aqueous Solutions and Metals

D. What Happens When...?

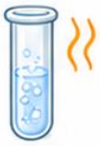
Let's think about what will happen!

1. Dilute hydrochloric acid is added to a small piece of aluminum?

2. Dilute hydrochloric acid is added to iron (steel wool)?

3. Aluminum continues to react with dilute HCl for a short time?

4. You touch the outside of the test tube carefully after the metal has reacted with dilute HCl?



5. Dilute HCl is poured onto aluminum and bubbles appear around the metal?

E. Give Reasons for the Following

Explain your answers!

1. Water is called the solvent in an aqueous solution.

2. Bubbles appear when dilute HCl reacts with aluminum or iron.

3. Aluminum and iron are placed in separate test tubes during the activity.

4. A pipette is used when adding dilute HCl to the test tubes.

Science helps us find the answers!

Lesson 10-2: What Is Left After Evaporation?

Activity Let's answer the following questions!

D What Happens When...?

1. Aluminum is added to dilute hydrochloric acid?

2. The clear solution formed after aluminum reacts with dilute HCl is heated gently in an evaporating dish?

3. The liquid in the evaporating dish evaporates?

4. Dilute hydrochloric acid is added again to the white powder left after evaporation?

5. The solution is heated on high heat until it becomes completely dry?

E Give Reasons for the Following

1. The white powder left after evaporation is considered a different substance from the original aluminum.

2. No bubbles are produced when dilute HCl is added to the white powder left after evaporation.

3. The liquid is evaporated after aluminum reacts with dilute hydrochloric acid.

4. Heating should be stopped while a little liquid is still left in the evaporating dish.

5. We can conclude that aluminum undergoes a chemical change when it reacts with dilute hydrochloric acid.